Assignment 6 extra

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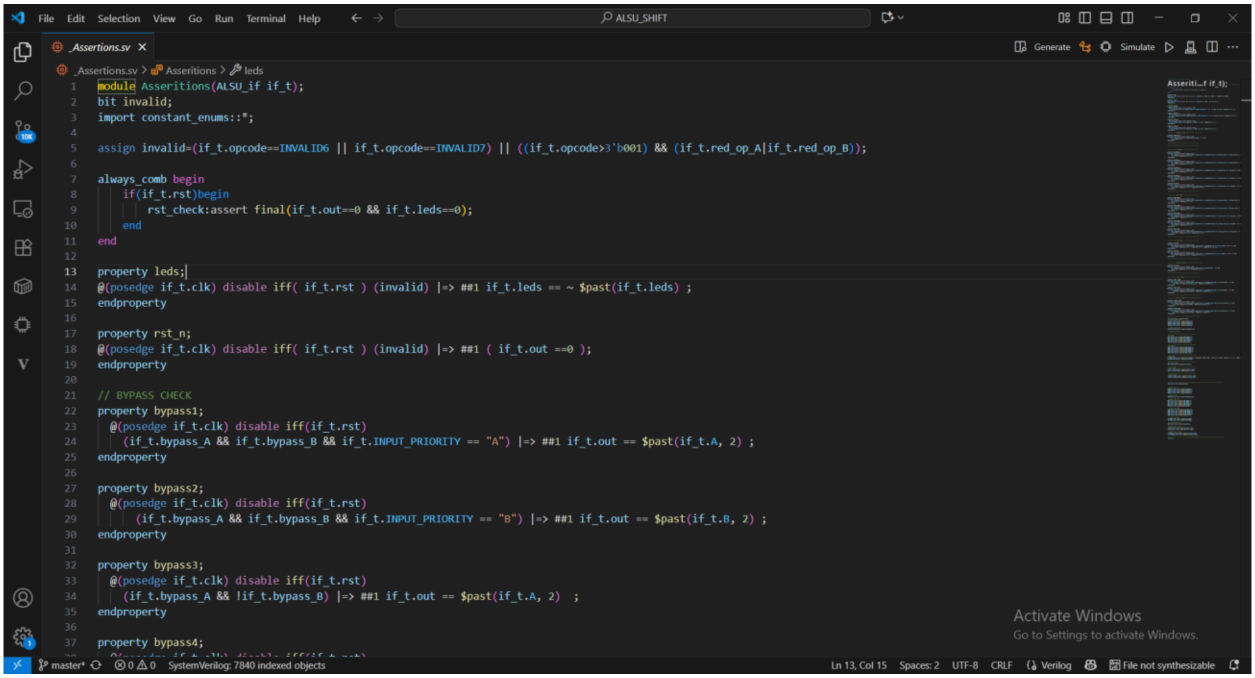
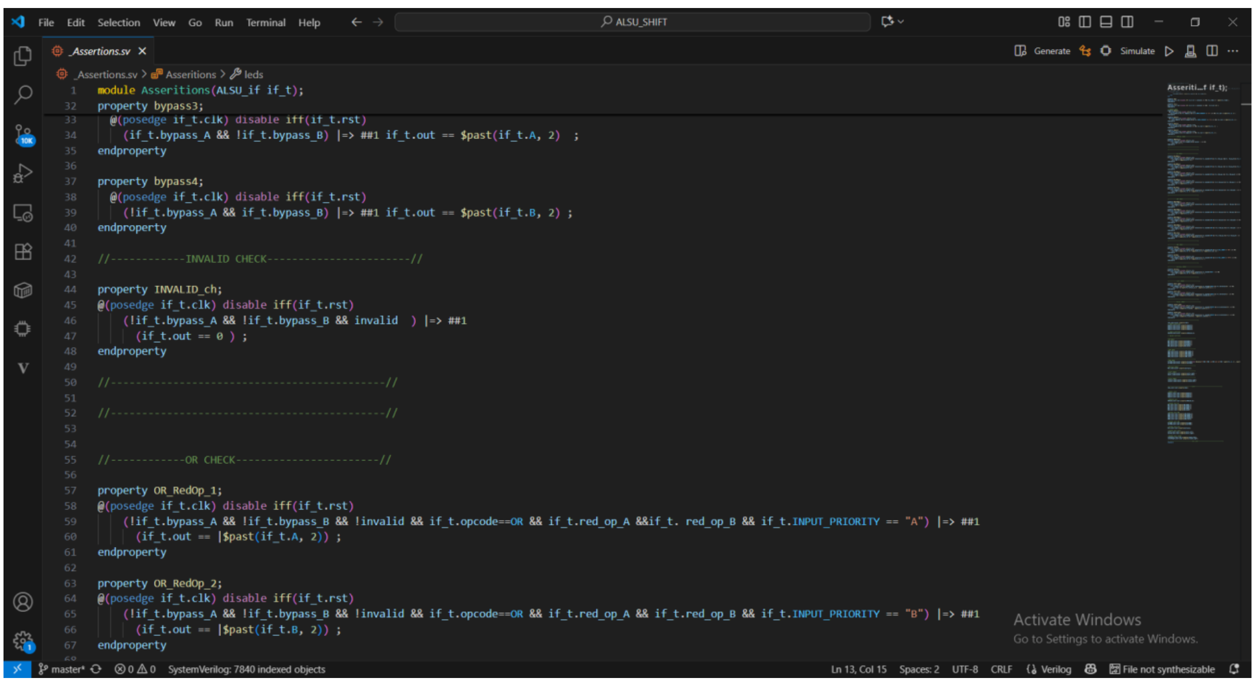
Verification Diploma

1. **Assertions**



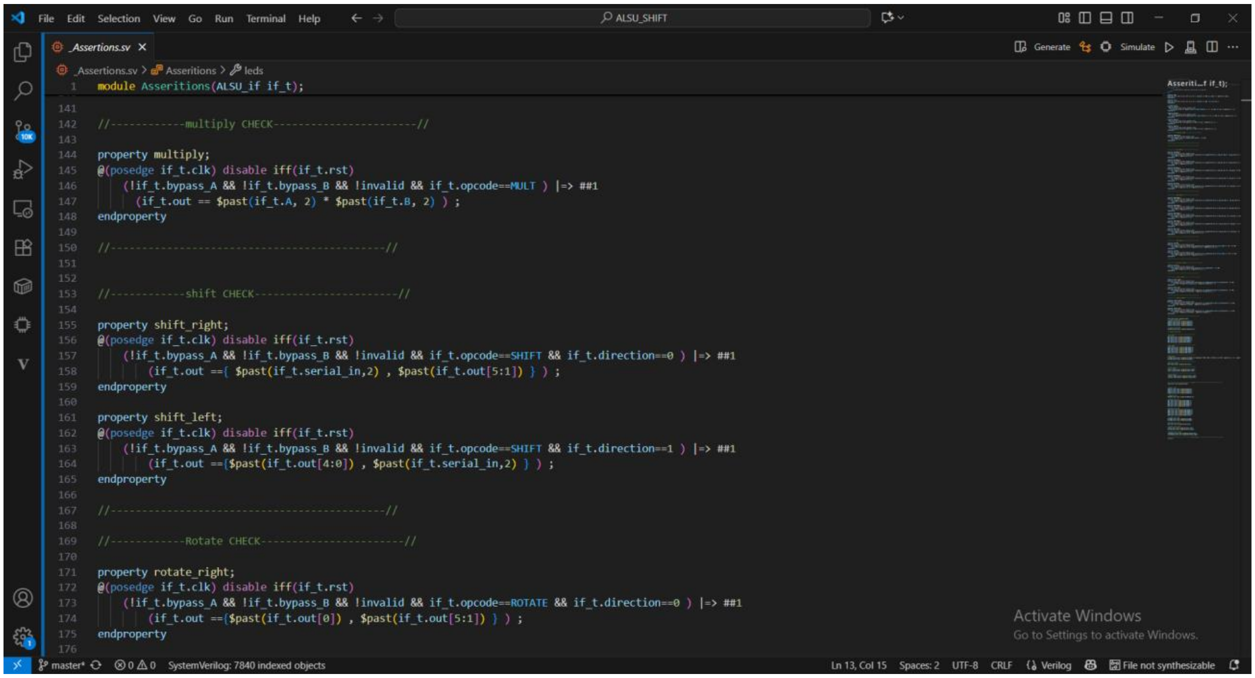
```
module Asseritions(ALSU_if if_t);
bit invalid;
import constant_enums::*;

assign invalid=(if_t.opcode==INVALID6 || if_t.opcode==INVALID7) || ((if_t.opcode>3'b001) && (if_t.red_op_A|if_t.red_op_B));

always_comb begin
    if(if_t.rst)begin
        rst_check:assert final(if_t.out==0 && if_t.leds==0);
    end
end

property leds;
@(posedge if_t.clk) disable iff( if_t.rst ) (invalid) |=> ##1 if_t.leds == ~ $past(if_t.leds) ;
endproperty

property rst_n;
@(posedge if_t.clk) disable iff( if_t.rst ) (invalid) |=> ##1 ( if_t.out ==0 );
endproperty

// BYPASS CHECK
property bypass1;
  @(posedge if_t.clk) disable iff(if_t.rst)
    (if_t.bypass_A && if_t.bypass_B && if_t.INPUT_PRIORITY == "A") |=> ##1 if_t.out == $past(if_t.A, 2) ;
endproperty

property bypass2;
  @(posedge if_t.clk) disable iff(if_t.rst)
      (if_t.bypass_A && if_t.bypass_B && if_t.INPUT_PRIORITY == "B") |=> ##1 if_t.out == $past(if_t.B, 2) ;
endproperty

property bypass3;
  @(posedge if_t.clk) disable iff(if_t.rst)
    (if_t.bypass_A && !if_t.bypass_B) |=> ##1 if_t.out == $past(if_t.A, 2)  ;
endproperty

property bypass4;
```



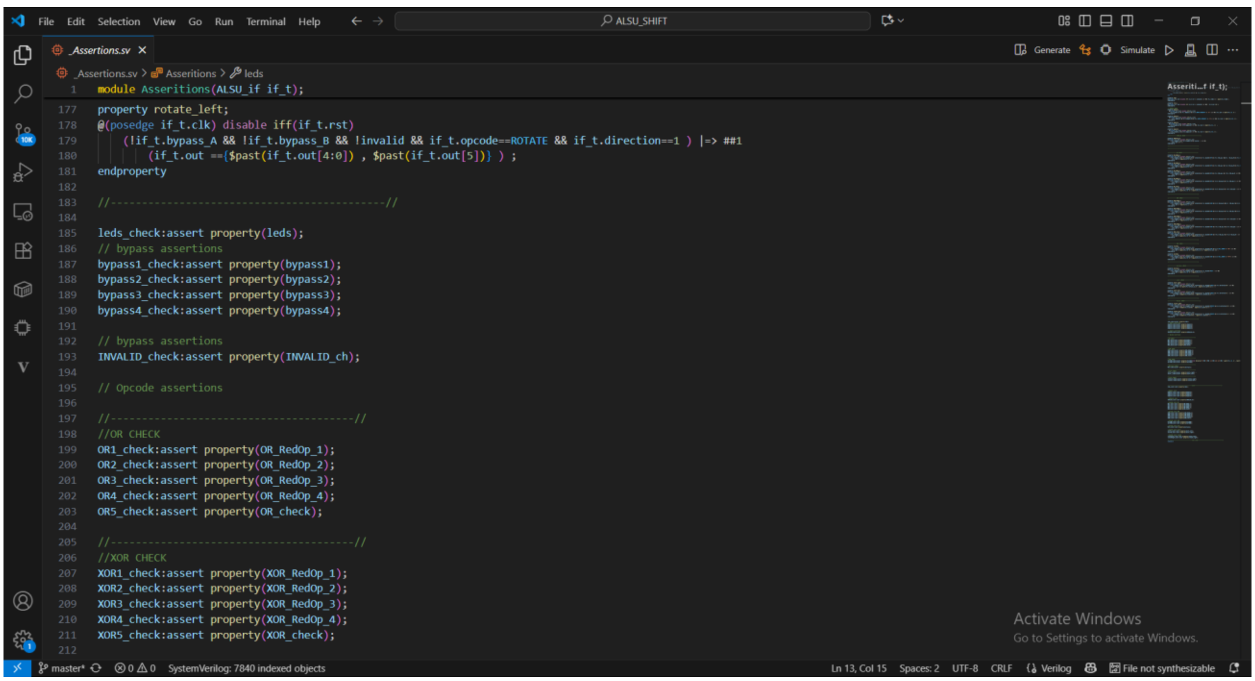
```
property bypass3;
  @(posedge if_t.clk) disable iff(if_t.rst)
    (if_t.bypass_A && !if_t.bypass_B) |=> ##1 if_t.out == $past(if_t.A, 2)  ;
endproperty

property bypass4;
  @(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && if_t.bypass_B) |=> ##1 if_t.out == $past(if_t.B, 2) ;
endproperty

//-----------INVALID CHECK---------------------//

property INVALID_ch;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && invalid  ) |=> ##1
      (if_t.out == 0 ) ;
endproperty

//----------------------------------------//

//----------------------------------------//


//-----------OR CHECK---------------------//

property OR_RedOp_1;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==OR && if_t.red_op_A &&if_t. red_op_B && if_t.INPUT_PRIORITY == "A") |=> ##1
      (if_t.out == |$past(if_t.A, 2)) ;
endproperty

property OR_RedOp_2;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==OR && if_t.red_op_A && if_t.red_op_B && if_t.INPUT_PRIORITY == "B") |=> ##1
      (if_t.out == |$past(if_t.B, 2)) ;
endproperty
```

```systemverilog
module Asseritions(ALSU_if if_t);

property OR_RedOp_3;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==OR && if_t.red_op_A && !if_t.red_op_B) |=> ##1
      (if_t.out == |$past(if_t.A, 2)) ;
endproperty

property OR_RedOp_4;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==OR && !if_t.red_op_A && if_t.red_op_B ) |=> ##1
      (if_t.out == |$past(if_t.B, 2)) ;
endproperty

property OR_check;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==OR && !if_t.red_op_A && !if_t.red_op_B ) |=> ##1
      (if_t.out == $past(if_t.A, 2) | $past(if_t.B, 2)) ;
endproperty

//-----------------------------------------//


//-----------XOR CHECK---------------------//

property XOR_RedOp_1;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==XOR && if_t.red_op_A &&if_t. red_op_B && if_t.INPUT_PRIORITY == "A") |=> ##1
      (if_t.out == ^$past(if_t.A, 2)) ;
endproperty

property XOR_RedOp_2;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==XOR && if_t.red_op_A && if_t.red_op_B && if_t.INPUT_PRIORITY == "B") |=> ##1
      (if_t.out == ^$past(if_t.B, 2)) ;
endproperty
```

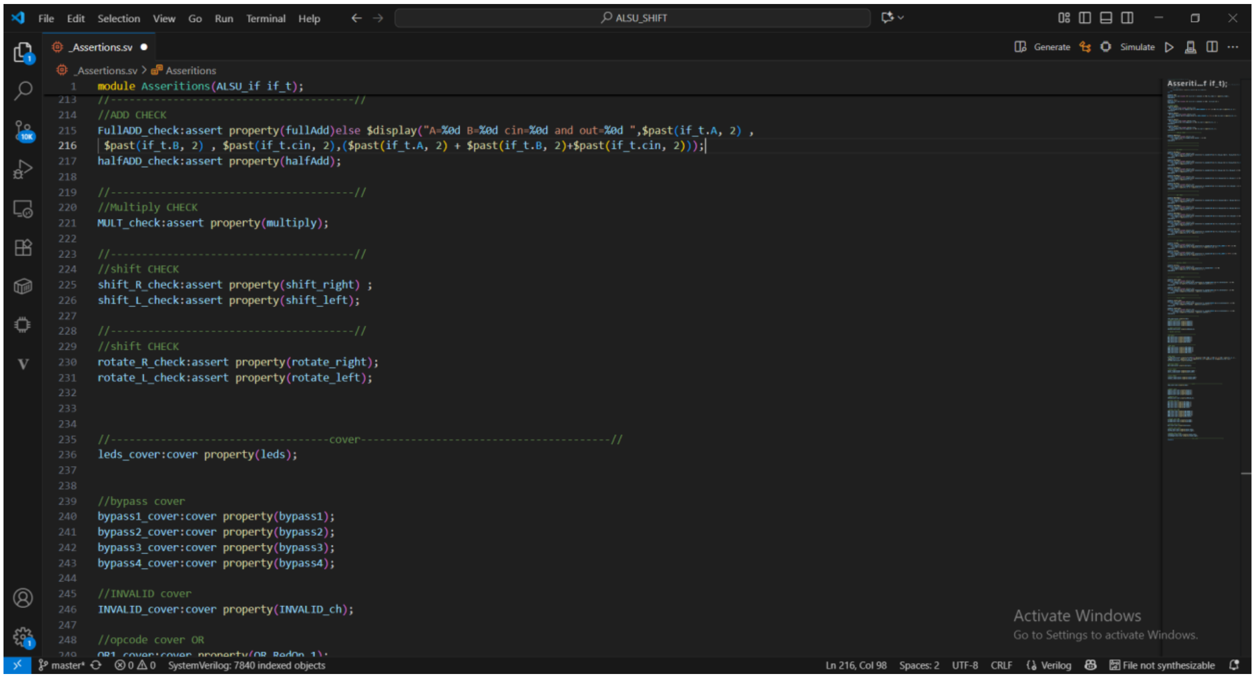
```systemverilog
module Asseritions(ALSU_if if_t);

property XOR_RedOp_3;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==XOR && if_t.red_op_A && !if_t.red_op_B) |=> ##1
      (if_t.out == ^$past(if_t.A, 2)) ;
endproperty

property XOR_RedOp_4;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==XOR && !if_t.red_op_A && if_t.red_op_B ) |=> ##1
      (if_t.out == ^$past(if_t.B, 2)) ;
endproperty

property XOR_check;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==XOR && !if_t.red_op_A && !if_t.red_op_B ) |=> ##1
      (if_t.out == $past(if_t.A, 2) ^ $past(if_t.B, 2)) ;
endproperty

//-----------------------------------------//


//-----------ADD CHECK---------------------//

property fullAdd;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==ADD && if_t.FULL_ADDER == "ON") |=> ##1
      (if_t.out == $past(if_t.A, 2) + $past(if_t.B, 2) + $past(if_t.cin, 2)) ;
endproperty

property halfAdd;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==ADD && if_t.FULL_ADDER == "OFF") |=> ##1
      (if_t.out == $past(if_t.A, 2) + $past(if_t.B, 2) ) ;
endproperty
```



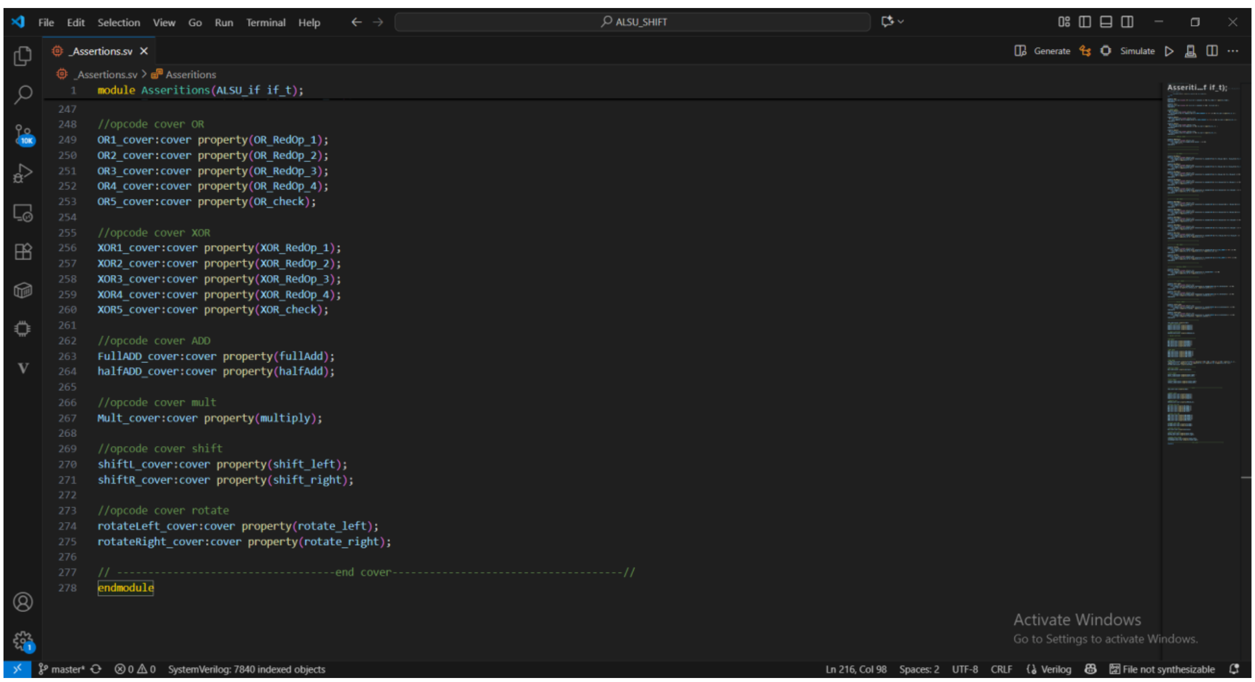
```systemverilog
module Aseritions(ALSU_if if_t);

//-----------multiply CHECK---------------------//

property multiply;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==MULT ) |=> ##1
      (if_t.out == $past(if_t.A, 2) * $past(if_t.B, 2) ) ;
endproperty

//------------------------------------------//


//-----------shift CHECK---------------------//

property shift_right;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==SHIFT && if_t.direction==0 ) |=> ##1
        (if_t.out =={ $past(if_t.serial_in,2) , $past(if_t.out[5:1]) } ) ;
endproperty

property shift_left;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==SHIFT && if_t.direction==1 ) |=> ##1
        (if_t.out =={$past(if_t.out[4:0]) , $past(if_t.serial_in,2) } ) ;
endproperty

//------------------------------------------//

//-----------Rotate CHECK---------------------//

property rotate_right;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==ROTATE && if_t.direction==0 ) |=> ##1
        (if_t.out =={$past(if_t.out[0]) , $past(if_t.out[5:1]) } ) ;
endproperty
```



```systemverilog
module Aseritions(ALSU_if if_t);
property rotate_left;
@(posedge if_t.clk) disable iff(if_t.rst)
    (!if_t.bypass_A && !if_t.bypass_B && !invalid && if_t.opcode==ROTATE && if_t.direction==1 ) |=> ##1
        (if_t.out =={$past(if_t.out[4:0]) , $past(if_t.out[5])} ) ;
endproperty

//------------------------------------------//

leds_check:assert property(leds);
// bypass assertions
bypass1_check:assert property(bypass1);
bypass2_check:assert property(bypass2);
bypass3_check:assert property(bypass3);
bypass4_check:assert property(bypass4);

// bypass assertions
INVALID_check:assert property(INVALID_ch);

// Opcode assertions

//------------------------------------//
//OR CHECK
OR1_check:assert property(OR_RedOp_1);
OR2_check:assert property(OR_RedOp_2);
OR3_check:assert property(OR_RedOp_3);
OR4_check:assert property(OR_RedOp_4);
OR5_check:assert property(OR_check);

//------------------------------------//
//XOR CHECK
XOR1_check:assert property(XOR_RedOp_1);
XOR2_check:assert property(XOR_RedOp_2);
XOR3_check:assert property(XOR_RedOp_3);
XOR4_check:assert property(XOR_RedOp_4);
XOR5_check:assert property(XOR_check);
```

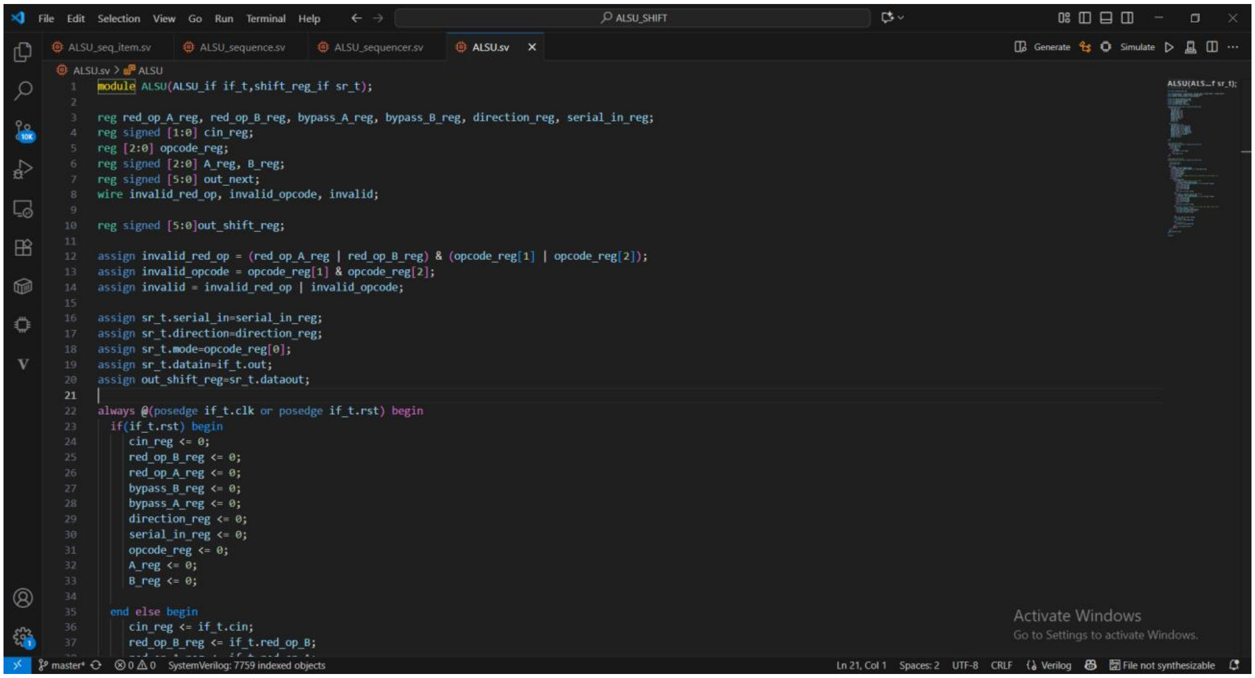


```
module Asseritions(ALSU_if if_t);
//-------------------------------------//
//ADD CHECK
FullADD_check:assert property(fullAdd)else $display("A=%0d B=%0d cin=%0d and out=%0d ",$past(if_t.A, 2) ,
 $past(if_t.B, 2) , $past(if_t.cin, 2),($past(if_t.A, 2) + $past(if_t.B, 2)+$past(if_t.cin, 2)));
halfADD_check:assert property(halfAdd);

//-------------------------------------//
//Multiply CHECK
MULT_check:assert property(multiply);

//-------------------------------------//
//shift CHECK
shift_R_check:assert property(shift_right) ;
shift_L_check:assert property(shift_left);

//-------------------------------------//
//shift CHECK
rotate_R_check:assert property(rotate_right);
rotate_L_check:assert property(rotate_left);



//--------------------------------cover-------------------------------------//
leds_cover:cover property(leds);


//bypass cover
bypass1_cover:cover property(bypass1);
bypass2_cover:cover property(bypass2);
bypass3_cover:cover property(bypass3);
bypass4_cover:cover property(bypass4);

//INVALID cover
INVALID_cover:cover property(INVALID_ch);

//opcode cover OR
```

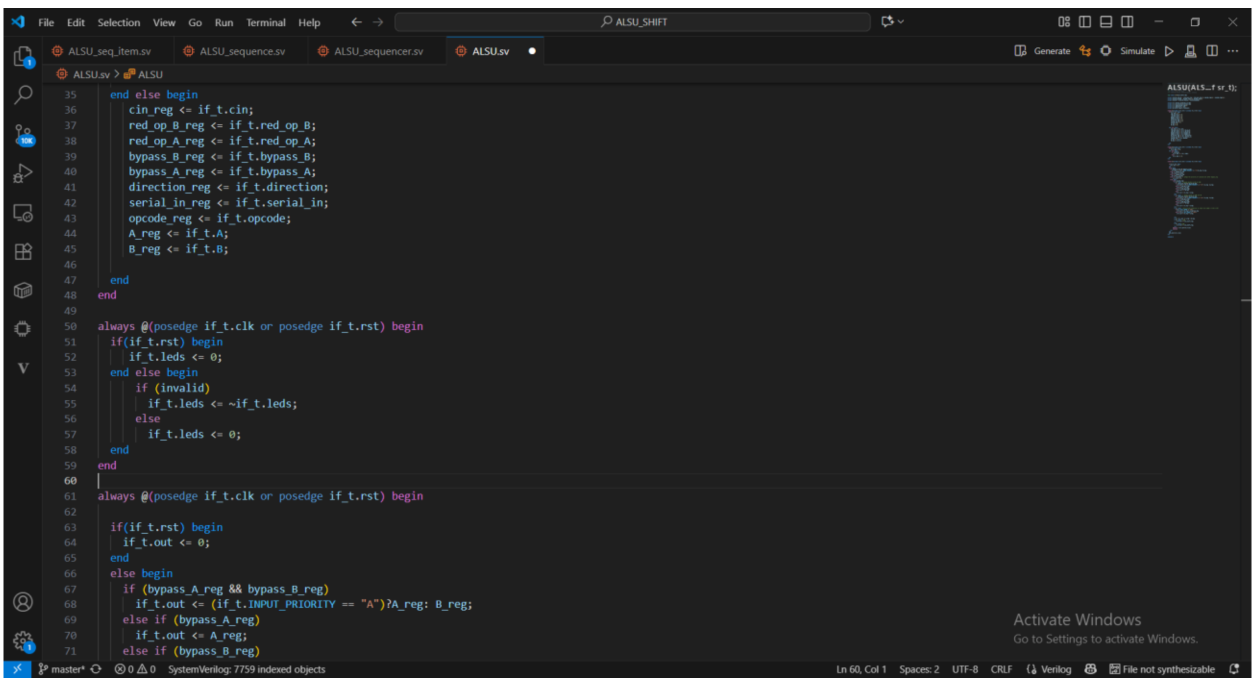


```
module Asseritions(ALSU_if if_t);

//opcode cover OR
OR1_cover:cover property(OR_RedOp_1);
OR2_cover:cover property(OR_RedOp_2);
OR3_cover:cover property(OR_RedOp_3);
OR4_cover:cover property(OR_RedOp_4);
OR5_cover:cover property(OR_check);

//opcode cover XOR
XOR1_cover:cover property(XOR_RedOp_1);
XOR2_cover:cover property(XOR_RedOp_2);
XOR3_cover:cover property(XOR_RedOp_3);
XOR4_cover:cover property(XOR_RedOp_4);
XOR5_cover:cover property(XOR_check);

//opcode cover ADD
FullADD_cover:cover property(fullAdd);
halfADD_cover:cover property(halfAdd);

//opcode cover mult
Mult_cover:cover property(multiply);

//opcode cover shift
shiftL_cover:cover property(shift_left);
shiftR_cover:cover property(shift_right);

//opcode cover rotate
rotateLeft_cover:cover property(rotate_left);
rotateRight_cover:cover property(rotate_right);

// ---------------------------------end cover-------------------------------------//
endmodule
```

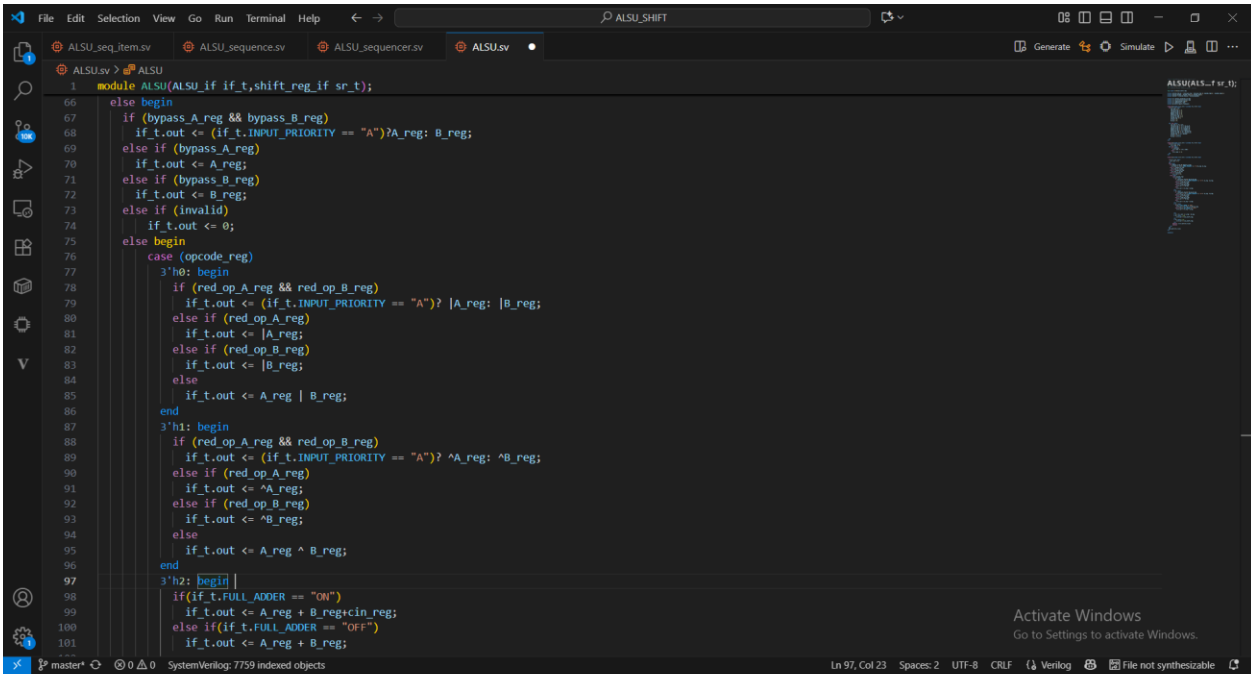
2. ALSU design



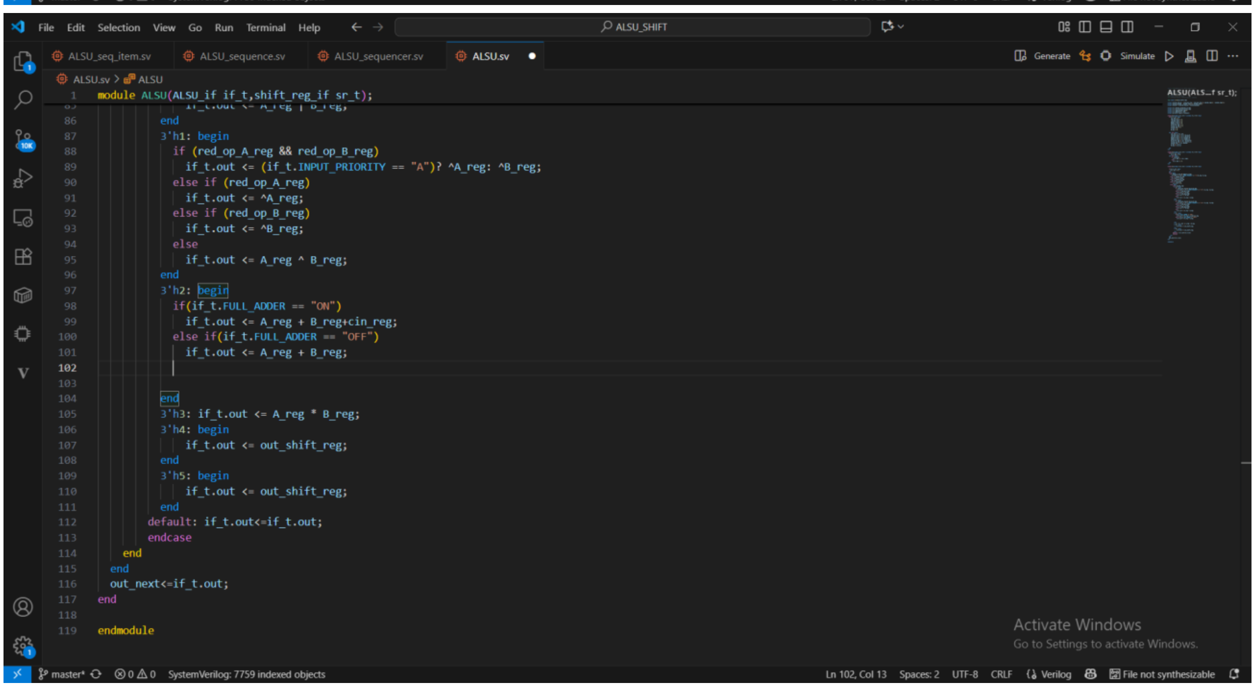
```
module ALSU(ALSU_if if_t,shift_reg_if sr_t);

reg red_op_A_reg, red_op_B_reg, bypass_A_reg, bypass_B_reg, direction_reg, serial_in_reg;
reg signed [1:0] cin_reg;
reg [2:0] opcode_reg;
reg signed [2:0] A_reg, B_reg;
reg signed [5:0] out_next;
wire invalid_red_op, invalid_opcode, invalid;

reg signed [5:0]out_shift_reg;

assign invalid_red_op = (red_op_A_reg | red_op_B_reg) & (opcode_reg[1] | opcode_reg[2]);
assign invalid_opcode = opcode_reg[1] & opcode_reg[2];
assign invalid = invalid_red_op | invalid_opcode;

assign sr_t.serial_in=serial_in_reg;
assign sr_t.direction=direction_reg;
assign sr_t.mode=opcode_reg[0];
assign sr_t.datain=if_t.out;
assign out_shift_reg=sr_t.dataout;

always @(posedge if_t.clk or posedge if_t.rst) begin
  if(if_t.rst) begin
     cin_reg <= 0;
     red_op_B_reg <= 0;
     red_op_A_reg <= 0;
     bypass_B_reg <= 0;
     bypass_A_reg <= 0;
     direction_reg <= 0;
     serial_in_reg <= 0;
     opcode_reg <= 0;
     A_reg <= 0;
     B_reg <= 0;

  end else begin
     cin_reg <= if_t.cin;
     red_op_B_reg <= if_t.red_op_B;
```



```
  end else begin
     cin_reg <= if_t.cin;
     red_op_B_reg <= if_t.red_op_B;
     red_op_A_reg <= if_t.red_op_A;
     bypass_B_reg <= if_t.bypass_B;
     bypass_A_reg <= if_t.bypass_A;
     direction_reg <= if_t.direction;
     serial_in_reg <= if_t.serial_in;
     opcode_reg <= if_t.opcode;
     A_reg <= if_t.A;
     B_reg <= if_t.B;

  end
end

always @(posedge if_t.clk or posedge if_t.rst) begin
  if(if_t.rst) begin
     if_t.leds <= 0;
  end else begin
      if (invalid)
        if_t.leds <= ~if_t.leds;
      else
        if_t.leds <= 0;
  end
end

always @(posedge if_t.clk or posedge if_t.rst) begin

  if(if_t.rst) begin
    if_t.out <= 0;
  end
  else begin
    if (bypass_A_reg && bypass_B_reg)
      if_t.out <= (if_t.INPUT_PRIORITY == "A")?A_reg: B_reg;
    else if (bypass_A_reg)
      if_t.out <= A_reg;
    else if (bypass_B_reg)
```

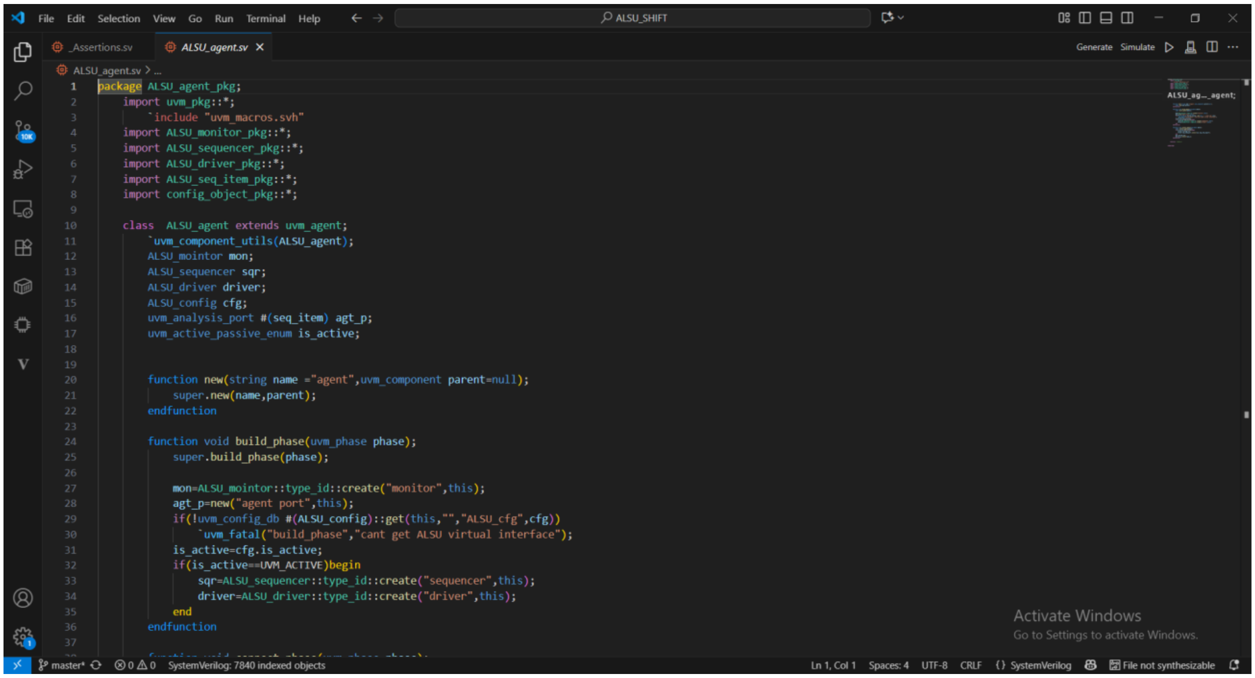


ALSU.sv
module ALSU(ALSU_if if_t,shift_reg_if sr_t);
else begin
if (bypass_A_reg && bypass_B_reg)
if_t.out <= (if_t.INPUT_PRIORITY == "A")?A_reg: B_reg;
else if (bypass_A_reg)
if_t.out <= A_reg;
else if (bypass_B_reg)
if_t.out <= B_reg;
else if (invalid)
if_t.out <= 0;
else begin
case (opcode_reg)
3'h0: begin
if (red_op_A_reg && red_op_B_reg)
if_t.out <= (if_t.INPUT_PRIORITY == "A")? |A_reg: |B_reg;
else if (red_op_A_reg)
if_t.out <= |A_reg;
else if (red_op_B_reg)
if_t.out <= |B_reg;
else
if_t.out <= A_reg | B_reg;
end
3'h1: begin
if (red_op_A_reg && red_op_B_reg)
if_t.out <= (if_t.INPUT_PRIORITY == "A")? ^A_reg: ^B_reg;
else if (red_op_A_reg)
if_t.out <= ^A_reg;
else if (red_op_B_reg)
if_t.out <= ^B_reg;
else
if_t.out <= A_reg ^ B_reg;
end
3'h2: begin
if(if_t.FULL_ADDER == "ON")
if_t.out <= A_reg + B_reg+cin_reg;
else if(if_t.FULL_ADDER == "OFF")
if_t.out <= A_reg + B_reg;

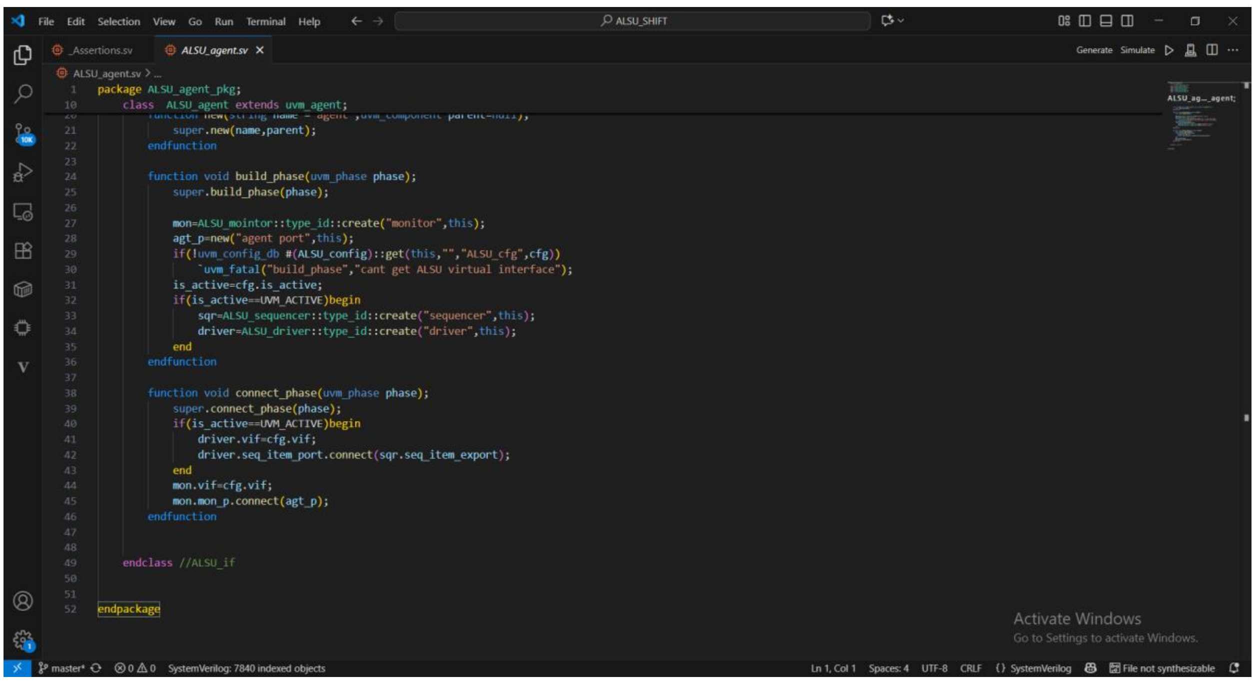


ALSU.sv
module ALSU(ALSU_if if_t,shift_reg_if sr_t);
end
3'h1: begin
if (red_op_A_reg && red_op_B_reg)
if_t.out <= (if_t.INPUT_PRIORITY == "A")? ^A_reg: ^B_reg;
else if (red_op_A_reg)
if_t.out <= ^A_reg;
else if (red_op_B_reg)
if_t.out <= ^B_reg;
else
if_t.out <= A_reg ^ B_reg;
end
3'h2: begin
if(if_t.FULL_ADDER == "ON")
if_t.out <= A_reg + B_reg+cin_reg;
else if(if_t.FULL_ADDER == "OFF")
if_t.out <= A_reg + B_reg;
end
3'h3: if_t.out <= A_reg * B_reg;
3'h4: begin
if_t.out <= out_shift_reg;
end
3'h5: begin
if_t.out <= out_shift_reg;
end
default: if_t.out<=if_t.out;
endcase
end
end
out_next<=if_t.out;
end
endmodule

3. ALSU_agent

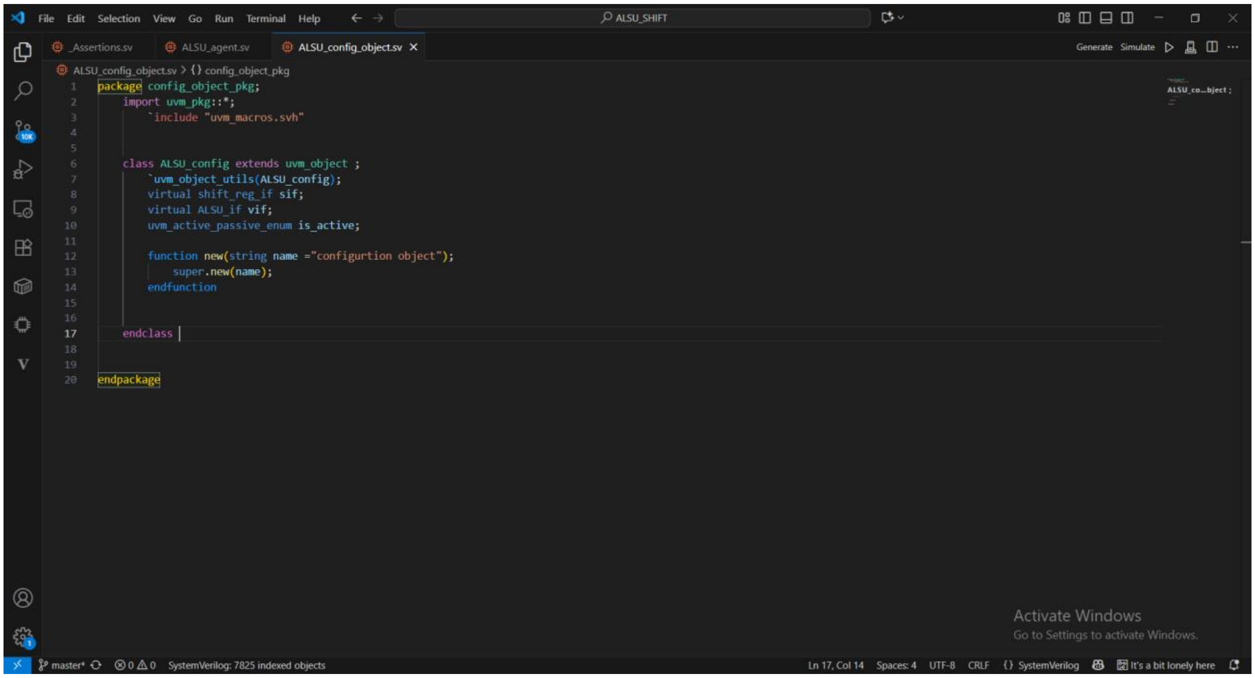


```systemverilog
package ALSU_agent_pkg;
    import uvm_pkg::*;
        `include "uvm_macros.svh"
    import ALSU_monitor_pkg::*;
    import ALSU_sequencer_pkg::*;
    import ALSU_driver_pkg::*;
    import ALSU_seq_item_pkg::*;
    import config_object_pkg::*;

    class  ALSU_agent extends uvm_agent;
        `uvm_component_utils(ALSU_agent);
        ALSU_mointor mon;
        ALSU_sequencer sqr;
        ALSU_driver driver;
        ALSU_config cfg;
        uvm_analysis_port #(seq_item) agt_p;
        uvm_active_passive_enum is_active;


        function new(string name ="agent",uvm_component parent=null);
            super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);

            mon=ALSU_mointor::type_id::create("monitor",this);
            agt_p=new("agent port",this);
            if(!uvm_config_db #(ALSU_config)::get(this,"","ALSU_cfg",cfg))
                `uvm_fatal("build_phase","cant get ALSU virtual interface");
            is_active=cfg.is_active;
            if(is_active==UVM_ACTIVE)begin
                sqr=ALSU_sequencer::type_id::create("sequencer",this);
                driver=ALSU_driver::type_id::create("driver",this);
            end
        endfunction

```



```systemverilog
package ALSU_agent_pkg;
    class  ALSU_agent extends uvm_agent;
            super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);

            mon=ALSU_mointor::type_id::create("monitor",this);
            agt_p=new("agent port",this);
            if(!uvm_config_db #(ALSU_config)::get(this,"","ALSU_cfg",cfg))
                `uvm_fatal("build_phase","cant get ALSU virtual interface");
            is_active=cfg.is_active;
            if(is_active==UVM_ACTIVE)begin
                sqr=ALSU_sequencer::type_id::create("sequencer",this);
                driver=ALSU_driver::type_id::create("driver",this);
            end
        endfunction

        function void connect_phase(uvm_phase phase);
            super.connect_phase(phase);
            if(is_active==UVM_ACTIVE)begin
                driver.vif=cfg.vif;
                driver.seq_item_port.connect(sqr.seq_item_export);
            end
            mon.vif=cfg.vif;
            mon.mon_p.connect(agt_p);
        endfunction


    endclass //ALSU_if


endpackage
```

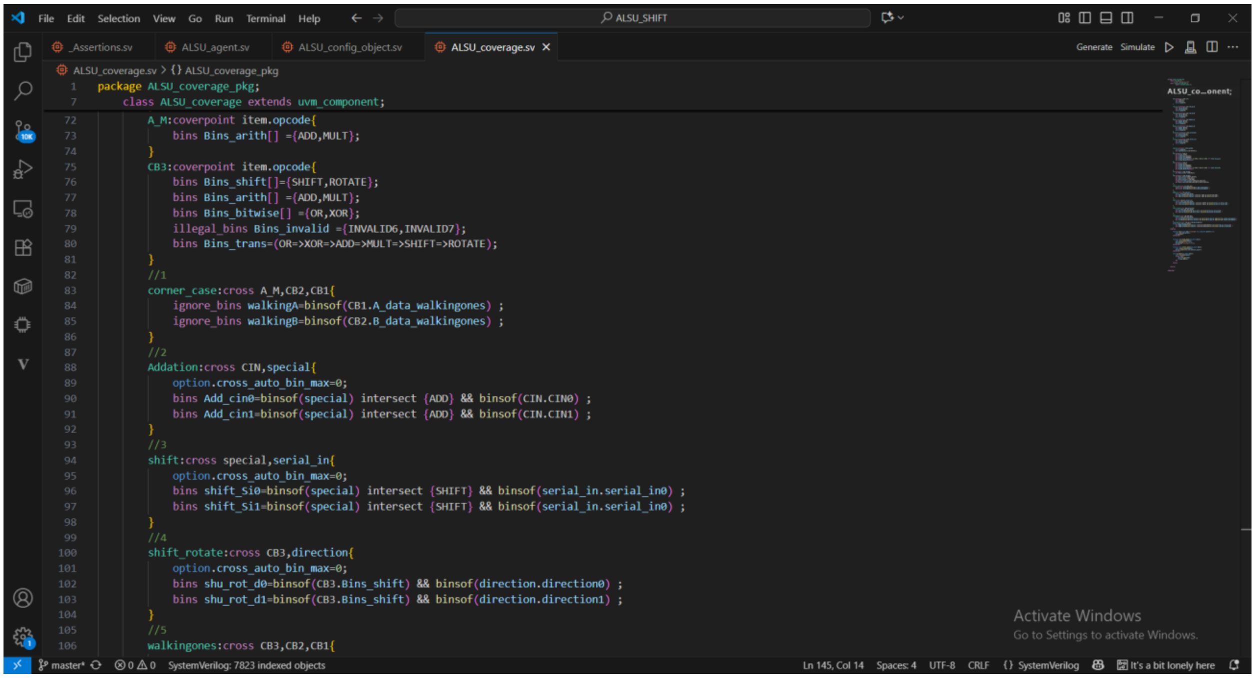
4. ALSU_config_object



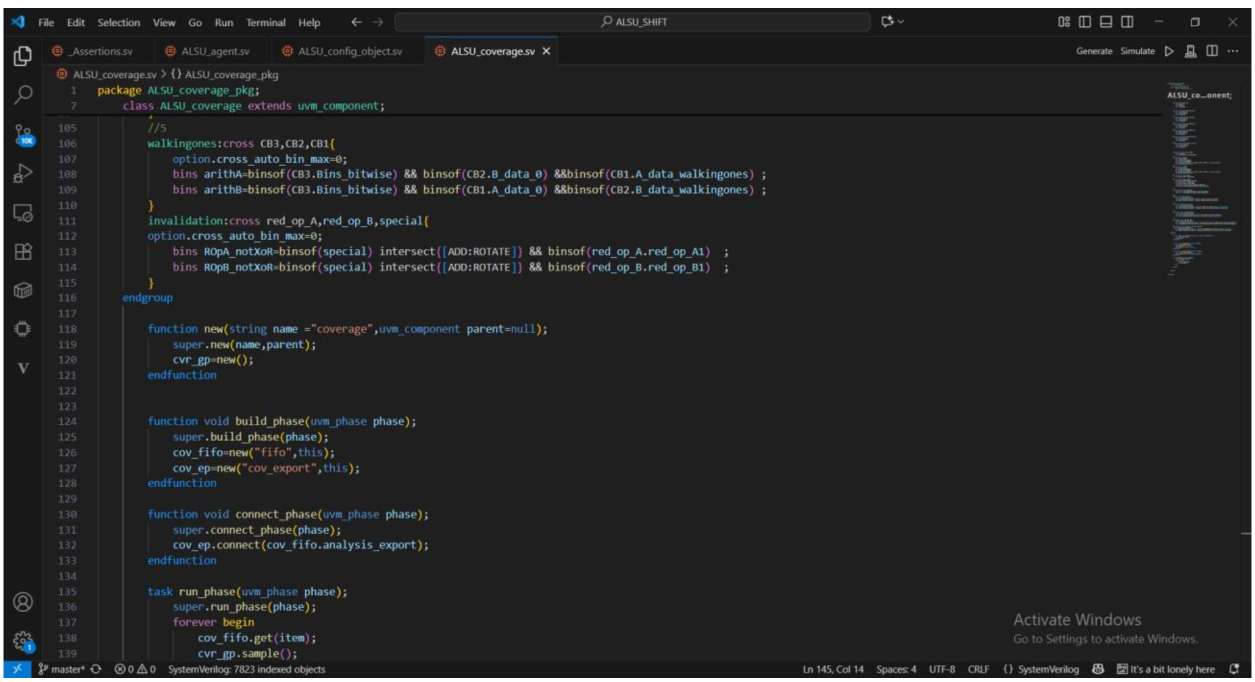
5. ALSU_coverage

```
package ALSU_coverage_pkg;
    import uvm_pkg::*;
        `include "uvm_macros.svh"
    import ALSU_seq_item_pkg::*;
            import constant_enums::*;

    class ALSU_coverage extends uvm_component;
        `uvm_component_utils(ALSU_coverage);

        seq_item item;
        uvm_analysis_export #(seq_item)cov_ep;
        uvm_tlm_analysis_fifo #(seq_item) cov_fifo;

        covergroup cvr_gp;

        CIN:coverpoint item.cin{
            bins CIN0={0};
            bins CIN1={1};
            option.weight=0;
        }

        red_op_A:coverpoint item.red_op_A{
            bins red_op_A0={0};
            bins red_op_A1={1};
            option.weight=0;
        }
        red_op_B:coverpoint item.red_op_B{
            bins red_op_B0={0};
            bins red_op_B1={1};
            option.weight=0;
        }
        bypass_A:coverpoint item.bypass_A{
            bins bypass_A0={0};
            bins bypass_A1={1};
            option.weight=0;
        }
        bypass_B:coverpoint item.bypass_B{
```

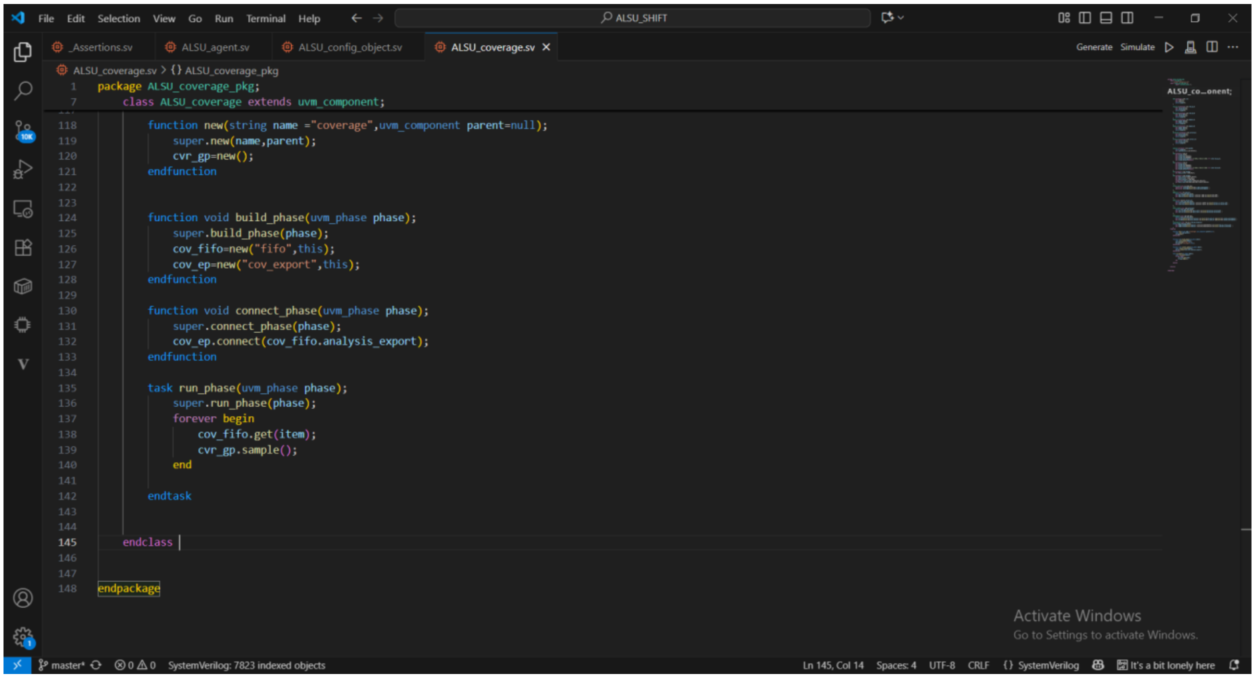
```
package ALSU_coverage_pkg;
    class ALSU_coverage extends uvm_component;
        bypass_B:coverpoint item.bypass_B{
            bins bypass_B0={0};
            bins bypass_B1={1};
            option.weight=0;
        }
        direction:coverpoint item.direction{
            bins direction0={0};
            bins direction1={1};
            option.weight=0;
        }
        serial_in:coverpoint item.serial_in{
            bins serial_in0={0};
            bins serial_in1={1};
            option.weight=0;
        }


        special:coverpoint item.opcode{
            option.weight=0;
            bins operataions[]={[OR:ROTATE]};
        }
        CB1:coverpoint item.A{
            bins A_data_0={0};
            bins A_data_max={MAXPOS};
            bins A_data_min={MAXNEG};
            bins A_data_walkingones[] ={3'b001,3'b010,3'b100} iff (item.red_op_A);
            bins A_data_default=default;
        }
        CB2:coverpoint item.B{
            bins B_data_0={0};
            bins B_data_max={MAXPOS};
            bins B_data_min={MAXNEG};
            bins B_data_walkingones[] ={3'b001,3'b010,3'b100} iff (item.red_op_B);
            bins B_data_default=default;
        }
```



```
package ALSU_coverage_pkg;
    class ALSU_coverage extends uvm_component;
            A_M:coverpoint item.opcode{
                bins Bins_arith[] ={ADD,MULT};
            }
            CB3:coverpoint item.opcode{
                bins Bins_shift[]={SHIFT,ROTATE};
                bins Bins_arith[] ={ADD,MULT};
                bins Bins_bitwise[] ={OR,XOR};
                illegal_bins Bins_invalid ={INVALID6,INVALID7};
                bins Bins_trans=(OR=>XOR=>ADD=>MULT=>SHIFT=>ROTATE);
            }
            //1
            corner_case:cross A_M,CB2,CB1{
                ignore_bins walkingA=binsof(CB1.A_data_walkingones) ;
                ignore_bins walkingB=binsof(CB2.B_data_walkingones) ;
            }
            //2
            Addation:cross CIN,special{
                option.cross_auto_bin_max=0;
                bins Add_cin0=binsof(special) intersect {ADD} && binsof(CIN.CIN0) ;
                bins Add_cin1=binsof(special) intersect {ADD} && binsof(CIN.CIN1) ;
            }
            //3
            shift:cross special,serial_in{
                option.cross_auto_bin_max=0;
                bins shift_Si0=binsof(special) intersect {SHIFT} && binsof(serial_in.serial_in0) ;
                bins shift_Si1=binsof(special) intersect {SHIFT} && binsof(serial_in.serial_in0) ;
            }
            //4
            shift_rotate:cross CB3,direction{
                option.cross_auto_bin_max=0;
                bins shu_rot_d0=binsof(CB3.Bins_shift) && binsof(direction.direction0) ;
                bins shu_rot_d1=binsof(CB3.Bins_shift) && binsof(direction.direction1) ;
            }
            //5
            walkingones:cross CB3,CB2,CB1{
```

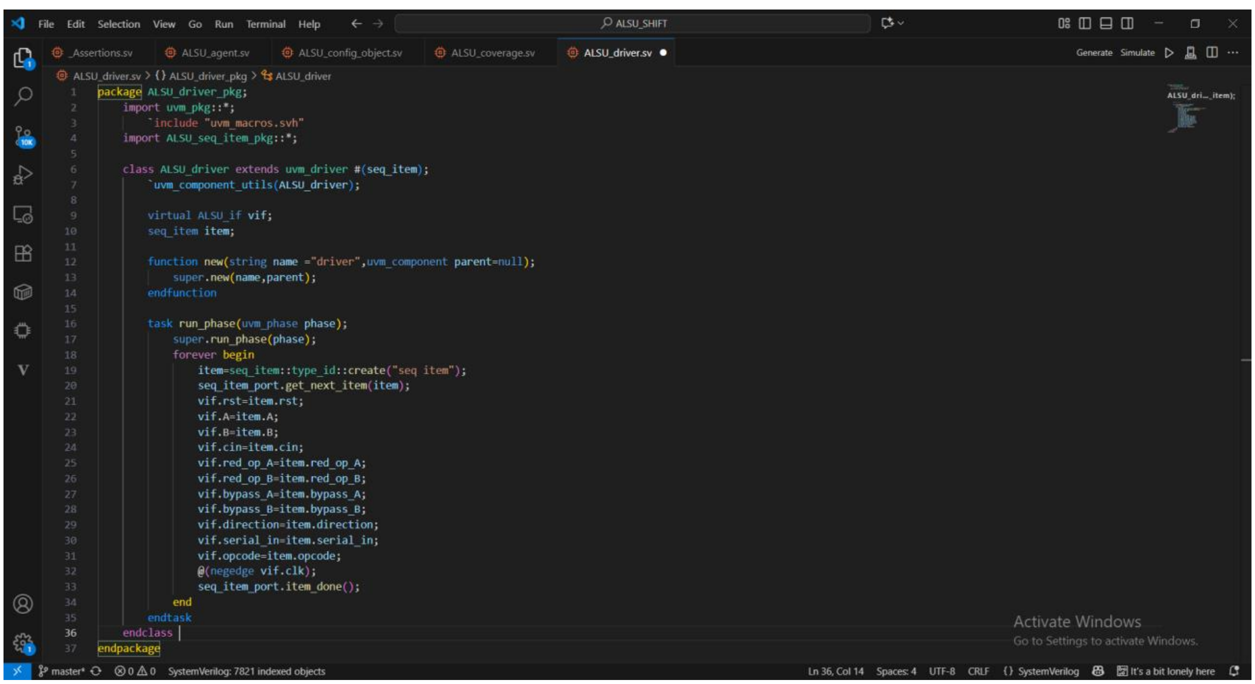


```
package ALSU_coverage_pkg;
    class ALSU_coverage extends uvm_component;
            //5
            walkingones:cross CB3,CB2,CB1{
                option.cross_auto_bin_max=0;
                bins arithA=binsof(CB3.Bins_bitwise) && binsof(CB2.B_data_0) &&binsof(CB1.A_data_walkingones) ;
                bins arithB=binsof(CB3.Bins_bitwise) && binsof(CB1.A_data_0) &&binsof(CB2.B_data_walkingones) ;
            }
            invalidation:cross red_op_A,red_op_B,special{
            option.cross_auto_bin_max=0;
                bins ROpA_notXoR=binsof(special) intersect{[ADD:ROTATE]} && binsof(red_op_A.red_op_A1)  ;
                bins ROpB_notXoR=binsof(special) intersect{[ADD:ROTATE]} && binsof(red_op_B.red_op_B1)  ;
            }
        endgroup

            function new(string name ="coverage",uvm_component parent=null);
                super.new(name,parent);
                cvr_gp=new();
            endfunction


            function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                cov_fifo=new("fifo",this);
                cov_ep=new("cov_export",this);
            endfunction

            function void connect_phase(uvm_phase phase);
                super.connect_phase(phase);
                cov_ep.connect(cov_fifo.analysis_export);
            endfunction

            task run_phase(uvm_phase phase);
                super.run_phase(phase);
                forever begin
                    cov_fifo.get(item);
                    cvr_gp.sample();
```



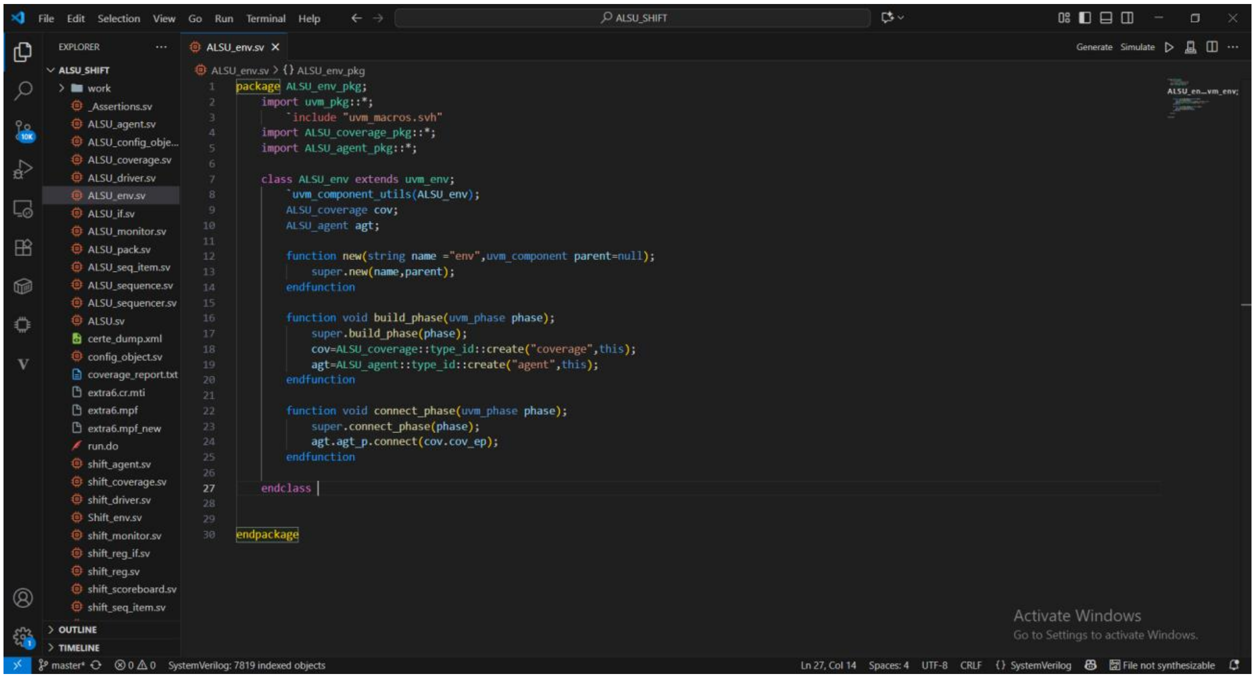
```
package ALSU_coverage_pkg;
    class ALSU_coverage extends uvm_component;
        function new(string name ="coverage",uvm_component parent=null);
            super.new(name,parent);
            cvr_gp=new();
        endfunction


        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            cov_fifo=new("fifo",this);
            cov_ep=new("cov_export",this);
        endfunction

        function void connect_phase(uvm_phase phase);
            super.connect_phase(phase);
            cov_ep.connect(cov_fifo.analysis_export);
        endfunction

        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            forever begin
                cov_fifo.get(item);
                cvr_gp.sample();
            end

        endtask


    endclass


endpackage
```

6. ALSU_driver

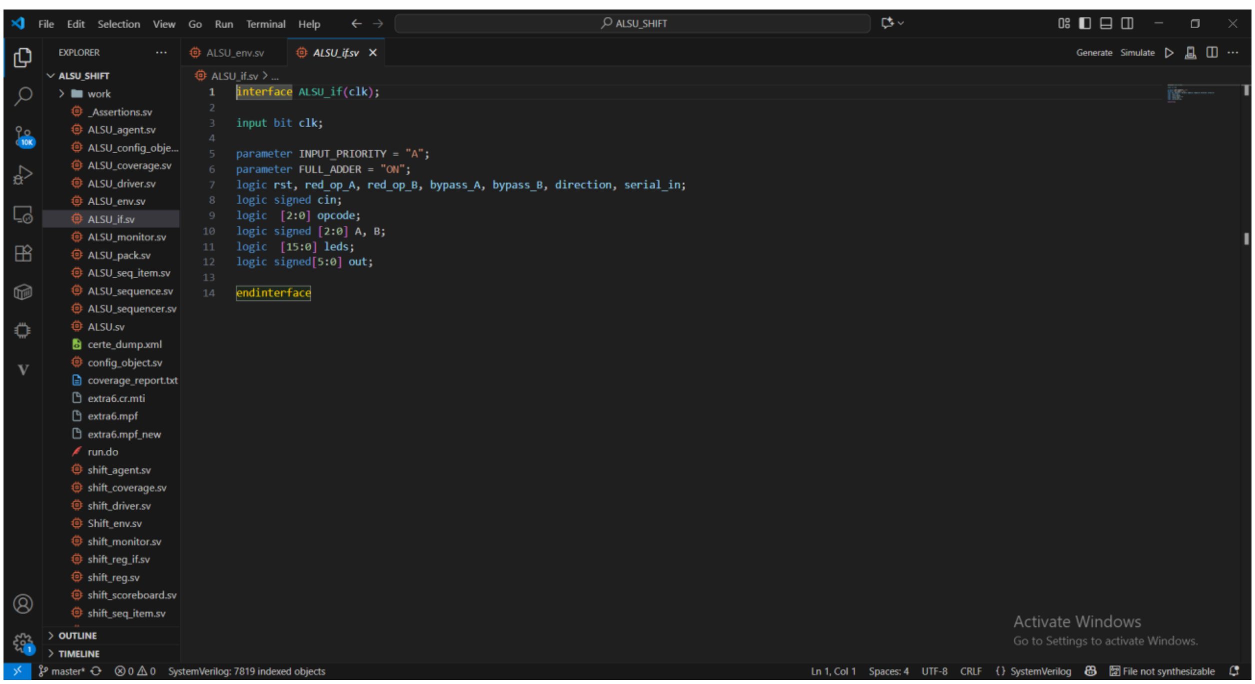


```
package ALSU_driver_pkg;
    import uvm_pkg::*;
        `include "uvm_macros.svh"
    import ALSU_seq_item_pkg::*;

    class ALSU_driver extends uvm_driver #(seq_item);
        `uvm_component_utils(ALSU_driver);

        virtual ALSU_if vif;
        seq_item item;

        function new(string name ="driver",uvm_component parent=null);
            super.new(name,parent);
        endfunction

        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            forever begin
                item=seq_item::type_id::create("seq item");
                seq_item_port.get_next_item(item);
                vif.rst=item.rst;
                vif.A=item.A;
                vif.B=item.B;
                vif.cin=item.cin;
                vif.red_op_A=item.red_op_A;
                vif.red_op_B=item.red_op_B;
                vif.bypass_A=item.bypass_A;
                vif.bypass_B=item.bypass_B;
                vif.direction=item.direction;
                vif.serial_in=item.serial_in;
                vif.opcode=item.opcode;
                @(negedge vif.clk);
                seq_item_port.item_done();
            end
        endtask
    endclass
endpackage
```

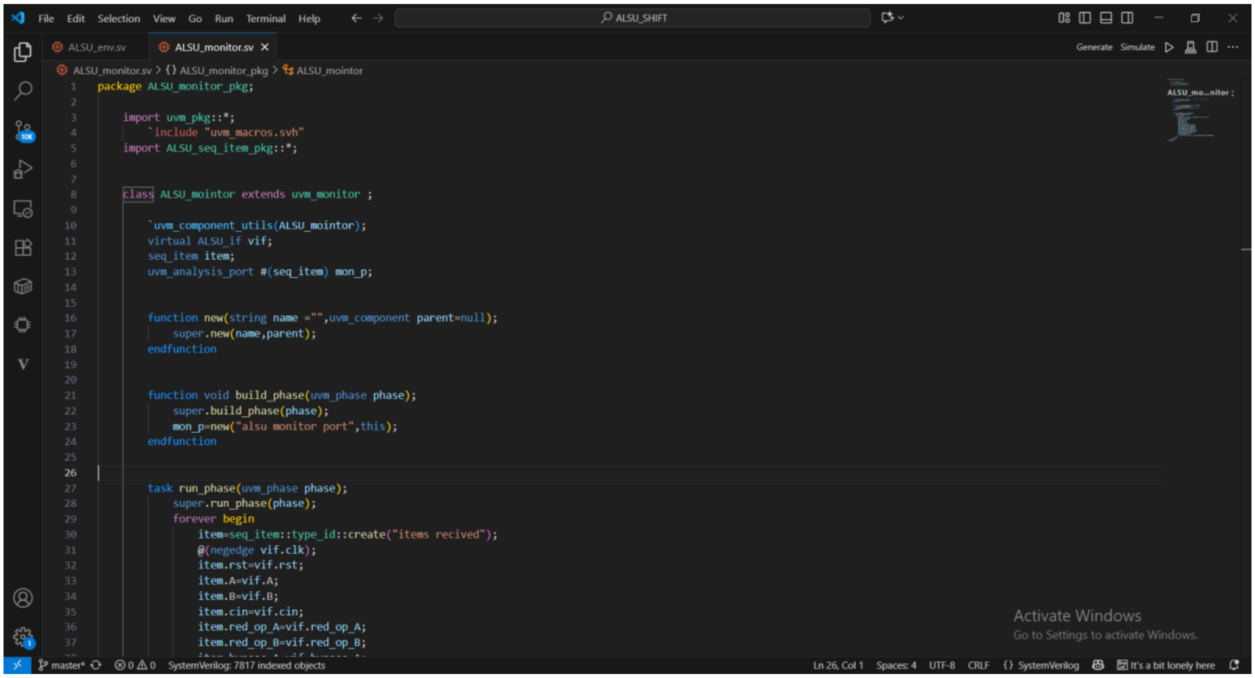
7. ALSU_env

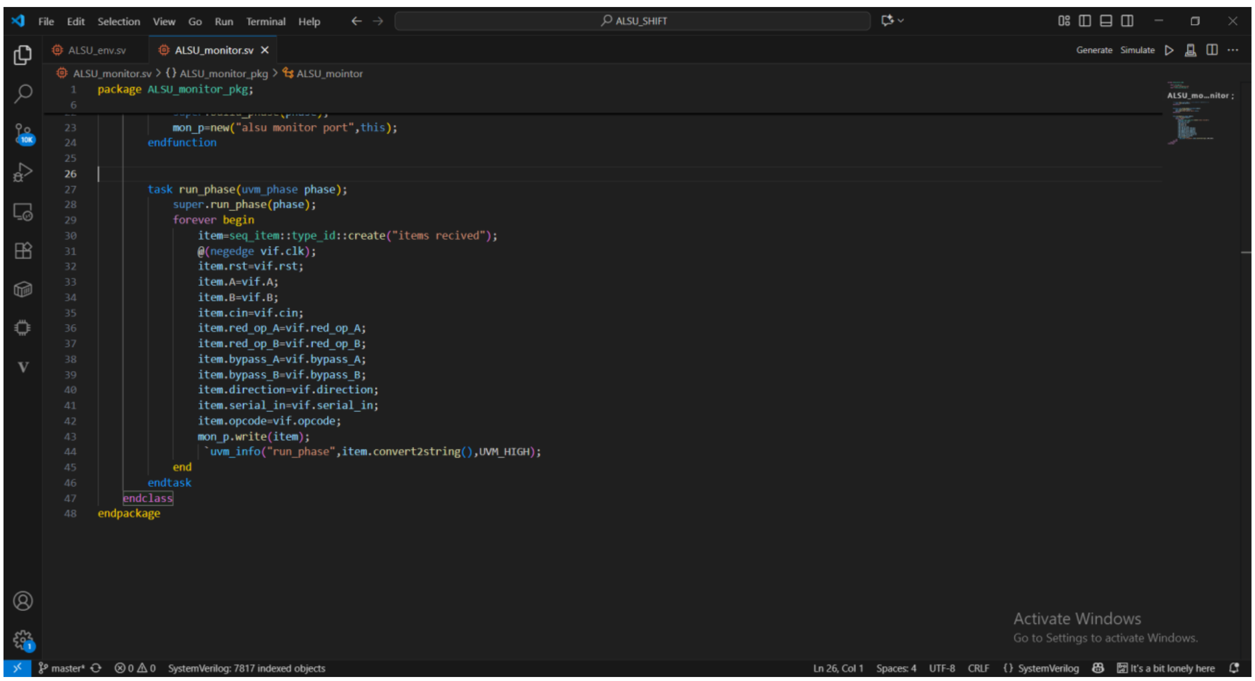


8. ALSU_if

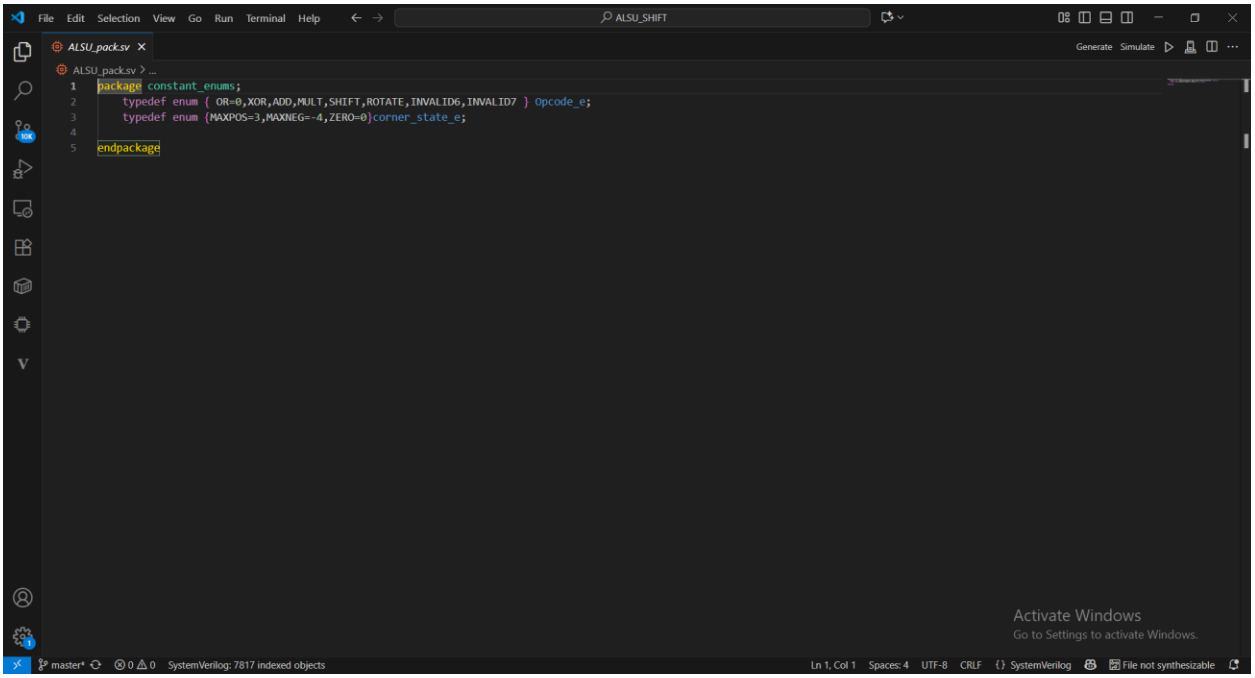


9. ALSU_monitor





10. ALSU_shared_package

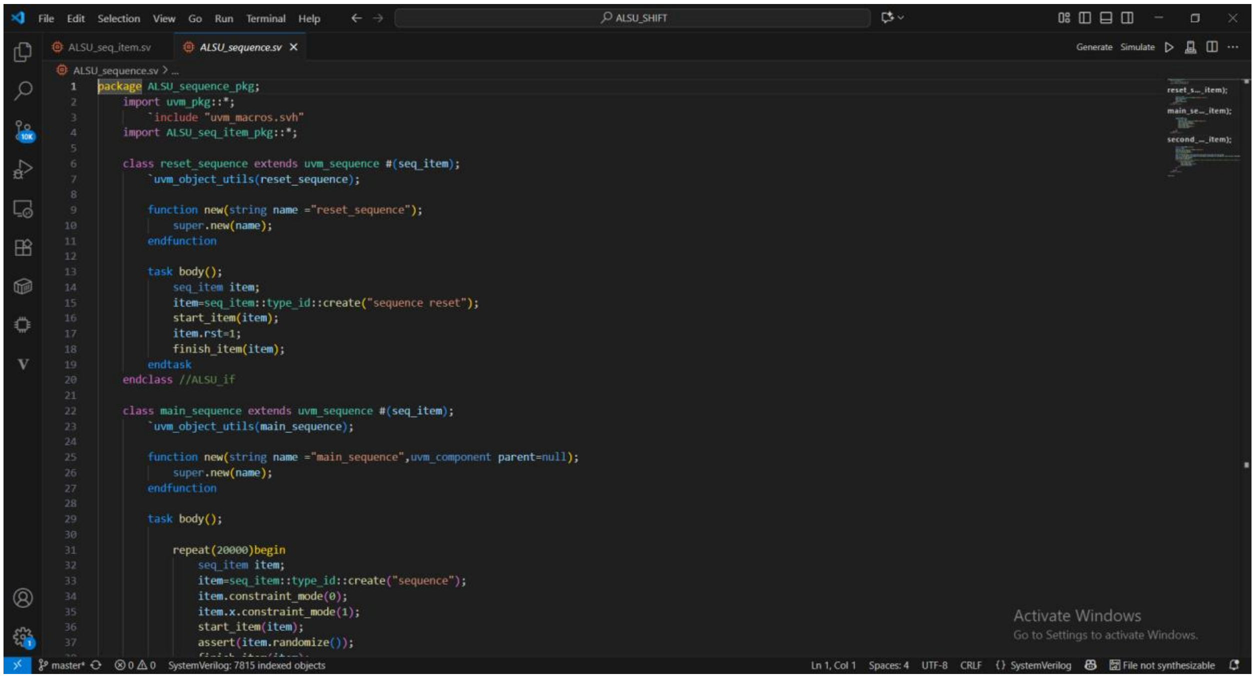


11. ALSU_seq_item

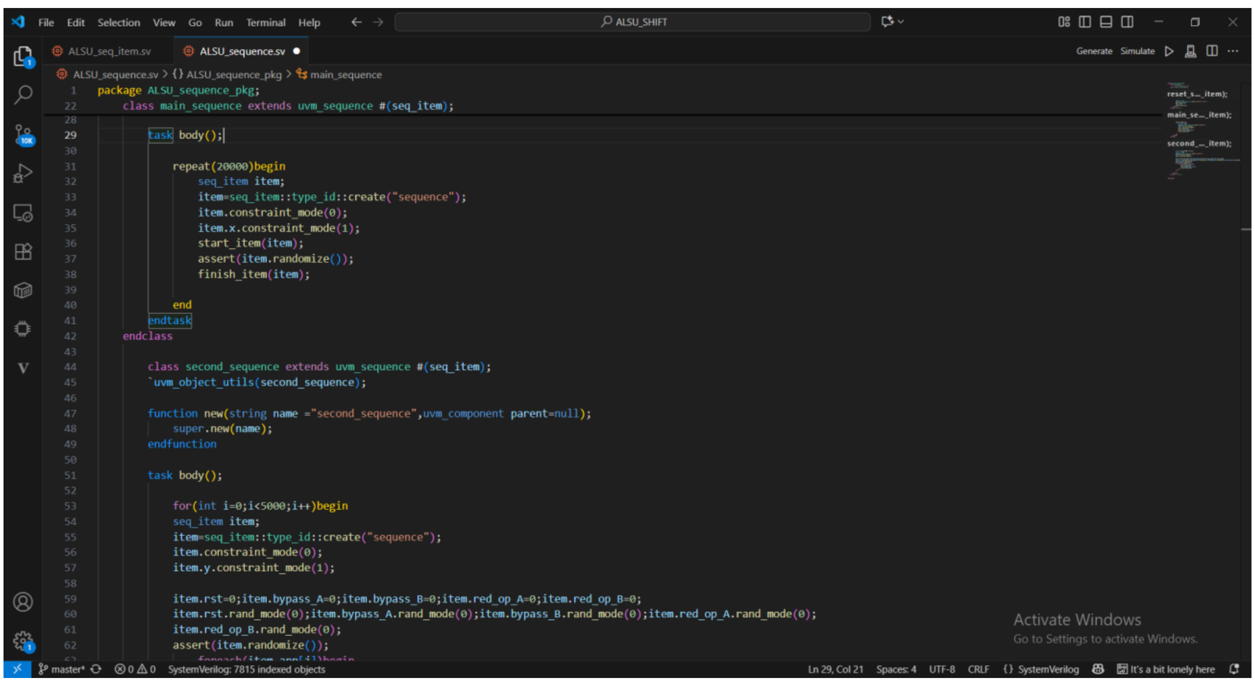
```systemverilog
package ALSU_seq_item_pkg;
    import uvm_pkg::*;
        `include "uvm_macros.svh"
        import constant_enums::*;

    class seq_item extends uvm_sequence_item;
        `uvm_object_utils(seq_item);
        rand bit  clk, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
        rand logic cin;
        rand logic  [2:0] opcode;
        rand bit signed [2:0] A, B;
        bit signed [5:0] out;
        bit [15:0] leds;

        bit [2:0] ones_number={3'b001,3'b010,3'b100};
        rand bit [2:0] found,notfound;
        rand corner_state_e a_state;
        rand bit [2:0] rem_numbers;
        rand Opcode_e arr[6];
        function new(string name ="seq_item");
            super.new(name);
        endfunction

        function string convert2string();
            return $sformatf("%s  reset =%0b cin=%0b ,red_op_A=%0b ,red_op_B=%0b ,bypass_A=%0b ,bypass_B=%0b ,
            direction=%0b ,serial_in=%0b ,opcode=%0b ,A=%0b,B=%0b ,out =%0b,leds=%0b ",
            super.convert2string(),rst,cin,red_op_A,red_op_B,bypass_A,bypass_B,direction,serial_in,opcode,A,B,out,leds);
        endfunction

        function string convert2string_stimulus();
            return $sformatf("%s  reset =%0b cin=%0b ,red_op_A=%0b ,red_op_B=%0b ,bypass_A=%0b ,bypass_B=%0b ,
            direction=%0b ,serial_in=%0b ,opcode=%0b ,A=%0b,B=%0b",
            super.convert2string(),rst,cin,red_op_A,red_op_B,bypass_A,bypass_B,direction,serial_in,opcode,A,B);
        endfunction

        constraint x {
            rem_numbers!= MAXPOS||MAXNEG||ZERO;
```

```systemverilog
    class seq_item extends uvm_sequence_item;
        constraint x {
            rem_numbers!= MAXPOS||MAXNEG||ZERO;
            rst dist {1:=5 , 0:=95};

            found inside {ones_number};
            !(notfound inside {ones_number});

            if (opcode ==ADD || opcode== MULT){
                A dist {a_state:=80,rem_numbers:=20};
                B dist {a_state:=80,rem_numbers:=20};
            }
            if (opcode ==OR || opcode== XOR ){
                if(red_op_A){
                    A dist {found:=80,notfound:=20};
                    B==3'b000;
                }
                else if (red_op_B){
                    B dist {found:=80,notfound:=20};
                    A==3'b000;
                }
            }


            opcode dist {[OR:ROTATE]:=80,[INVALID6:INVALID7]};

            bypass_A dist {0:=90,1:=10};
            bypass_B dist {0:=90,1:=10};
        }
        constraint y{
            unique{arr};
            foreach(arr[i])
                arr[i] inside {[OR:ROTATE]};
        }
    endclass
endpackage
```

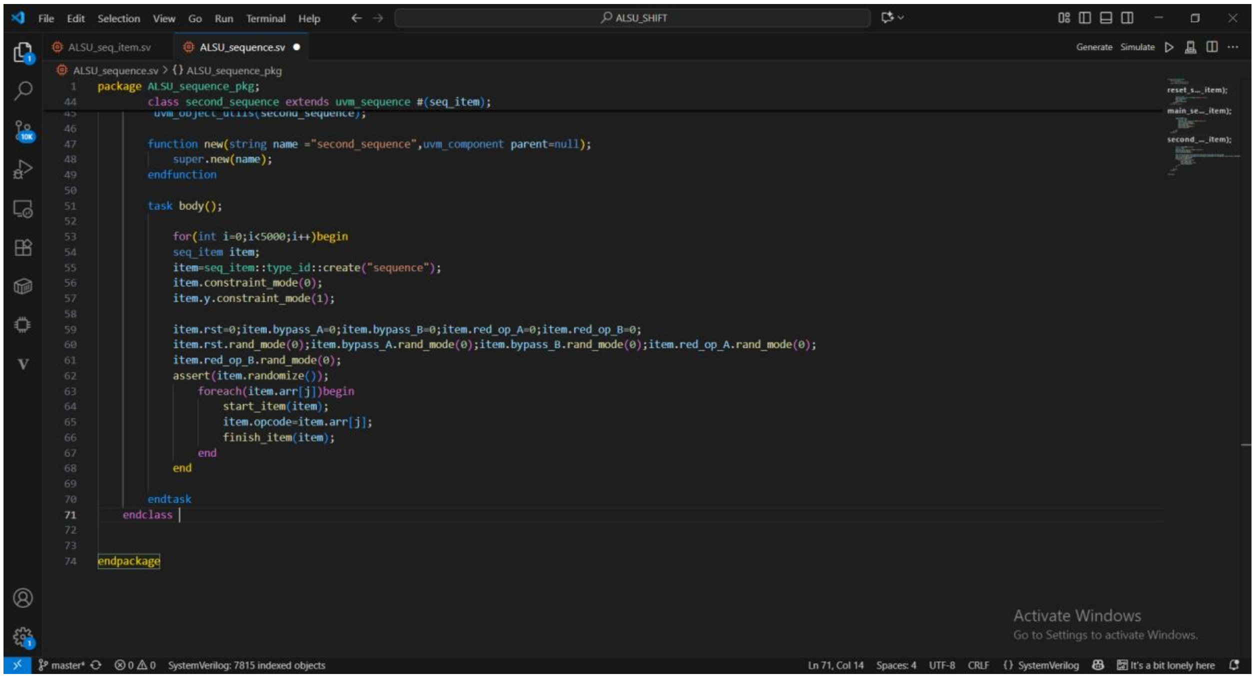
12. ALSU_sequence



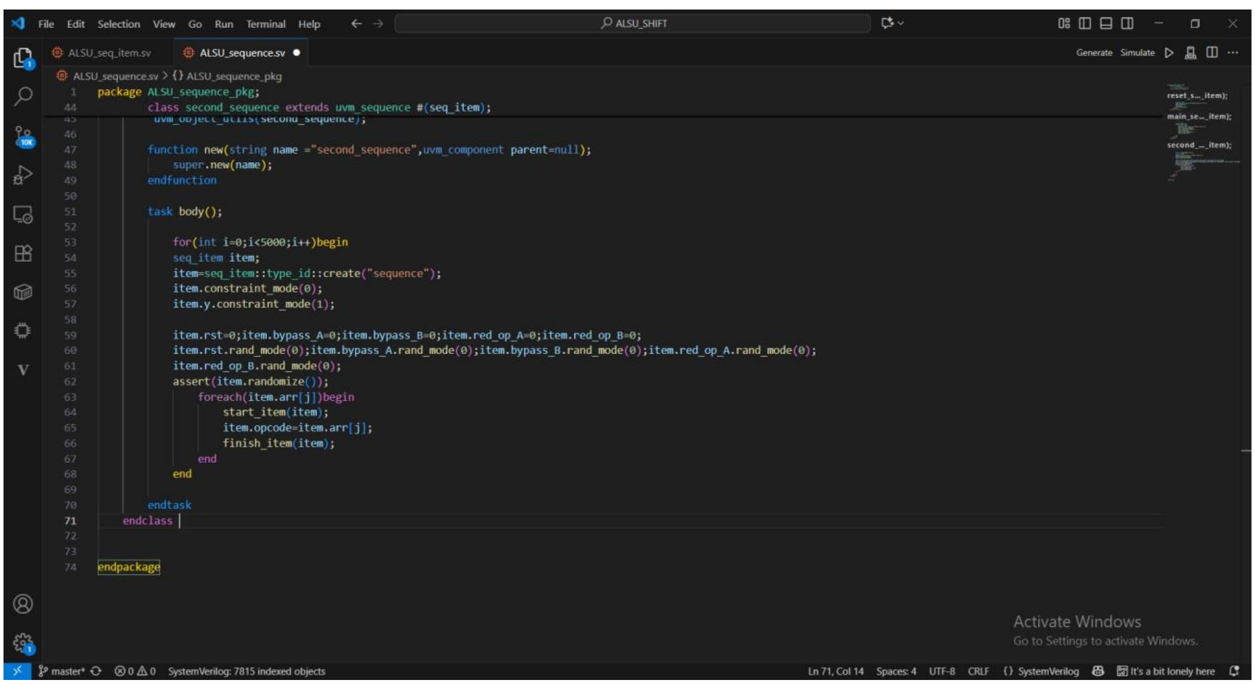
ALSU_seq_item.sv
ALSU_sequence.sv
ALSU_SHIFT
Generate Simulate
package ALSU_sequence_pkg;
import uvm_pkg::*;
`include "uvm_macros.svh"
import ALSU_seq_item_pkg::*;
class reset_sequence extends uvm_sequence #(seq_item);
`uvm_object_utils(reset_sequence);
function new(string name ="reset_sequence");
super.new(name);
endfunction
task body();
seq_item item;
item=seq_item::type_id::create("sequence reset");
start_item(item);
item.rst=1;
finish_item(item);
endtask
endclass //ALSU_if
class main_sequence extends uvm_sequence #(seq_item);
`uvm_object_utils(main_sequence);
function new(string name ="main_sequence",uvm_component parent=null);
super.new(name);
endfunction
task body();
repeat(20000)begin
seq_item item;
item=seq_item::type_id::create("sequence");
item.constraint_mode(0);
item.x.constraint_mode(1);
start_item(item);
assert(item.randomize());
Activate Windows
Go to Settings to activate Windows.
master* 0 0 SystemVerilog: 7815 indexed objects
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} SystemVerilog File not synthesizable



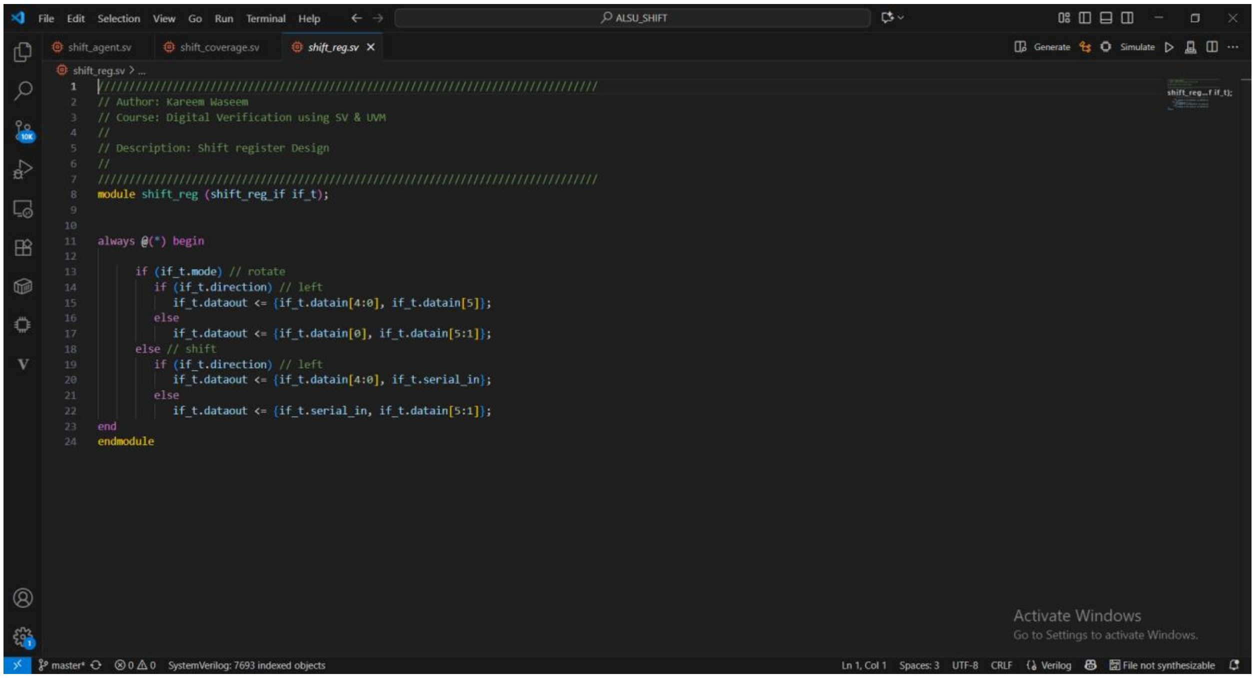
ALSU_seq_item.sv
ALSU_sequence.sv
ALSU_SHIFT
Generate Simulate
ALSU_sequence.sv > {} ALSU_sequence_pkg > main_sequence
package ALSU_sequence_pkg;
class main_sequence extends uvm_sequence #(seq_item);
task body();
repeat(20000)begin
seq_item item;
item=seq_item::type_id::create("sequence");
item.constraint_mode(0);
item.x.constraint_mode(1);
start_item(item);
assert(item.randomize());
finish_item(item);
end
endtask
endclass
class second_sequence extends uvm_sequence #(seq_item);
`uvm_object_utils(second_sequence);
function new(string name ="second_sequence",uvm_component parent=null);
super.new(name);
endfunction
task body();
for(int i=0;i<5000;i++)begin
seq_item item;
item=seq_item::type_id::create("sequence");
item.constraint_mode(0);
item.y.constraint_mode(1);
item.rst=0;item.bypass_A=0;item.bypass_B=0;item.red_op_A=0;item.red_op_B=0;
item.rst.rand_mode(0);item.bypass_A.rand_mode(0);item.bypass_B.rand_mode(0);item.red_op_A.rand_mode(0);
item.red_op_B.rand_mode(0);
assert(item.randomize());
Activate Windows
Go to Settings to activate Windows.
master* 0 0 SystemVerilog: 7815 indexed objects
Ln 29, Col 21 Spaces: 4 UTF-8 CRLF {} SystemVerilog It's a bit lonely here



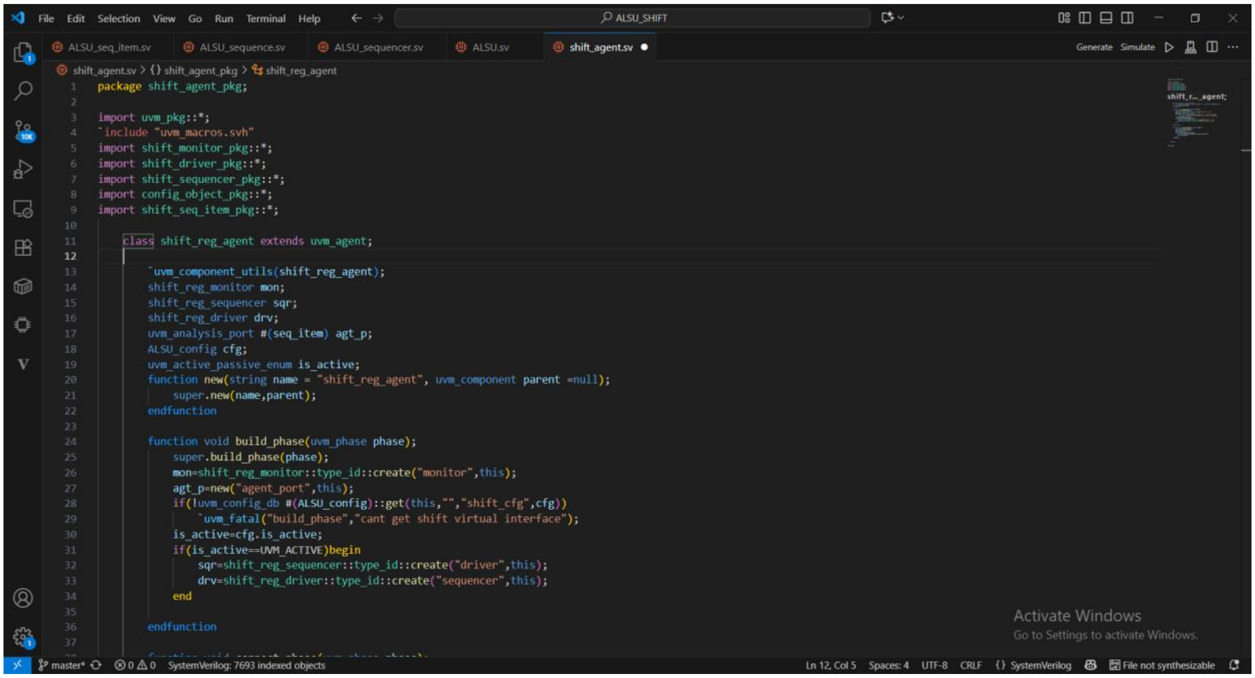
13. ALSU_sequencer



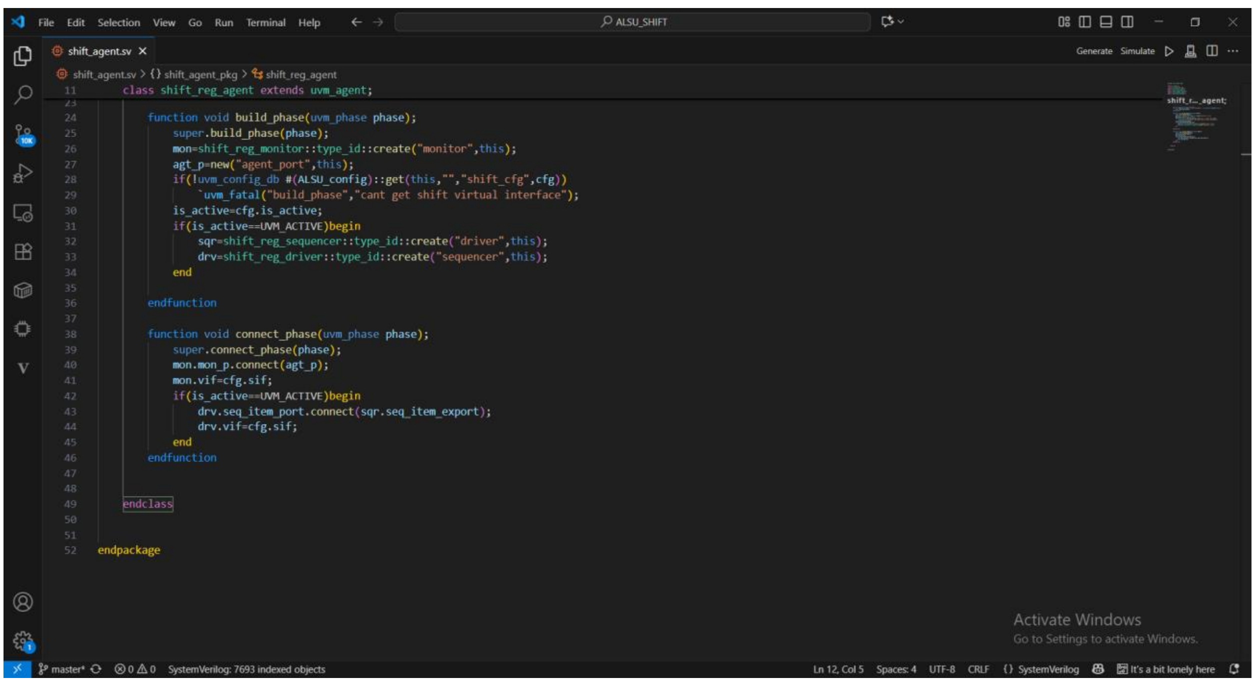
14. shift_reg_design



15. shift_reg_agent

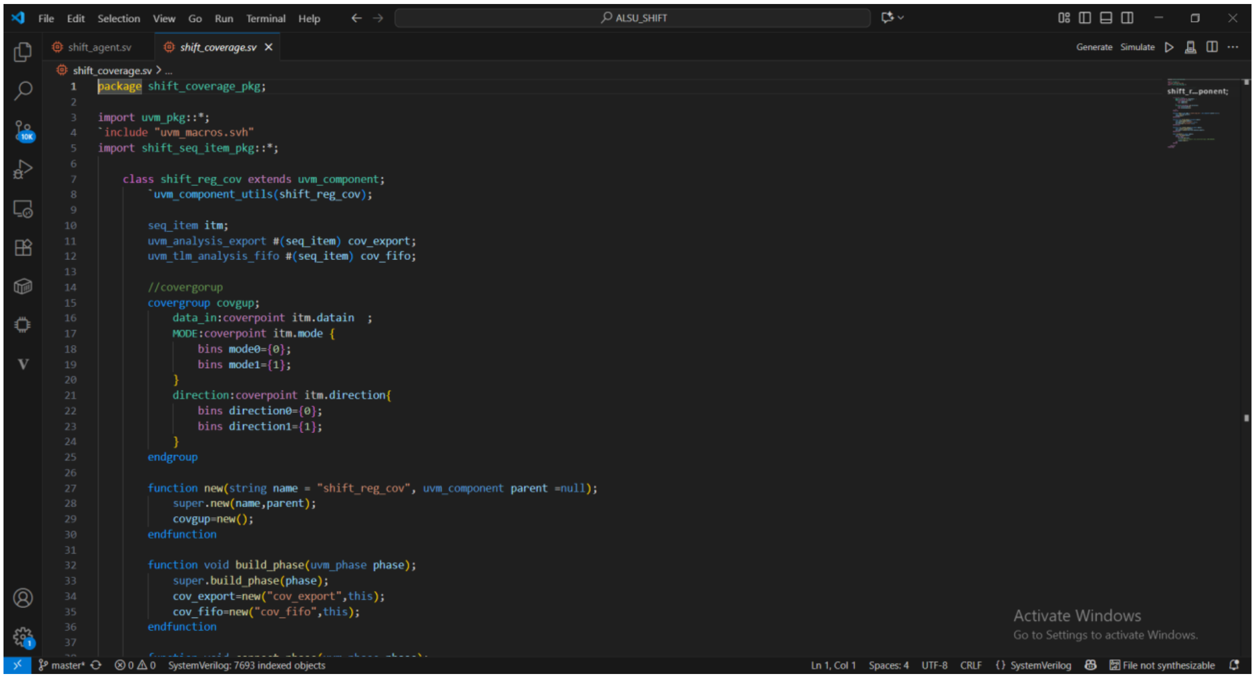


```systemverilog
package shift_agent_pkg;

import uvm_pkg::*;
`include "uvm_macros.svh"
import shift_monitor_pkg::*;
import shift_driver_pkg::*;
import shift_sequencer_pkg::*;
import config_object_pkg::*;
import shift_seq_item_pkg::*;

    class shift_reg_agent extends uvm_agent;

        `uvm_component_utils(shift_reg_agent);
        shift_reg_monitor mon;
        shift_reg_sequencer sqr;
        shift_reg_driver drv;
        uvm_analysis_port #(seq_item) agt_p;
        ALSU_config cfg;
        uvm_active_passive_enum is_active;
        function new(string name = "shift_reg_agent", uvm_component parent =null);
            super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            mon=shift_reg_monitor::type_id::create("monitor",this);
            agt_p=new("agent_port",this);
            if(!uvm_config_db #(ALSU_config)::get(this,"","shift_cfg",cfg))
                `uvm_fatal("build_phase","cant get shift virtual interface");
            is_active=cfg.is_active;
            if(is_active==UVM_ACTIVE)begin
                sqr=shift_reg_sequencer::type_id::create("driver",this);
                drv=shift_reg_driver::type_id::create("sequencer",this);
            end

        endfunction
```

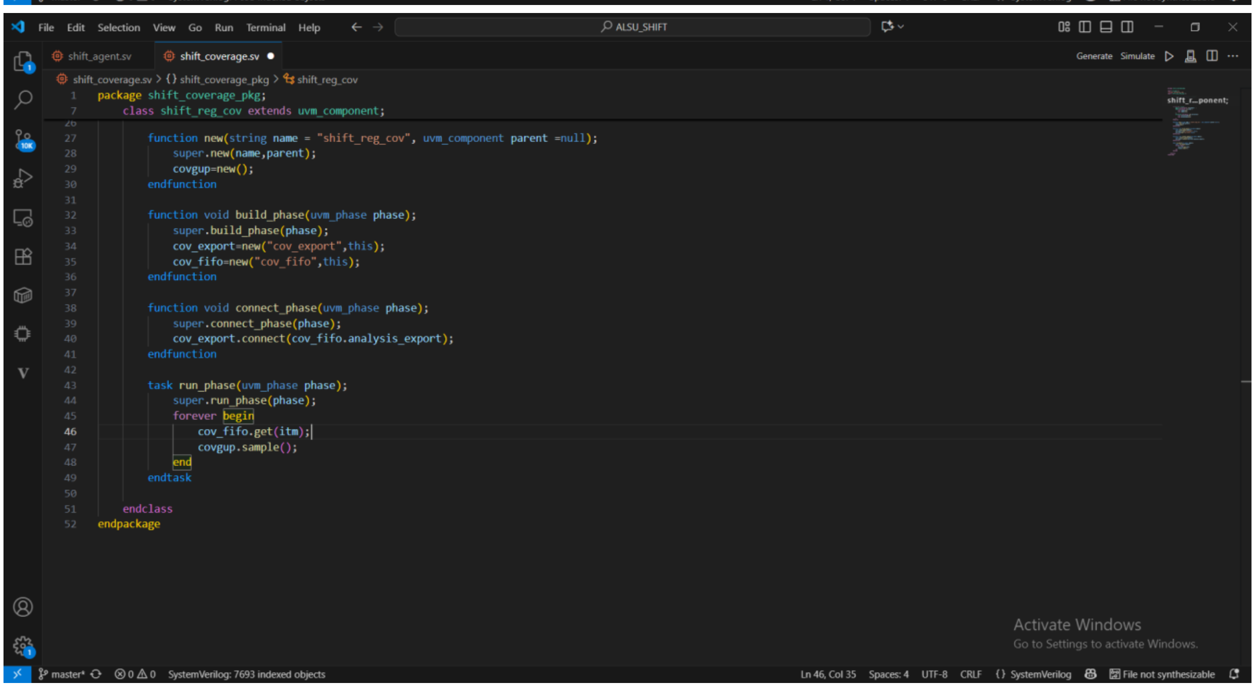


```systemverilog
    class shift_reg_agent extends uvm_agent;
        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            mon=shift_reg_monitor::type_id::create("monitor",this);
            agt_p=new("agent_port",this);
            if(!uvm_config_db #(ALSU_config)::get(this,"","shift_cfg",cfg))
                `uvm_fatal("build_phase","cant get shift virtual interface");
            is_active=cfg.is_active;
            if(is_active==UVM_ACTIVE)begin
                sqr=shift_reg_sequencer::type_id::create("driver",this);
                drv=shift_reg_driver::type_id::create("sequencer",this);
            end

        endfunction

        function void connect_phase(uvm_phase phase);
            super.connect_phase(phase);
            mon.mon_p.connect(agt_p);
            mon.vif=cfg.sif;
            if(is_active==UVM_ACTIVE)begin
                drv.seq_item_port.connect(sqr.seq_item_export);
                drv.vif=cfg.sif;
            end
        endfunction


    endclass


endpackage
```

16. shift_reg_coverage



```
package shift_coverage_pkg;

import uvm_pkg::*;
`include "uvm_macros.svh"
import shift_seq_item_pkg::*;

    class shift_reg_cov extends uvm_component;
        `uvm_component_utils(shift_reg_cov);

        seq_item itm;
        uvm_analysis_export #(seq_item) cov_export;
        uvm_tlm_analysis_fifo #(seq_item) cov_fifo;

        //covergorup
        covergroup covgup;
            data_in:coverpoint itm.datain  ;
            MODE:coverpoint itm.mode {
                bins mode0={0};
                bins mode1={1};
            }
            direction:coverpoint itm.direction{
                bins direction0={0};
                bins direction1={1};
            }
        endgroup

        function new(string name = "shift_reg_cov", uvm_component parent =null);
            super.new(name,parent);
            covgup=new();
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            cov_export=new("cov_export",this);
            cov_fifo=new("cov_fifo",this);
        endfunction

```



```
        function new(string name = "shift_reg_cov", uvm_component parent =null);
            super.new(name,parent);
            covgup=new();
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            cov_export=new("cov_export",this);
            cov_fifo=new("cov_fifo",this);
        endfunction

        function void connect_phase(uvm_phase phase);
            super.connect_phase(phase);
            cov_export.connect(cov_fifo.analysis_export);
        endfunction

        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            forever begin
                cov_fifo.get(itm);
                covgup.sample();
            end
        endtask

    endclass
endpackage
```

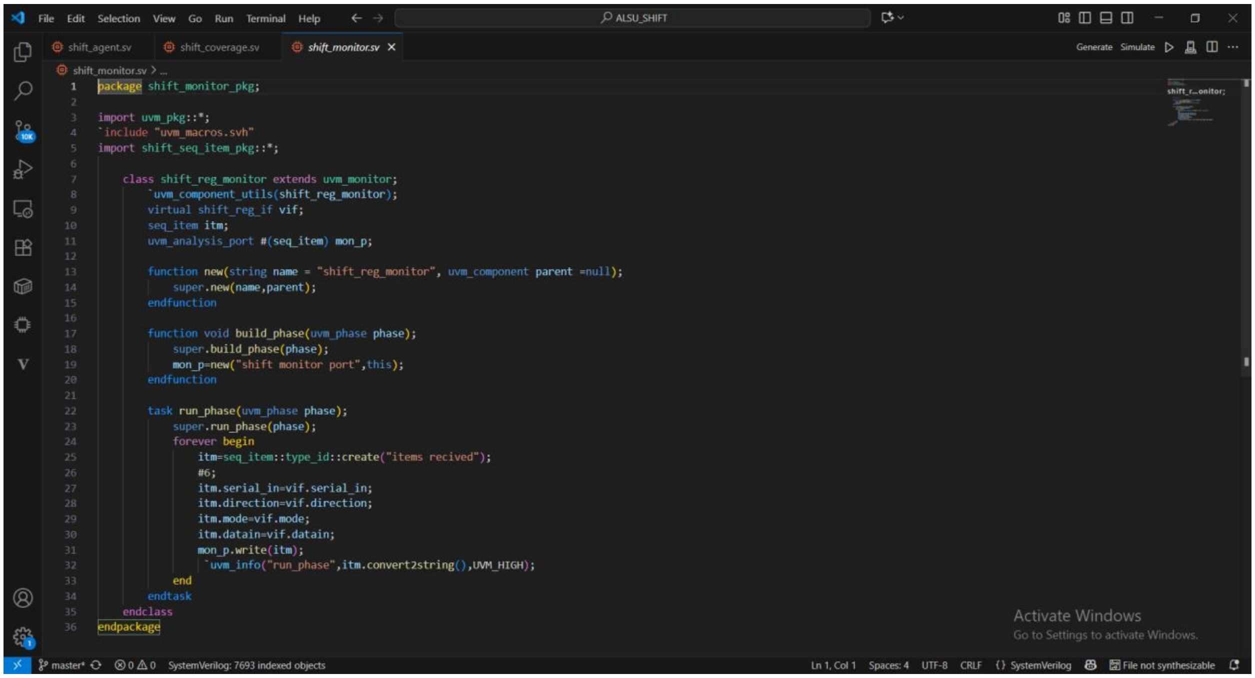
17. shift_reg_driver

```
package shift_driver_pkg;

import uvm_pkg::*;
`include "uvm_macros.svh"
import shift_seq_item_pkg::*;

    class shift_reg_driver extends uvm_driver #(seq_item);
        `uvm_component_utils(shift_reg_driver);
        virtual shift_reg_if vif;
        seq_item itm;

        function new(string name = "shift_reg_driver", uvm_component parent =null);
            super.new(name,parent);
        endfunction

        task run_phase(uvm_phase phase);
            super.run_phase(phase);

            forever begin
                itm=seq_item::type_id::create("item_send");
                seq_item_port.get_next_item(itm);
                vif.serial_in=itm.serial_in;
                vif.direction=itm.direction;
                vif.mode=itm.mode;
                vif.datain=itm.datain;
                #2;

                seq_item_port.item_done();
                `uvm_info("run_phase",itm.convert2string(),UVM_HIGH);

            end
        endtask
    endclass
endpackage
```

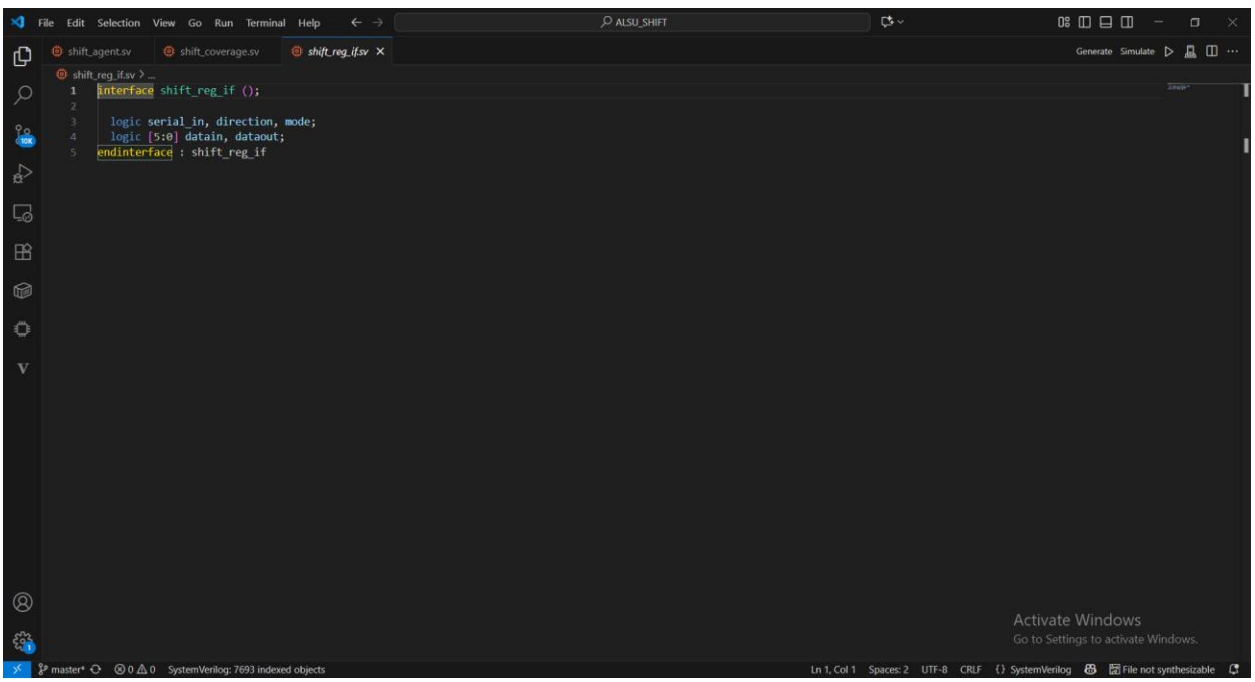
18. Shift_env

```


package shift_env_pkg;
import uvm_pkg::*;
`include "uvm_macros.svh"
    import shift_scoreboard_pkg::*;
    import shift_coverage_pkg::*;
    import shift_agent_pkg::*;

    class shit_reg_env extends uvm_component ;
        `uvm_component_utils(shit_reg_env);
        shift_reg_sb sb;
        shift_reg_cov cov;
        shift_reg_agent agent;

        function new(string name = "shift_reg_env", uvm_component parent =null);
            super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            sb=shift_reg_sb::type_id::create("scoreboard",this);
            cov=shift_reg_cov::type_id::create("coverage collector",this);
            agent=shift_reg_agent::type_id::create("Agent",this);
        endfunction

        function void connect_phase(uvm_phase phase);
            super.connect_phase(phase);
            agent.agt_p.connect(sb.sb_export);
            agent.agt_p.connect(cov.cov_export);
        endfunction
    endclass
endpackage
```

19. shift_monitor



20. shift_reg_if

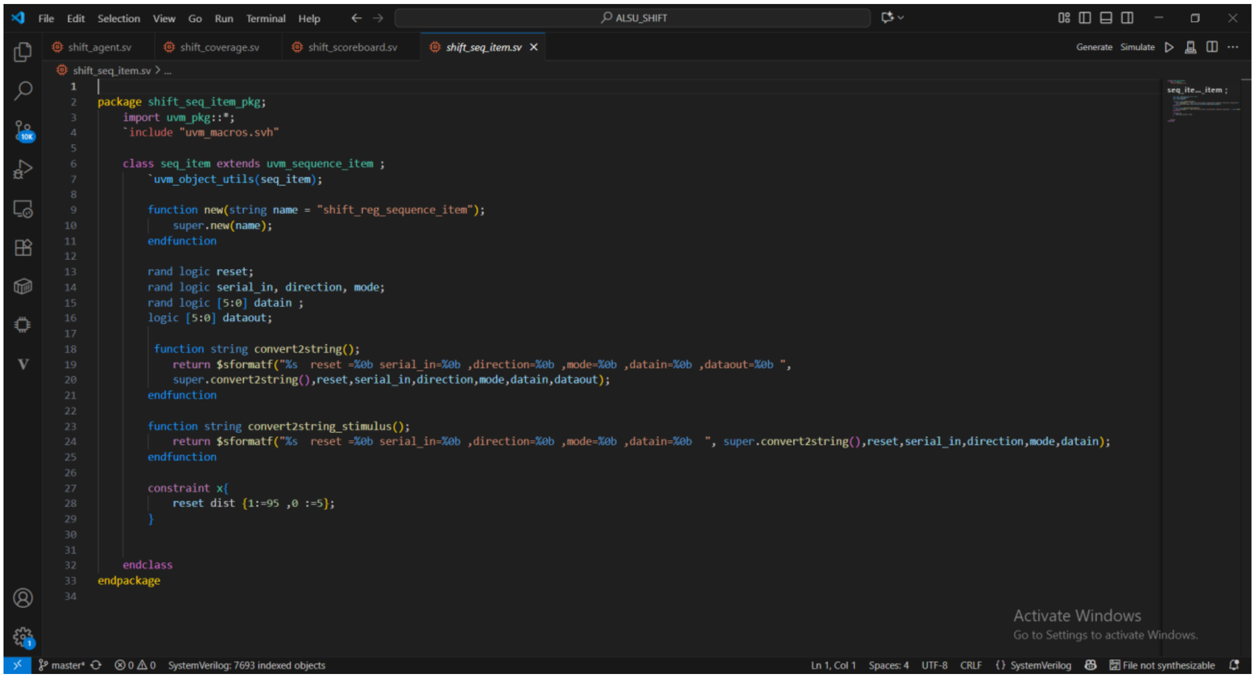


21. shift_scoreboard

```systemverilog
package shift_scoreboard_pkg;

import uvm_pkg::*;
`include "uvm_macros.svh"
import shift_seq_item_pkg::*;

    class shift_reg_sb extends uvm_scoreboard;
        `uvm_component_utils(shift_reg_sb);

        seq_item itm;
        logic [5:0] dataout_ref;
        int error,correct;
        uvm_analysis_export #(seq_item) sb_export;
        uvm_tlm_analysis_fifo #(seq_item) sb_fifo;
        function new(string name = "shift_reg_sb", uvm_component parent =null);
            super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
            super.build_phase(phase);
            sb_export=new("sb_export",this);
            sb_fifo=new("sb_fifo",this);
        endfunction

        function void connect_phase(uvm_phase phase);
            super.connect_phase(phase);
            sb_export.connect(sb_fifo.analysis_export);
        endfunction

        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            forever begin
                sb_fifo.get(itm);
                refrence_model(itm);
                if(itm.dataout != dataout_ref)begin
                    `uvm_error("run_phase",$sformatf("Trascaion recieved is %s and data_out ref =%0d",itm.convert2string(),dataout_ref));
                    error++;
```

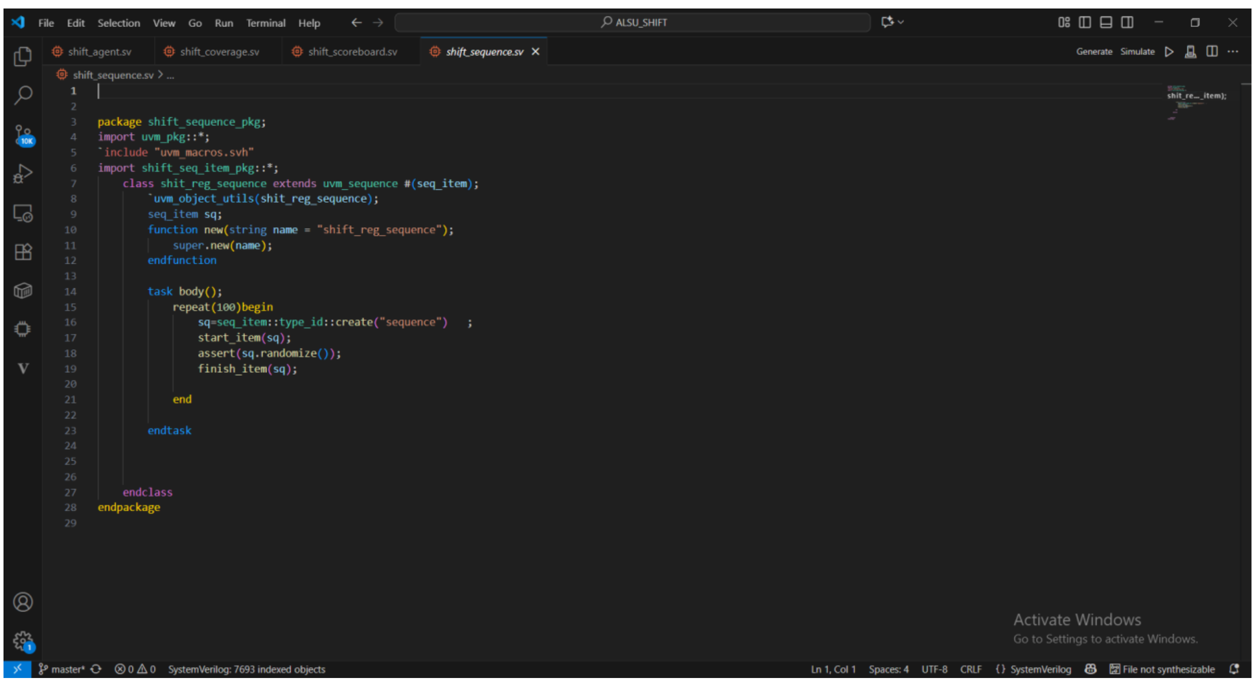
```systemverilog
        task run_phase(uvm_phase phase);
            super.run_phase(phase);
            forever begin
                sb_fifo.get(itm);
                refrence_model(itm);
                if(itm.dataout != dataout_ref)begin
                    `uvm_error("run_phase",$sformatf("Trascaion recieved is %s and data_out ref =%0d",itm.convert2string(),dataout_ref));
                    error++;
                end
                else correct++;
            end
        endtask

        task refrence_model(seq_item itms);
                if (itms.mode) // rotate
                    if (itms.direction) // left
                        dataout_ref = {itms.datain[4:0], itms.datain[5]};
                    else
                        dataout_ref = {itms.datain[0], itms.datain[5:1]};
                else // shift
                    if (itms.direction) // left
                        dataout_ref = {itms.datain[4:0], itms.serial_in};
                    else
                        dataout_ref = {itms.serial_in, itms.datain[5:1]};

        endtask

        function void report_phase (uvm_phase phase);
            super.report_phase(phase);
            `uvm_info("report_phase",$sformatf("cases passed %0d,failed cases %0d",correct,error),UVM_MEDIUM);
        endfunction
    endclass
endpackage
```

22. shift_seq_item



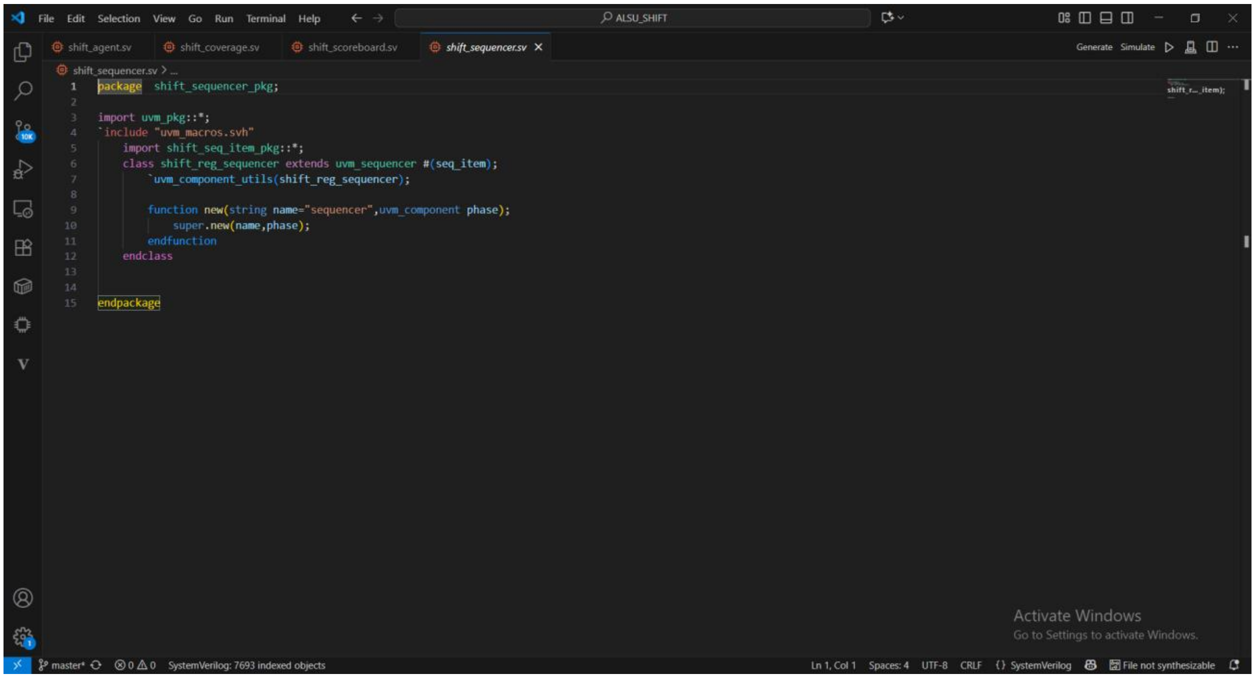
```
package shift_seq_item_pkg;
    import uvm_pkg::*;
    `include "uvm_macros.svh"

    class seq_item extends uvm_sequence_item ;
        `uvm_object_utils(seq_item);

        function new(string name = "shift_reg_sequence_item");
            super.new(name);
        endfunction

        rand logic reset;
        rand logic serial_in, direction, mode;
        rand logic [5:0] datain ;
        logic [5:0] dataout;

         function string convert2string();
            return $sformatf("%s  reset =%0b serial_in=%0b ,direction=%0b ,mode=%0b ,datain=%0b ,dataout=%0b ",
            super.convert2string(),reset,serial_in,direction,mode,datain,dataout);
        endfunction

        function string convert2string_stimulus();
            return $sformatf("%s  reset =%0b serial_in=%0b ,direction=%0b ,mode=%0b ,datain=%0b  ", super.convert2string(),reset,serial_in,direction,mode,datain);
        endfunction

        constraint x{
            reset dist {1:=95 ,0 :=5};
        }


    endclass
endpackage
```

23. shift_sequence



```

package shift_sequence_pkg;
import uvm_pkg::*;
`include "uvm_macros.svh"
import shift_seq_item_pkg::*;
    class shit_reg_sequence extends uvm_sequence #(seq_item);
        `uvm_object_utils(shit_reg_sequence);
        seq_item sq;
        function new(string name = "shift_reg_sequence");
            super.new(name);
        endfunction

        task body();
            repeat(100)begin
                sq=seq_item::type_id::create("sequence")    ;
                start_item(sq);
                assert(sq.randomize());
                finish_item(sq);

            end

        endtask



    endclass
endpackage
```

24. shift_sequencer

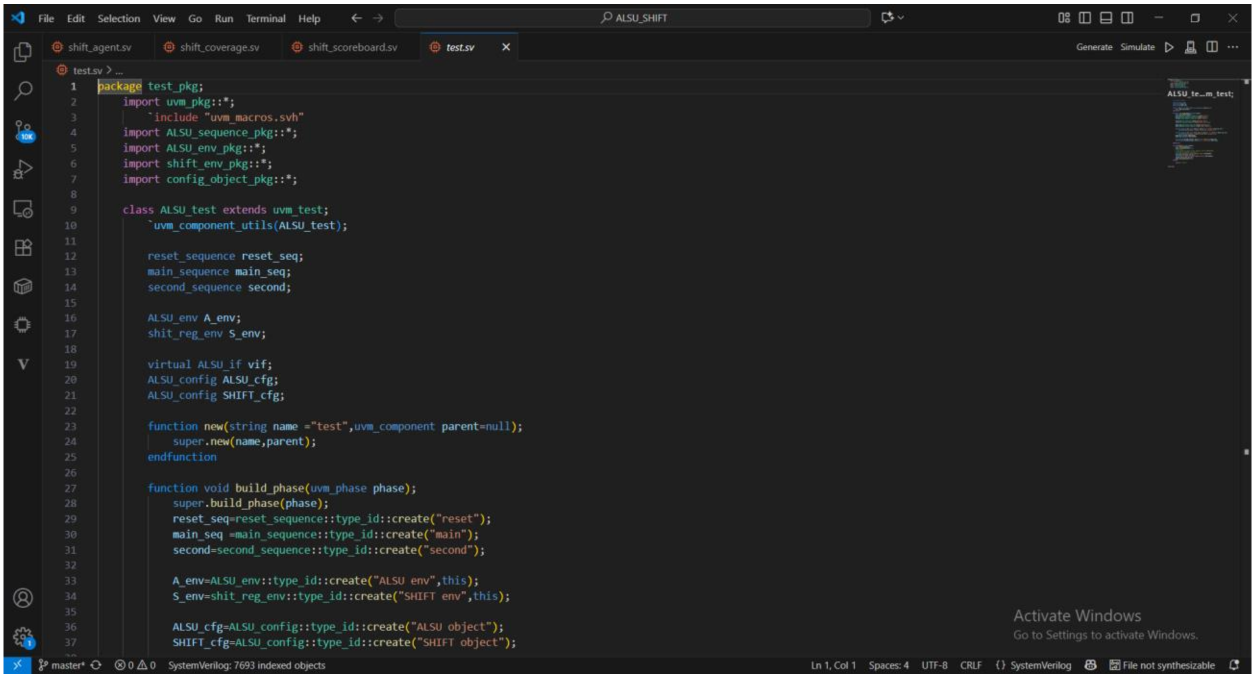

File Edit Selection View Go Run Terminal Help
ALSU_SHIFT
shift_agent.sv
shift_coverage.sv
shift_scoreboard.sv
shift_sequencer.sv
Generate Simulate
shift_sequencer.sv > ...
package shift_sequencer_pkg;

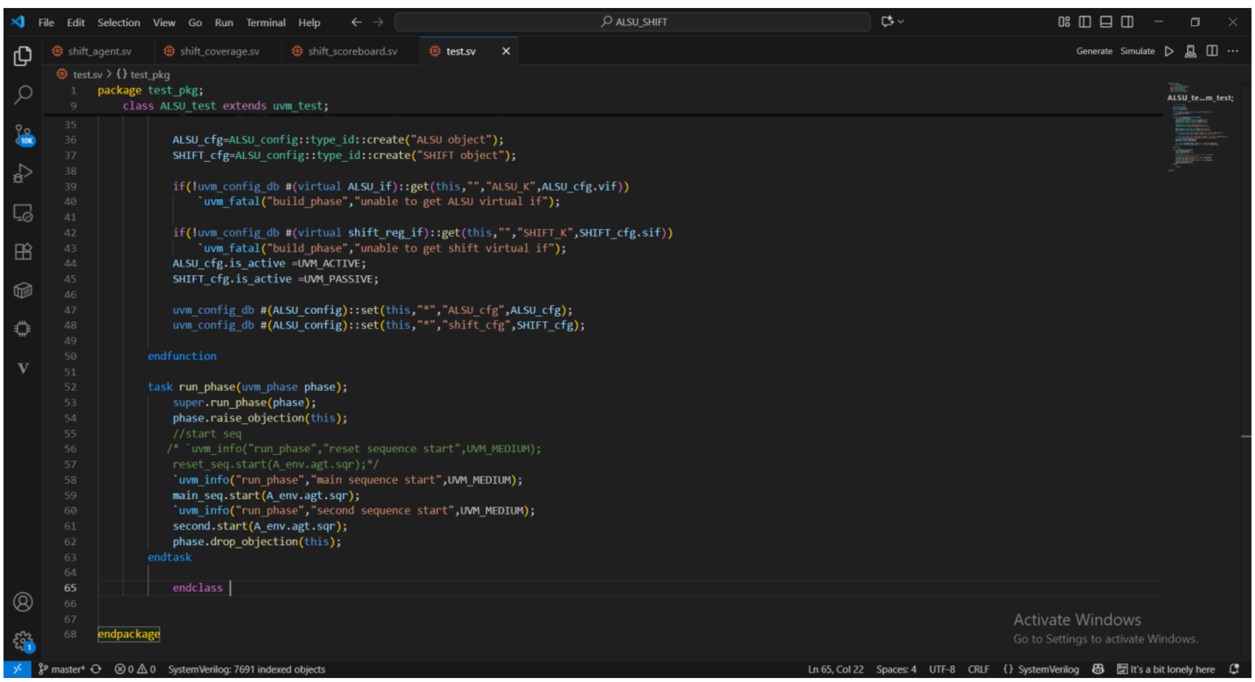
import uvm_pkg::*;
`include "uvm_macros.svh"
import shift_seq_item_pkg::*;
class shift_reg_sequencer extends uvm_sequencer #(seq_item);
`uvm_component_utils(shift_reg_sequencer);

function new(string name="sequencer",uvm_component phase);
super.new(name,phase);
endfunction
endclass

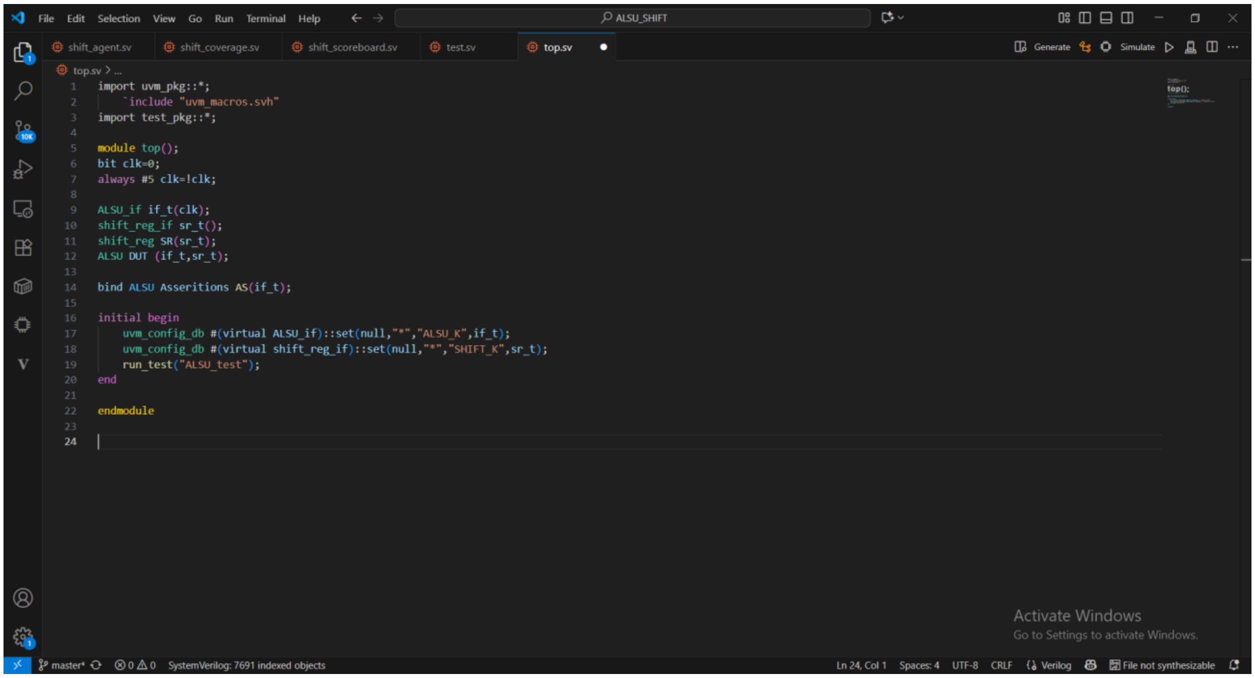
endpackage
Activate Windows
Go to Settings to activate Windows.
master*
SystemVerilog: 7693 indexed objects
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF SystemVerilog File not synthesizable

25. test

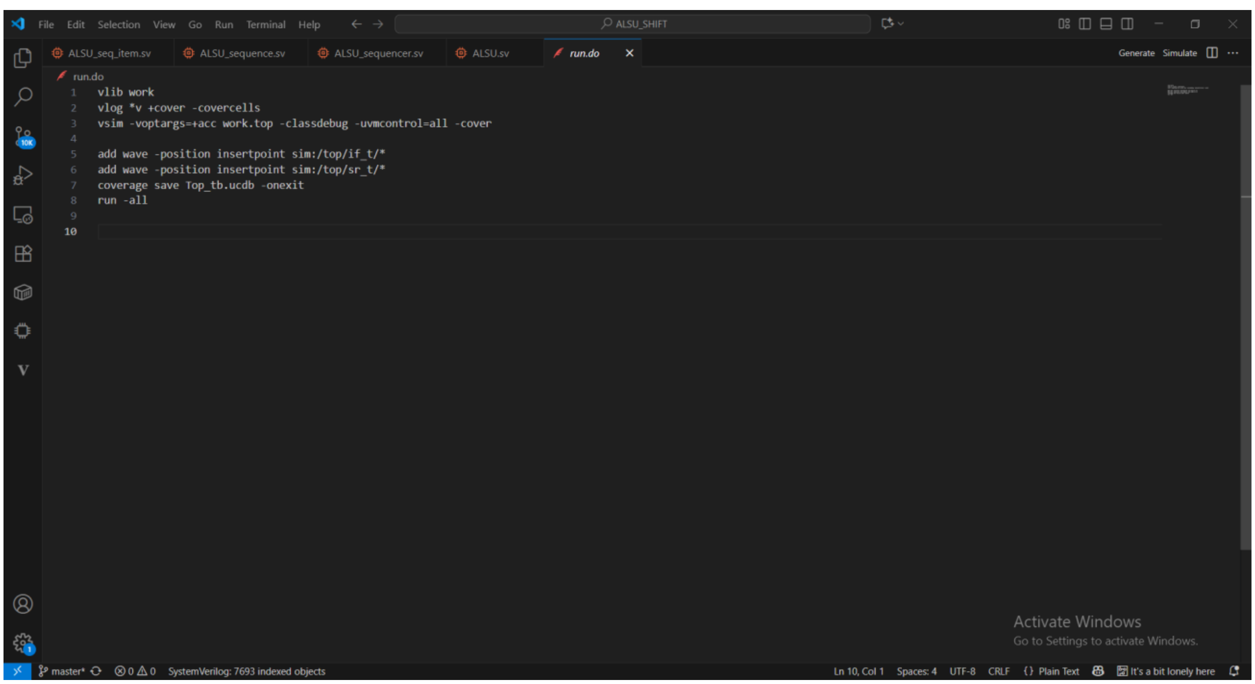

File Edit Selection View Go Run Terminal Help
ALSU_SHIFT
shift_agent.sv
shift_coverage.sv
shift_scoreboard.sv
test.sv
Generate Simulate
package test_pkg;
import uvm_pkg::*;
`include "uvm_macros.svh"
import ALSU_sequence_pkg::*;
import ALSU_env_pkg::*;
import shift_env_pkg::*;
import config_object_pkg::*;
class ALSU_test extends uvm_test;
`uvm_component_utils(ALSU_test);
reset_sequence reset_seq;
main_sequence main_seq;
second_sequence second;
ALSU_env A_env;
shit_reg_env S_env;
virtual ALSU_if vif;
ALSU_config ALSU_cfg;
ALSU_config SHIFT_cfg;
function new(string name ="test",uvm_component parent=null);
super.new(name,parent);
endfunction
function void build_phase(uvm_phase phase);
super.build_phase(phase);
reset_seq=reset_sequence::type_id::create("reset");
main_seq =main_sequence::type_id::create("main");
second=second_sequence::type_id::create("second");
A_env=ALSU_env::type_id::create("ALSU env",this);
S_env=shit_reg_env::type_id::create("SHIFT env",this);
ALSU_cfg=ALSU_config::type_id::create("ALSU object");
SHIFT_cfg=ALSU_config::type_id::create("SHIFT object");
Activate Windows
Go to Settings to activate Windows.
master* 0 0 SystemVerilog: 7693 indexed objects
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF SystemVerilog File not synthesizable


File Edit Selection View Go Run Terminal Help
ALSU_SHIFT
shift_agent.sv
shift_coverage.sv
shift_scoreboard.sv
test.sv
Generate Simulate
test.sv > {} test_pkg
package test_pkg;
class ALSU_test extends uvm_test;
ALSU_cfg=ALSU_config::type_id::create("ALSU object");
SHIFT_cfg=ALSU_config::type_id::create("SHIFT object");
if(!uvm_config_db #(virtual ALSU_if)::get(this,"","ALSU_K",ALSU_cfg.vif))
`uvm_fatal("build_phase","unable to get ALSU virtual if");
if(!uvm_config_db #(virtual shift_reg_if)::get(this,"","SHIFT_K",SHIFT_cfg.sif))
`uvm_fatal("build_phase","unable to get shift virtual if");
ALSU_cfg.is_active =UVM_ACTIVE;
SHIFT_cfg.is_active =UVM_PASSIVE;
uvm_config_db #(ALSU_config)::set(this,"*","ALSU_cfg",ALSU_cfg);
uvm_config_db #(ALSU_config)::set(this,"*","shift_cfg",SHIFT_cfg);
endfunction
task run_phase(uvm_phase phase);
super.run_phase(phase);
phase.raise_objection(this);
//start seq
/* `uvm_info("run_phase","reset sequence start",UVM_MEDIUM);
reset_seq.start(A_env.agt.sqr);*/
`uvm_info("run_phase","main sequence start",UVM_MEDIUM);
main_seq.start(A_env.agt.sqr);
`uvm_info("run_phase","second sequence start",UVM_MEDIUM);
second.start(A_env.agt.sqr);
phase.drop_objection(this);
endtask
endclass
endpackage
Activate Windows
Go to Settings to activate Windows.
master* 0 0 SystemVerilog: 7691 indexed objects
Ln 65, Col 22 Spaces: 4 UTF-8 CRLF SystemVerilog It's a bit lonely here

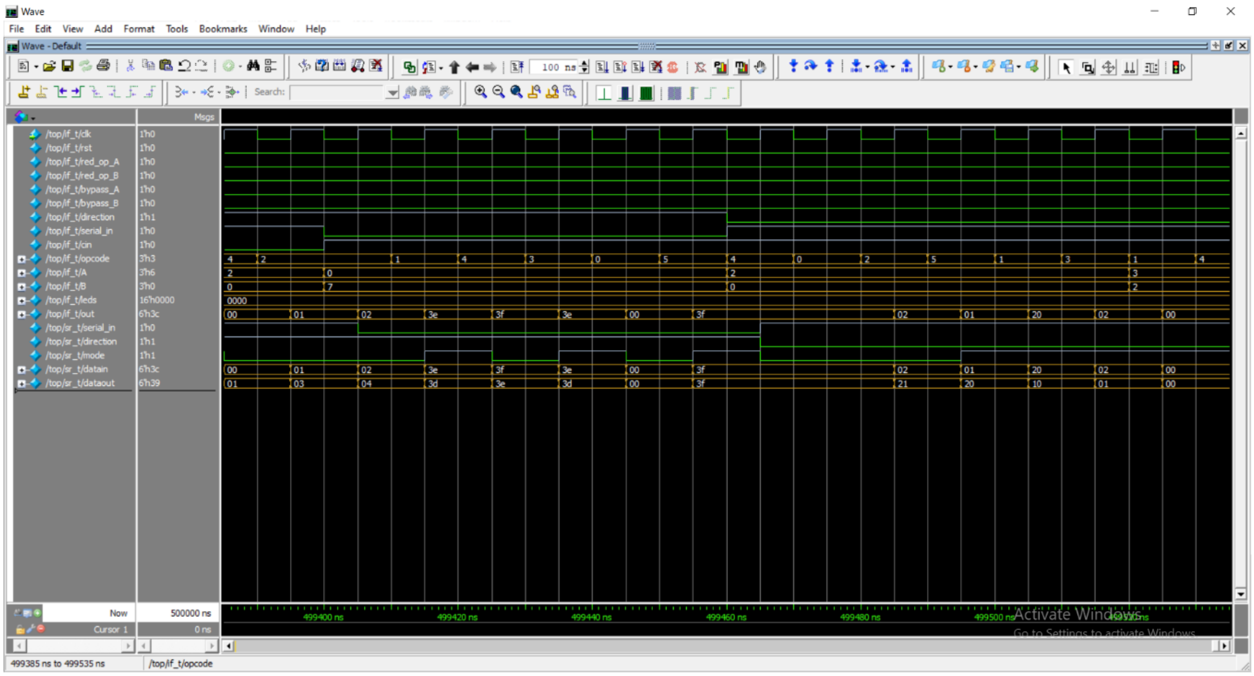
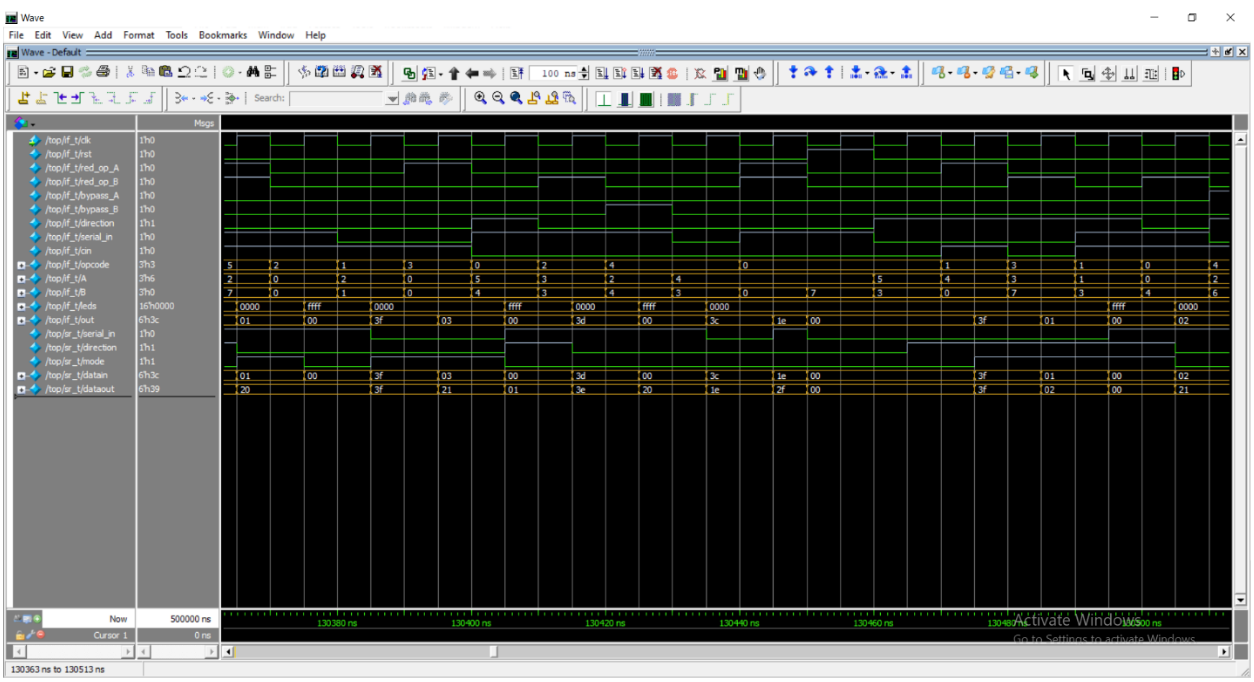
26. top

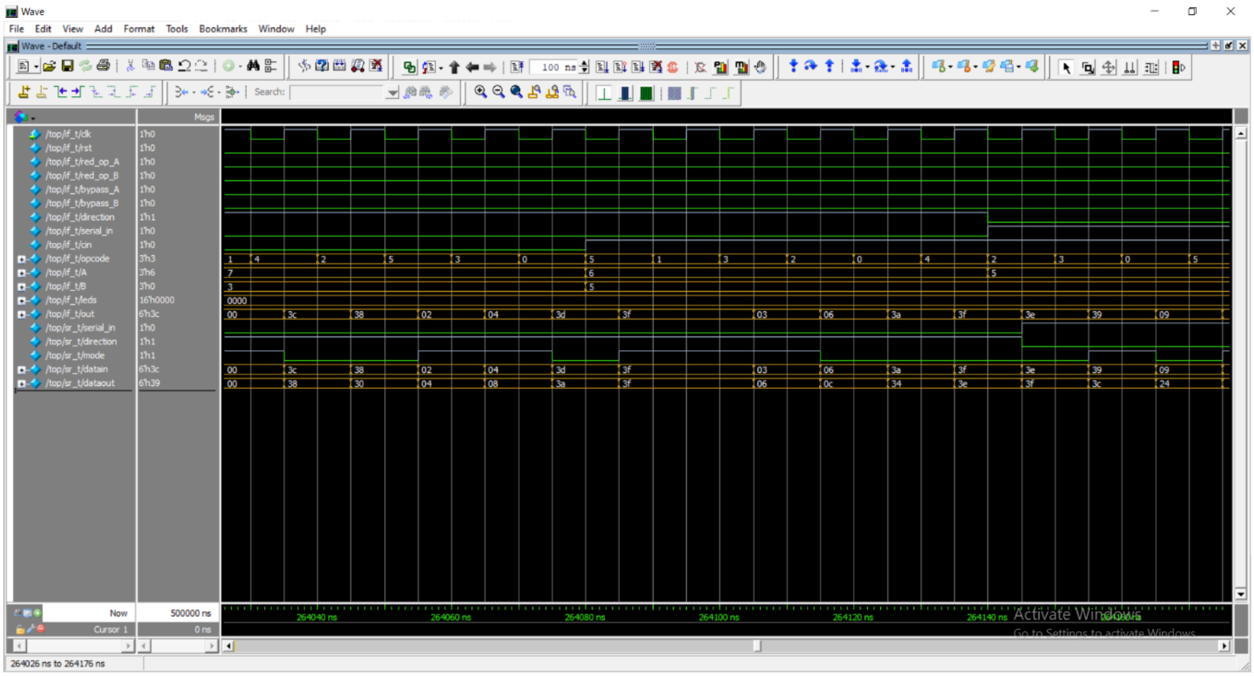


Do file



Waveform


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/top/if_t/clk 1'h0
/top/if_t/rst 1'h0
/top/if_t/red_op_A 1'h0
/top/if_t/red_op_B 1'h0
/top/if_t/bypass_A 1'h0
/top/if_t/bypass_B 1'h0
/top/if_t/direction 1'h1
/top/if_t/serial_in 1'h0
/top/if_t/cin 1'h0
/top/if_t/opcode 3'h3
/top/if_t/A 3'h6
/top/if_t/B 3'h0
/top/if_t/leds 16'h0000
/top/if_t/out 6'h3c
/top/sr_t/serial_in 1'h0
/top/sr_t/direction 1'h1
/top/sr_t/mode 1'h1
/top/sr_t/datain 6'h3c
/top/sr_t/dataout 6'h39
Now 500000 ns
Cursor 1 0 ns
499400 ns
499420 ns
499440 ns
499460 ns
499480 ns
499500 ns
499385 ns to 499535 ns
/top/if_t/opcode

Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/top/if_t/clk 1'h0
/top/if_t/rst 1'h0
/top/if_t/red_op_A 1'h0
/top/if_t/red_op_B 1'h0
/top/if_t/bypass_A 1'h0
/top/if_t/bypass_B 1'h0
/top/if_t/direction 1'h1
/top/if_t/serial_in 1'h0
/top/if_t/cin 1'h0
/top/if_t/opcode 3'h3
/top/if_t/A 3'h6
/top/if_t/B 3'h0
/top/if_t/leds 16'h0000
/top/if_t/out 6'h3c
/top/sr_t/serial_in 1'h0
/top/sr_t/direction 1'h1
/top/sr_t/mode 1'h1
/top/sr_t/datain 6'h3c
/top/sr_t/dataout 6'h39
Now 500000 ns
Cursor 1 0 ns
130380 ns
130400 ns
130420 ns
130440 ns
130460 ns
130363 ns to 130513 ns



Reports

assert

coverage_report - Notepad

File Edit Format View Help

```
Assertion Coverage:
    Assertions                  24        20         4    83.33%
--------------------------------------------------------------------
Name                 File(Line)                  Failure      Pass
                                                 Count        Count
--------------------------------------------------------------------
/top/DUT/AS/rst_check
                     _Assertions.sv(9)                 0          1
/top/DUT/AS/leds_check
                     _Assertions.sv(185)               0          1
/top/DUT/AS/bypass1_check
                     _Assertions.sv(187)               0          1
/top/DUT/AS/bypass2_check
                     _Assertions.sv(188)               0          0
/top/DUT/AS/bypass3_check
                     _Assertions.sv(189)               0          1
/top/DUT/AS/bypass4_check
                     _Assertions.sv(190)               0          1
/top/DUT/AS/INVALID_check
                     _Assertions.sv(193)               0          1
/top/DUT/AS/OR1_check
                     _Assertions.sv(199)               0          1
/top/DUT/AS/OR2_check
                     _Assertions.sv(200)               0          0
/top/DUT/AS/OR3_check
                     _Assertions.sv(201)               0          1
/top/DUT/AS/OR4_check
                     _Assertions.sv(202)               0          1
/top/DUT/AS/OR5_check
                     _Assertions.sv(203)               0          1
/top/DUT/AS/XOR1_check
                     _Assertions.sv(207)               0          1
/top/DUT/AS/XOR2_check
                     _Assertions.sv(208)               0          0
/top/DUT/AS/XOR3_check
                     _Assertions.sv(209)               0          1
/top/DUT/AS/XOR4_check
                     _Assertions.sv(210)               0          1
/top/DUT/AS/XOR5_check
                     _Assertions.sv(211)               0          1
/top/DUT/AS/FullADD_check
                     _Assertions.sv(215)               0          1
```


Ln 43, Col 17 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
                     _Assertions.sv(203)               0          1
/top/DUT/AS/XOR1_check
                     _Assertions.sv(207)               0          1
/top/DUT/AS/XOR2_check
                     _Assertions.sv(208)               0          0
/top/DUT/AS/XOR3_check
                     _Assertions.sv(209)               0          1
/top/DUT/AS/XOR4_check
                     _Assertions.sv(210)               0          1
/top/DUT/AS/XOR5_check
                     _Assertions.sv(211)               0          1
/top/DUT/AS/FullADD_check
                     _Assertions.sv(215)               0          1
/top/DUT/AS/halfADD_check
                     _Assertions.sv(216)               0          0
/top/DUT/AS/MULT_check
                     _Assertions.sv(220)               0          1
/top/DUT/AS/shift_R_check
                     _Assertions.sv(224)               0          1
/top/DUT/AS/shift_L_check
                     _Assertions.sv(225)               0          1
/top/DUT/AS/rotate_R_check
                     _Assertions.sv(229)               0          1
/top/DUT/AS/rotate_L_check
                     _Assertions.sv(230)               0          1
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                         2         2         0   100.00%

================================Branch Details================================

Branch Coverage for instance /top/DUT/AS

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File _Assertions.sv
------------------------------------IF Branch------------------------------------
    8                                      44600     Count coming in to IF
    8              1                        1648
                                           42952     All False Count
Branch totals: 2 hits of 2 branches = 100.00%
```


Ln 160, Col 51 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
Expression Coverage:
    Enabled Coverage              Bins    Covered    Misses  Coverage
    ----------------              ----    -------    ------  --------
    Expressions                      5          5         0   100.00%

================================Expression Details================================

Expression Coverage for instance /top/DUT/AS --

  File _Assertions.sv
----------------Focused Expression View-----------------
Line       5 Item    1  (((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) || ((if_t.opcode > 1) && (if_t.red_op_A | if_t.red_op_B)))
Expression totals: 5 of 5 input terms covered = 100.00%

                 Input Term   Covered  Reason for no coverage   Hint
                -----------  --------  -----------------------  --------------
  (if_t.opcode == INVALID6)         Y
  (if_t.opcode == INVALID7)         Y
          (if_t.opcode > 1)         Y
              if_t.red_op_A         Y
              if_t.red_op_B         Y

     Rows:       Hits  FEC Target                  Non-masking condition(s)
 ---------  ---------  --------------------        -------------------------
  Row   1:          1  (if_t.opcode == INVALID6)_0  (~((if_t.opcode > 1) && (if_t.red_op_A | if_t.red_op_B)) && ~(if_t.opcode == INVALID7))
  Row   2:          1  (if_t.opcode == INVALID6)_1  -
  Row   3:          1  (if_t.opcode == INVALID7)_0  (~((if_t.opcode > 1) && (if_t.red_op_A | if_t.red_op_B)) && ~(if_t.opcode == INVALID6))
  Row   4:          1  (if_t.opcode == INVALID7)_1  ~(if_t.opcode == INVALID6)
  Row   5:          1  (if_t.opcode > 1)_0          ~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7))
  Row   6:          1  (if_t.opcode > 1)_1          (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.red_op_A | if_t.red_op_B))
  Row   7:          1  if_t.red_op_A_0              (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_B)
  Row   8:          1  if_t.red_op_A_1              (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_B)
  Row   9:          1  if_t.red_op_B_0              (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_A)
 Row  10:          1  if_t.red_op_B_1              (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_A)


Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       2         2         0   100.00%

================================Statement Details================================
```

Activate Windows
Go to Settings to activate Windows.

Ln 160, Col 51 | 100% | Windows (CRLF) | UTF-8

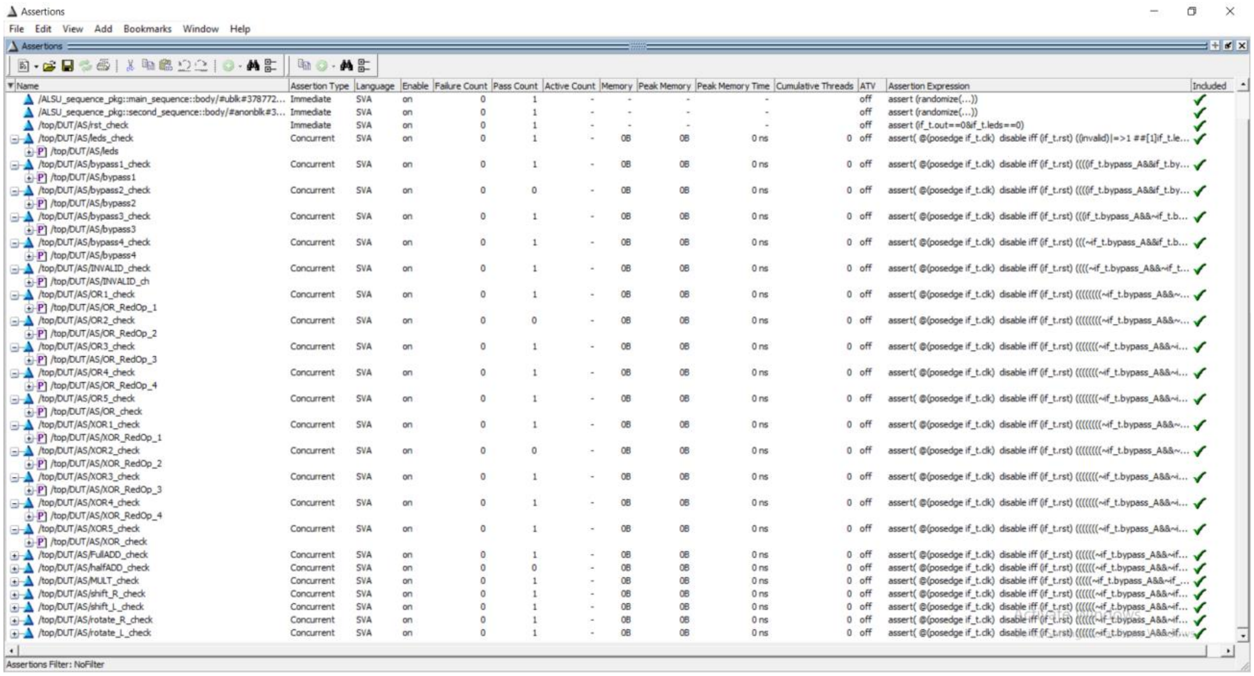
coverage_report - Notepad

File Edit Format View Help

```
 Row  10:          1  if_t.red_op_B_1              (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_A)


Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       2         2         0   100.00%

================================Statement Details================================

Statement Coverage for instance /top/DUT/AS --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File _Assertions.sv
    5               1                      48005
    7               1                      44600
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                          2         2         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /top/DUT/AS --

                                              Node      1H->0L      0L->1H  "Coverage"
                                              ---------------------------------------
                                           invalid           1           1      100.00

Total Node Count     =          1
Toggled Node Count   =          1
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (2 of 2 bins)

=================================================================================
=== Instance: /top/DUT
=== Design Unit: work.ALSU
=================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
```

Activate Windows
Go to Settings to activate Windows.

Ln 160, Col 51 | 100% | Windows (CRLF) | UTF-8



Assertions

File Edit View Add Bookmarks Window Help

Name	Assertion Type	Language	Enable	Failure Count	Pass Count	Active Count	Memory	Peak Memory	Peak Memory Time	Cumulative Threads	ATV	Assertion Expression	Included
/ALSU_sequence_pkg::main_sequence::body/#ublk#378772...	Immediate	SVA	on	0	1	-	-	-	-		off	assert (randomize(...))	✓
/ALSU_sequence_pkg::second_sequence::body/#anonblk#3...	Immediate	SVA	on	0	1	-	-	-	-		off	assert (randomize(...))	✓
/top/DUT/AS/rst_check	Immediate	SVA	on	0	1	-	-	-	-		off	assert (if_t.out==0&if_t.leds==0)	✓
/top/DUT/AS/leds_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((invalid)|=>1 ##[1]if_t.le...	✓
P /top/DUT/AS/leds													
/top/DUT/AS/bypass1_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((if_t.bypass_A&&if_t.by...	✓
P /top/DUT/AS/bypass1													
/top/DUT/AS/bypass2_check	Concurrent	SVA	on	0	0	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((if_t.bypass_A&&if_t.by...	✓
P /top/DUT/AS/bypass2													
/top/DUT/AS/bypass3_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((if_t.bypass_A&&~if_t.b...	✓
P /top/DUT/AS/bypass3													
/top/DUT/AS/bypass4_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((~if_t.bypass_A&&if_t.b...	✓
P /top/DUT/AS/bypass4													
/top/DUT/AS/INVALID_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((~if_t.bypass_A&&~if_t...	✓
P /top/DUT/AS/INVALID_ch													
/top/DUT/AS/OR1_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((((~if_t.bypass_A&&~...	✓
P /top/DUT/AS/OR_RedOp_1													
/top/DUT/AS/OR2_check	Concurrent	SVA	on	0	0	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((((~if_t.bypass_A&&~...	✓
P /top/DUT/AS/OR_RedOp_2													
/top/DUT/AS/OR3_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((((((~if_t.bypass_A&&~i...	✓
P /top/DUT/AS/OR_RedOp_3													
/top/DUT/AS/OR4_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((((((~if_t.bypass_A&&~i...	✓
P /top/DUT/AS/OR_RedOp_4													
/top/DUT/AS/OR5_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((((((~if_t.bypass_A&&~i...	✓
P /top/DUT/AS/OR_check													
/top/DUT/AS/XOR1_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((((~if_t.bypass_A&&~...	✓
P /top/DUT/AS/XOR_RedOp_1													
/top/DUT/AS/XOR2_check	Concurrent	SVA	on	0	0	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((((~if_t.bypass_A&&~...	✓
P /top/DUT/AS/XOR_RedOp_2													
/top/DUT/AS/XOR3_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((((((~if_t.bypass_A&&~i...	✓
P /top/DUT/AS/XOR_RedOp_3													
/top/DUT/AS/XOR4_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((((((~if_t.bypass_A&&~i...	✓
P /top/DUT/AS/XOR_RedOp_4													
/top/DUT/AS/XOR5_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((((((~if_t.bypass_A&&~i...	✓
P /top/DUT/AS/XOR_check													
/top/DUT/AS/FullADD_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((~if_t.bypass_A&&~if...	✓
/top/DUT/AS/halfADD_check	Concurrent	SVA	on	0	0	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((~if_t.bypass_A&&~if...	✓
/top/DUT/AS/MULT_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) ((((((~if_t.bypass_A&&~if_...	✓
/top/DUT/AS/shift_R_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((~if_t.bypass_A&&~if...	✓
/top/DUT/AS/shift_L_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((~if_t.bypass_A&&~if...	✓
/top/DUT/AS/rotate_R_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((~if_t.bypass_A&&~if...	✓
/top/DUT/AS/rotate_L_check	Concurrent	SVA	on	0	1	-	0B	0B	0 ns	0	off	assert(@(posedge if_t.clk) disable iff (if_t.rst) (((((((~if_t.bypass_A&&~if...	✓

Assertions Filter: NoFilter

Cvr

Fsm

coverage_report - Notepad

File Edit Format View Help

```
Coverage Report by instance with details

================================================================================
=== Instance: /top/if_t
=== Design Unit: work.ALSU_if
================================================================================
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         80        80         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /top/if_t --

                                              Node      1H->0L      0L->1H  "Coverage"
                                              ---------------------------------------
                                            A[2-0]           1           1      100.00
                                            B[2-0]           1           1      100.00
                                          bypass_A           1           1      100.00
                                          bypass_B           1           1      100.00
                                               cin           1           1      100.00
                                               clk           1           1      100.00
                                         direction           1           1      100.00
                                        leds[15-0]           1           1      100.00
                                       opcode[2-0]           1           1      100.00
                                          out[5-0]           1           1      100.00
                                          red_op_A           1           1      100.00
                                          red_op_B           1           1      100.00
                                               rst           1           1      100.00
                                         serial_in           1           1      100.00

Total Node Count     =         40
Toggled Node Count   =         40
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (80 of 80 bins)

================================================================================
=== Instance: /top/sr_t
=== Design Unit: work.shift_reg_if
================================================================================
Toggle Coverage:
```

Ln 1, Col 1 | 100% | Windows (CRLF) | UTF-8

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File Edit Format View Help

```
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         30        30         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /top/sr_t --

                                              Node      1H->0L      0L->1H  "Coverage"
                                              ---------------------------------------
                                       datain[5-0]           1           1      100.00
                                      dataout[5-0]           1           1      100.00
                                         direction           1           1      100.00
                                              mode           1           1      100.00
                                         serial_in           1           1      100.00

Total Node Count     =         15
Toggled Node Count   =         15
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (30 of 30 bins)

================================================================================
=== Instance: /top/SR
=== Design Unit: work.shift_reg
================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                         5         5         0   100.00%

================================Branch Details================================

Branch Coverage for instance /top/SR

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File shift_reg.sv
------------------------------------IF Branch------------------------------------
    13                                     47255     Count coming in to IF
    13              1                      23201
    19              1                      11515
```

Ln 43, Col 17 | 100% | Windows (CRLF) | UTF-8

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File Edit Format View Help

```
    Branches                      5         5         0   100.00%

================================Branch Details================================

Branch Coverage for instance /top/SR

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File shift_reg.sv
------------------------------------IF Branch------------------------------------
    13                                     47255     Count coming in to IF
    13              1                      23201
    19              1                      11515
    21              1                      12539
Branch totals: 3 hits of 3 branches = 100.00%

------------------------------------IF Branch------------------------------------
    14                                     23201     Count coming in to IF
    14              1                      11481
    16              1                      11720
Branch totals: 2 hits of 2 branches = 100.00%


Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       5         5         0   100.00%

================================Statement Details================================

Statement Coverage for instance /top/SR --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File shift_reg.sv
    11              1                          1
    15              1                      11481
    17              1                      11720
    20              1                      11515
    22              1                      12539

=================================================================================
=== Instance: /top/DUT/AS
```


Ln 43, Col 17 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
=================================================================================
=== Instance: /top/DUT
=== Design Unit: work.ALSU
=================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        28        28         0   100.00%

================================Branch Details================================

Branch Coverage for instance /top/DUT

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ALSU.sv
------------------------------------IF Branch------------------------------------
    28                                     50948     Count coming in to IF
    28              1                       1937
    40              1                      49011
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    57                                     50948     Count coming in to IF
    57              1                       1937
    59              1                      49011
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    60                                     49011     Count coming in to IF
    60              1                       8696
    62              1                      40315
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    70                                     50948     Count coming in to IF
    70              1                       1937
    73              1                      49011
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    74                                     49011     Count coming in to IF
    74              1                        176
```


Ln 160, Col 51 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
-----------------------------------IF Branch----------------------------------
    74                                    49011     Count coming in to IF
    74              1                       176
    76              1                      1560
    78              1                      1566
    80              1                      7123
    82              1                     38586
Branch totals: 5 hits of 5 branches = 100.00%

-----------------------------------CASE Branch----------------------------------
    83                                    38586     Count coming in to CASE
    84              1                      8604
    94              1                      7650
    104             1                      5535
    112             1                      5478
    113             1                      5674
    116             1                      5644
    119             1                         1
Branch totals: 7 hits of 7 branches = 100.00%

-----------------------------------IF Branch----------------------------------
    85                                     8604     Count coming in to IF
    85              1                       367
    87              1                       222
    89              1                       285
    91              1                      7730
Branch totals: 4 hits of 4 branches = 100.00%

-----------------------------------IF Branch----------------------------------
    95                                     7650     Count coming in to IF
    95              1                       368
    97              1                       246
    99              1                       284
    101             1                      6752
Branch totals: 4 hits of 4 branches = 100.00%


Condition Coverage:
    Enabled Coverage              Bins    Covered    Misses  Coverage
    ----------------              ----    ----       ------  --------
    Conditions                       6          6         0   100.00%
```

Activate Windows
Go to Settings to activate Windows.

Ln 358, Col 54 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
================================Condition Details================================

Condition Coverage for instance /top/DUT --

  File ALSU.sv
----------------Focused Condition View------------------
Line       74 Item    1  (bypass_A_reg && bypass_B_reg)
Condition totals: 2 of 2 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
   -----------  --------  -----------------------  --------------
  bypass_A_reg         Y
  bypass_B_reg         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  bypass_A_reg_0        -
  Row   2:          1  bypass_A_reg_1        bypass_B_reg
  Row   3:          1  bypass_B_reg_0        bypass_A_reg
  Row   4:          1  bypass_B_reg_1        bypass_A_reg

----------------Focused Condition View------------------
Line       85 Item    1  (red_op_A_reg && red_op_B_reg)
Condition totals: 2 of 2 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
   -----------  --------  -----------------------  --------------
  red_op_A_reg         Y
  red_op_B_reg         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  red_op_A_reg_0        -
  Row   2:          1  red_op_A_reg_1        red_op_B_reg
  Row   3:          1  red_op_B_reg_0        red_op_A_reg
  Row   4:          1  red_op_B_reg_1        red_op_A_reg

----------------Focused Condition View------------------
Line       95 Item    1  (red_op_A_reg && red_op_B_reg)
Condition totals: 2 of 2 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
```

Activate Windows
Go to Settings to activate Windows.

Ln 396, Col 69 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                      50        50         0   100.00%

================================Statement Details================================

Statement Coverage for instance /top/DUT --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ALSU.sv
    13             1                       47833
    14             1                       45634
    15             1                        9886
    23             1                       37668
    24             1                       40560
    27             1                       50948
    29             1                        1937
    30             1                        1937
    31             1                        1937
    32             1                        1937
    33             1                        1937
    34             1                        1937
    35             1                        1937
    36             1                        1937
    37             1                        1937
    38             1                        1937
    41             1                       49011
    42             1                       49011
    43             1                       49011
    44             1                       49011
    45             1                       49011
    46             1                       49011
    47             1                       49011
    48             1                       49011
    49             1                       49011
    50             1                       49011
    56             1                       50948
    58             1                        1937
    61             1                        8696
    63             1                       40315
    68             1                       50948
```


Ln 396, Col 69 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
    68             1                       50948
    71             1                        1937
    75             1                         176
    77             1                        1560
    79             1                        1566
    81             1                        7123
    86             1                         367
    88             1                         222
    90             1                         285
    92             1                        7730
    96             1                         368
    98             1                         246
    100            1                         284
    102            1                        6752
    106            1                        5535
    112            1                        5478
    114            1                        5674
    117            1                        5644
    119            1                           1
    123            1                       50948
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         64        64         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /top/DUT --

                                              Node      1H->0L      0L->1H  "Coverage"
                                              ---------------------------------------
                                        A_reg[2-0]           1           1      100.00
                                        B_reg[2-0]           1           1      100.00
                                      bypass_A_reg           1           1      100.00
                                      bypass_B_reg           1           1      100.00
                                      cin_reg[1-0]           1           1      100.00
                                     direction_reg           1           1      100.00
                                           invalid           1           1      100.00
                                    invalid_opcode           1           1      100.00
                                    invalid_red_op           1           1      100.00
                                   opcode_reg[2-0]           1           1      100.00
                                     out_next[5-0]           1           1      100.00
                                out_shift_reg[5-0]           1           1      100.00
```


Ln 559, Col 51 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
                                  invalid_red_op           1           1      100.00
                                 opcode_reg[2-0]           1           1      100.00
                                   out_next[5-0]           1           1      100.00
                              out_shift_reg[5-0]           1           1      100.00
                                    red_op_A_reg           1           1      100.00
                                    red_op_B_reg           1           1      100.00
                                   serial_in_reg           1           1      100.00

Total Node Count     =         32
Toggled Node Count   =         32
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (64 of 64 bins)

=================================================================================
=== Instance: /top
=== Design Unit: work.top
=================================================================================
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       5         5         0   100.00%

================================Statement Details================================

Statement Coverage for instance /top --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File top.sv
    7               1                     100001
    7               2                     100000
    17              1                          1
    18              1                          1
    19              1                          1
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                          2         2         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /top --
```

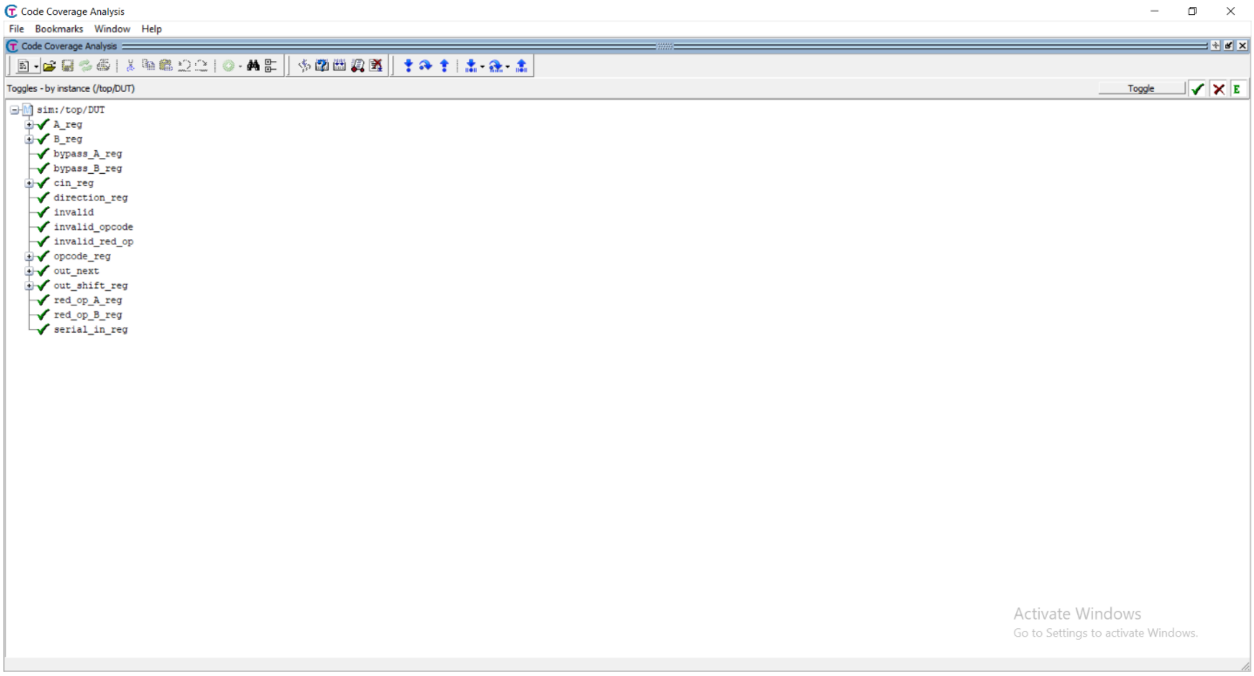

Ln 601, Col 71 | 100% | Windows (CRLF) | UTF-8

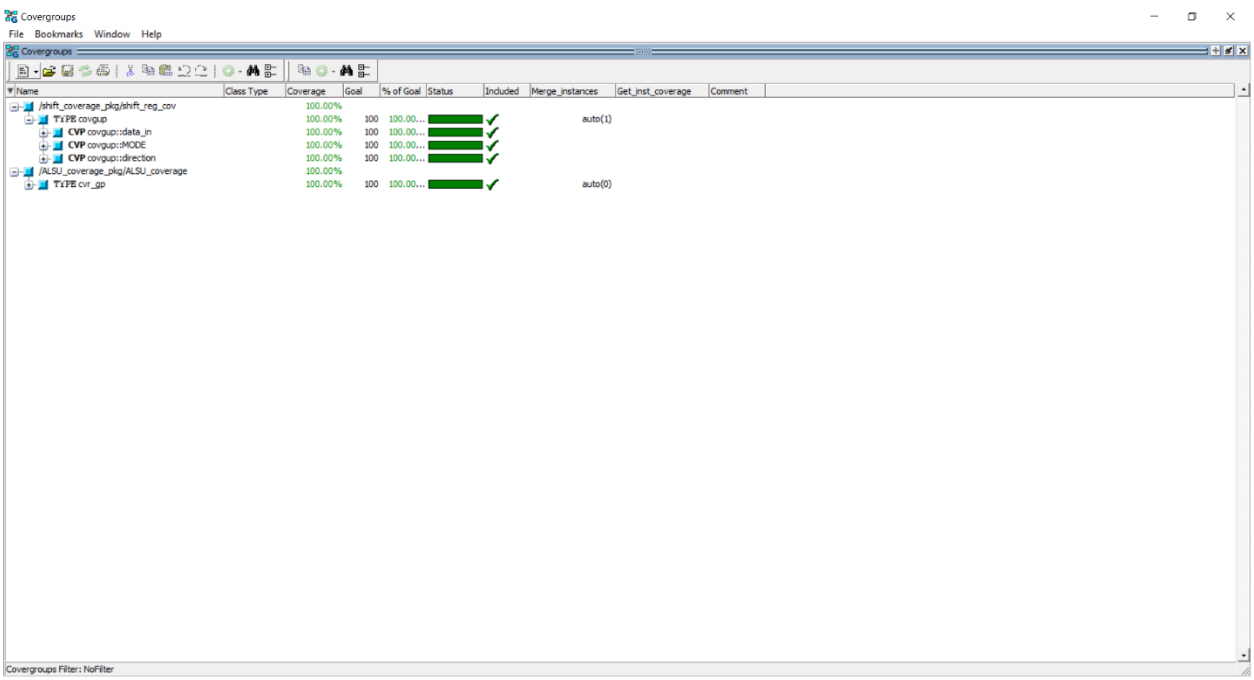
coverage_report - Notepad

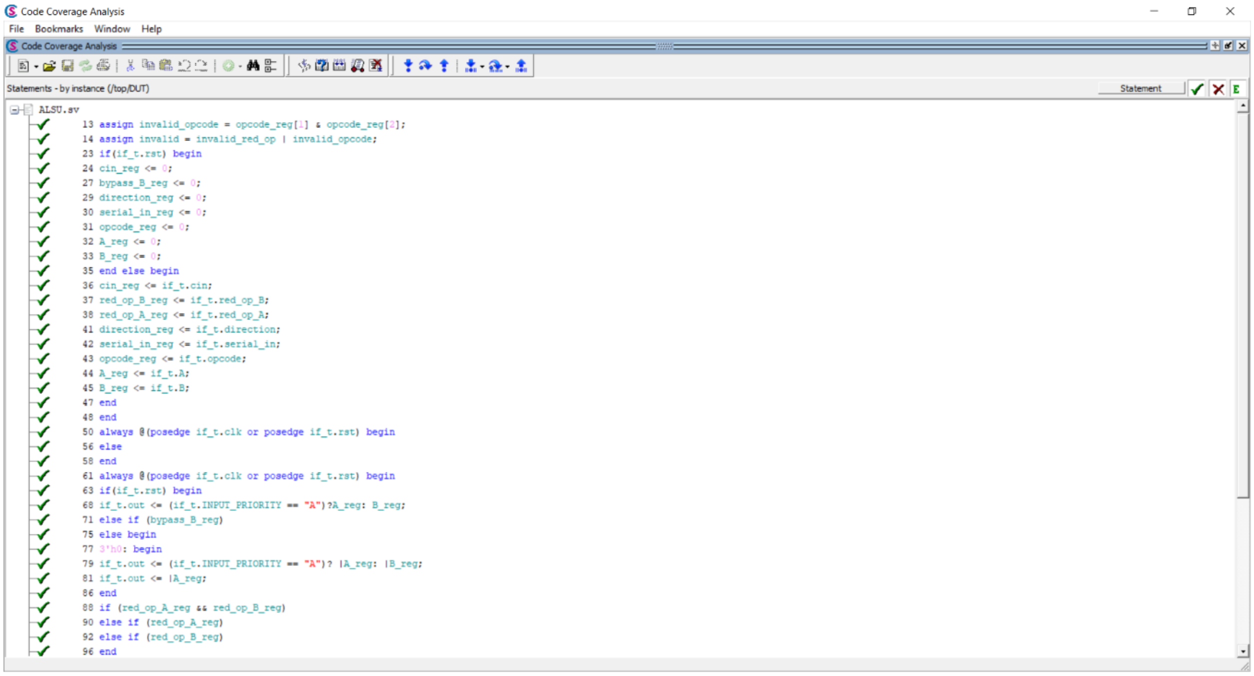
File Edit Format View Help

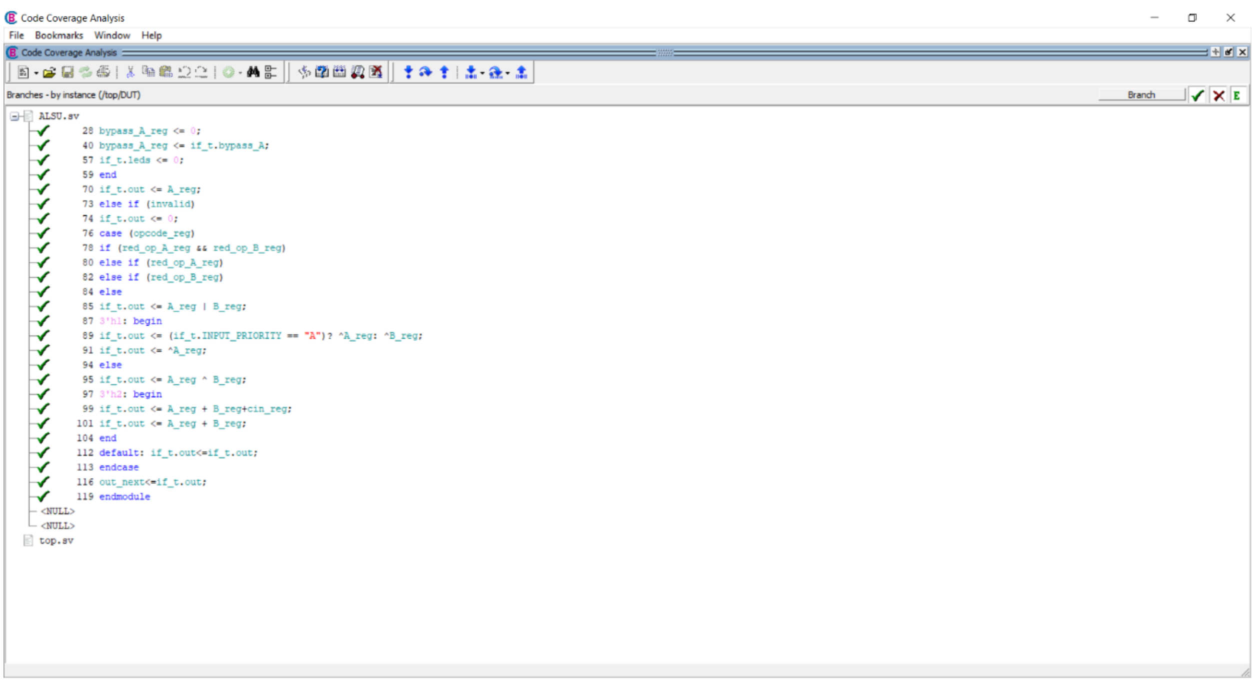
```
    Statements                       5         5         0   100.00%

================================Statement Details================================

Statement Coverage for instance /top --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File top.sv
    7               1                     100001
    7               2                     100000
    17              1                          1
    18              1                          1
    19              1                          1
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                          2         2         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /top --

                                              Node      1H->0L      0L->1H  "Coverage"
                                              ---------------------------------------
                                               clk           1           1      100.00

Total Node Count     =          1
Toggled Node Count   =          1
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (2 of 2 bins)

=================================================================================
=== Instance: /config_object_pkg
=== Design Unit: work.config_object_pkg
=================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        10         0        10     0.00%

================================Branch Details================================
```


Ln 601, Col 71 | 100% | Windows (CRLF) | UTF-8


Code Coverage Analysis
File Bookmarks Window Help
Code Coverage Analysis
Toggles - by instance (/top/DUT)
Toggle
sim:/top/DUT
A_reg
B_reg
bypass_A_reg
bypass_B_reg
cin_reg
direction_reg
invalid
invalid_opcode
invalid_red_op
opcode_reg
out_next
out_shift_reg
red_op_A_reg
red_op_B_reg
serial_in_reg
Activate Windows
Go to Settings to activate Windows.


Covergroups
File Bookmarks Window Help
Covergroups
Name Class Type Coverage Goal % of Goal Status Included Merge_instances Get_inst_coverage Comment
/shift_coverage_pkg/shift_reg_cov 100.00%
TYPE covgup 100.00% 100 100.00... auto(1)
CVP covgup::data_in 100.00% 100 100.00...
CVP covgup::MODE 100.00% 100 100.00...
CVP covgup::direction 100.00% 100 100.00...
/ALSU_coverage_pkg/ALSU_coverage 100.00%
TYPE cvr_gp 100.00% 100 100.00... auto(0)
Covergroups Filter: NoFilter


Code Coverage Analysis
File Bookmarks Window Help
Code Coverage Analysis
Statements - by instance (/top/DUT)
Statement
E
ALSU.sv
13 assign invalid_opcode = opcode_reg[1] & opcode_reg[2];
14 assign invalid = invalid_red_op | invalid_opcode;
23 if(if_t.rst) begin
24 cin_reg <= 0;
27 bypass_B_reg <= 0;
29 direction_reg <= 0;
30 serial_in_reg <= 0;
31 opcode_reg <= 0;
32 A_reg <= 0;
33 B_reg <= 0;
35 end else begin
36 cin_reg <= if_t.cin;
37 red_op_B_reg <= if_t.red_op_B;
38 red_op_A_reg <= if_t.red_op_A;
41 direction_reg <= if_t.direction;
42 serial_in_reg <= if_t.serial_in;
43 opcode_reg <= if_t.opcode;
44 A_reg <= if_t.A;
45 B_reg <= if_t.B;
47 end
48 end
50 always @(posedge if_t.clk or posedge if_t.rst) begin
56 else
58 end
61 always @(posedge if_t.clk or posedge if_t.rst) begin
63 if(if_t.rst) begin
68 if_t.out <= (if_t.INPUT_PRIORITY == "A")?A_reg: B_reg;
71 else if (bypass_B_reg)
75 else begin
77 3'h0: begin
79 if_t.out <= (if_t.INPUT_PRIORITY == "A")? |A_reg: |B_reg;
81 if_t.out <= |A_reg;
86 end
88 if (red_op_A_reg && red_op_B_reg)
90 else if (red_op_A_reg)
92 else if (red_op_B_reg)
96 end


Code Coverage Analysis
File Bookmarks Window Help
Code Coverage Analysis
Branches - by instance (/top/DUT)
Branch
E
ALSU.sv
28 bypass_A_reg <= 0;
40 bypass_A_reg <= if_t.bypass_A;
57 if_t.leds <= 0;
59 end
70 if_t.out <= A_reg;
73 else if (invalid)
74 if_t.out <= 0;
76 case (opcode_reg)
78 if (red_op_A_reg && red_op_B_reg)
80 else if (red_op_A_reg)
82 else if (red_op_B_reg)
84 else
85 if_t.out <= A_reg | B_reg;
87 3'h1: begin
89 if_t.out <= (if_t.INPUT_PRIORITY == "A")? ^A_reg: ^B_reg;
91 if_t.out <= ^A_reg;
94 else
95 if_t.out <= A_reg ^ B_reg;
97 3'h2: begin
99 if_t.out <= A_reg + B_reg+cin_reg;
101 if_t.out <= A_reg + B_reg;
104 end
112 default: if_t.out<=if_t.out;
113 endcase
116 out_next<=if_t.out;
119 endmodule
<NULL>
<NULL>
top.sv

coverage_report - Notepad

File Edit Format View Help

```
===============================================================================
=== Instance: /shift_coverage_pkg
=== Design Unit: work.shift_coverage_pkg
===============================================================================

Covergroup Coverage:
    Covergroups                      1         na         na   100.00%
        Coverpoints/Crosses          3         na         na        na
            Covergroup Bins         68         68          0   100.00%
----------------------------------------------------------------------------------------------------------
Covergroup                                             Metric       Goal       Bins    Status


----------------------------------------------------------------------------------------------------------
 TYPE /shift_coverage_pkg/shift_reg_cov/covgup        100.00%        100          -    Covered
    covered/total bins:                                    68         68          -
    missing/total bins:                                     0         68          -
    % Hit:                                            100.00%        100          -
    Coverpoint data_in                                100.00%        100          -    Covered
        covered/total bins:                                64         64          -
        missing/total bins:                                 0         64          -
        % Hit:                                        100.00%        100          -
        bin auto[0]                                     24403          1          -    Covered
        bin auto[1]                                      7150          1          -    Covered
        bin auto[2]                                      4225          1          -    Covered
        bin auto[3]                                      5207          1          -    Covered
        bin auto[4]                                      1326          1          -    Covered
        bin auto[5]                                       461          1          -    Covered
        bin auto[6]                                      1318          1          -    Covered
        bin auto[7]                                       174          1          -    Covered
        bin auto[8]                                       429          1          -    Covered
        bin auto[9]                                       577          1          -    Covered
        bin auto[10]                                       53          1          -    Covered
        bin auto[11]                                        6          1          -    Covered
        bin auto[12]                                      423          1          -    Covered
        bin auto[13]                                       54          1          -    Covered
        bin auto[14]                                       56          1          -    Covered
        bin auto[15]                                      152          1          -    Covered
        bin auto[16]                                      326          1          -    Covered
        bin auto[17]                                       32          1          -    Covered
        bin auto[18]                                       57          1          -    Covered
        bin auto[19]                                       32          1          -    Covered
        bin auto[20]                                        5          1          -    Covered
```


Ln 601, Col 71 100% Windows (CRLF) UTF-8

coverage_report - Notepad

File Edit Format View Help

```
        bin auto[20]                                        5          1          -    Covered
        bin auto[21]                                        3          1          -    Covered
        bin auto[22]                                        1          1          -    Covered
        bin auto[23]                                        6          1          -    Covered
        bin auto[24]                                       79          1          -    Covered
        bin auto[25]                                       23          1          -    Covered
        bin auto[26]                                       48          1          -    Covered
        bin auto[27]                                       18          1          -    Covered
        bin auto[28]                                       86          1          -    Covered
        bin auto[29]                                      242          1          -    Covered
        bin auto[30]                                      640          1          -    Covered
        bin auto[31]                                     1089          1          -    Covered
        bin auto[32]                                      996          1          -    Covered
        bin auto[33]                                      712          1          -    Covered
        bin auto[34]                                       71          1          -    Covered
        bin auto[35]                                       55          1          -    Covered
        bin auto[36]                                       68          1          -    Covered
        bin auto[37]                                       13          1          -    Covered
        bin auto[38]                                       26          1          -    Covered
        bin auto[39]                                       27          1          -    Covered
        bin auto[40]                                       25          1          -    Covered
        bin auto[41]                                       50          1          -    Covered
        bin auto[42]                                        8          1          -    Covered
        bin auto[43]                                       16          1          -    Covered
        bin auto[44]                                        4          1          -    Covered
        bin auto[45]                                       25          1          -    Covered
        bin auto[46]                                       75          1          -    Covered
        bin auto[47]                                      267          1          -    Covered
        bin auto[48]                                      154          1          -    Covered
        bin auto[49]                                      108          1          -    Covered
        bin auto[50]                                       49          1          -    Covered
        bin auto[51]                                       99          1          -    Covered
        bin auto[52]                                      337          1          -    Covered
        bin auto[53]                                      199          1          -    Covered
        bin auto[54]                                       80          1          -    Covered
        bin auto[55]                                      523          1          -    Covered
        bin auto[56]                                      635          1          -    Covered
        bin auto[57]                                      745          1          -    Covered
        bin auto[58]                                     1305          1          -    Covered
        bin auto[59]                                     1272          1          -    Covered
        bin auto[60]                                     4091          1          -    Covered
        bin auto[61]                                     5825          1          -    Covered
        bin auto[62]                                     6202          1          -    Covered
```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coverage_report - Notepad

File Edit Format View Help

```
        bin auto[62]                              6202          1          -    Covered
        bin auto[63]                             10568          1          -    Covered
    Coverpoint MODE                            100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
        bin mode0                                42886          1          -    Covered
        bin mode1                                40447          1          -    Covered
    Coverpoint direction                       100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
        bin direction0                           43349          1          -    Covered
        bin direction1                           39984          1          -    Covered
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                      14        12         2    85.71%

================================Statement Details================================

Statement Coverage for instance /shift_coverage_pkg --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File shift_coverage.sv
    8               1                    ***0***
    8               2                    ***0***
    8               3                          2
    28              1                          1
    29              1                          1
    33              1                          1
    34              1                          1
    35              1                          1
    39              1                          1
    40              1                          1
    44              1                          1
    45              1                          1
    46              1                      83334
    48              1                      83333

================================================================================
=== Instance: /shift_scoreboard_pkg
```

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```
        missing/total bins:                          0          6          -
        % Hit:                                 100.00%        100          -
    Coverpoint CB1                             100.00%        100          -    Covered
        covered/total bins:                          5          5          -
        missing/total bins:                          0          5          -
        % Hit:                                 100.00%        100          -
    Coverpoint CB2                             100.00%        100          -    Covered
        covered/total bins:                          5          5          -
        missing/total bins:                          0          5          -
        % Hit:                                 100.00%        100          -
    Coverpoint A_M                             100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
    Coverpoint CB3                             100.00%        100          -    Covered
        covered/total bins:                          7          7          -
        missing/total bins:                          0          7          -
        % Hit:                                 100.00%        100          -
    Cross corner_case                          100.00%        100          -    Covered
        covered/total bins:                         18         18          -
        missing/total bins:                          0         18          -
        % Hit:                                 100.00%        100          -
    Cross Addation                             100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
    Cross shift                                100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
    Cross shift_rotate                         100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
    Cross walkingones                          100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
    Cross invalidation                         100.00%        100          -    Covered
        covered/total bins:                          2          2          -
        missing/total bins:                          0          2          -
        % Hit:                                 100.00%        100          -
 Covergroup instance \/ALSU_coverage_pkg::ALSU_coverage::cvr_gp
```

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coverage_report - Notepad

File Edit Format View Help

```
        % Hit:                              100.00%        100          -
    Cross Addation                          100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Cross shift                             100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Cross shift_rotate                      100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Cross walkingones                       100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Cross invalidation                      100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
 Covergroup instance \/ALSU_coverage_pkg::ALSU_coverage::cvr_gp
                                            100.00%        100          -     Covered
    covered/total bins:                          63         67          -
    missing/total bins:                           4         67          -
    % Hit:                                   94.02%        100          -
    Coverpoint CIN [1]                      100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
        bin CIN0                              25143          1          -     Covered
        bin CIN1                              24857          1          -     Covered
    Coverpoint red_op_A [1]                 100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
        bin red_op_A0                         42007          1          -     Covered
        bin red_op_A1                          7993          1          -     Covered
    Coverpoint red_op_B [1]                 100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
        bin red_op_B0                         41786          1          -     Covered
```

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coverage_report - Notepad

File Edit Format View Help

```
        bin direction0                        25342          1          -     Covered
        bin direction1                        24658          1          -     Covered
    Coverpoint serial_in [1]                100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
        bin serial_in0                        24998          1          -     Covered
        bin serial_in1                        25002          1          -     Covered
    Coverpoint special [1]                  100.00%        100          -     Covered
        covered/total bins:                       6          6          -
        missing/total bins:                       0          6          -
        % Hit:                              100.00%        100          -
        bin operataions[0]                     8589          1          -     Covered
        bin operataions[1]                     8588          1          -     Covered
        bin operataions[2]                     7720          1          -     Covered
        bin operataions[3]                     7660          1          -     Covered
        bin operataions[4]                     8639          1          -     Covered
        bin operataions[5]                     8725          1          -     Covered
    Coverpoint CB1                          100.00%        100          -     Covered
        covered/total bins:                       5          5          -
        missing/total bins:                       0          5          -
        % Hit:                              100.00%        100          -
        bin A_data_0                           8210          1          -     Covered
        bin A_data_max                         7616          1          -     Covered
        bin A_data_min                         6873          1          -     Covered
        bin A_data_walkingones[1]               581          1          -     Covered
        bin A_data_walkingones[2]               604          1          -     Covered
        default bin A_data_default            16292                     -     Occurred
    Coverpoint CB2                          100.00%        100          -     Covered
        covered/total bins:                       5          5          -
        missing/total bins:                       0          5          -
        % Hit:                              100.00%        100          -
        bin B_data_0                           9111          1          -     Covered
        bin B_data_max                         7576          1          -     Covered
        bin B_data_min                         5739          1          -     Covered
        bin B_data_walkingones[1]               586          1          -     Covered
        bin B_data_walkingones[2]               567          1          -     Covered
        default bin B_data_default            16654                     -     Occurred
    Coverpoint A_M                          100.00%        100          -     Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
        bin Bins_arith[2]                      7720          1          -     Covered
```

Activate Windows
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Ln 1709, Col 44 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
        bin A_data_min                          6873          1          -    Covered
        bin A_data_walkingones[1]               581           1          -    Covered
        bin A_data_walkingones[2]               604           1          -    Covered
        default bin A_data_default              16292                    -    Occurred
    Coverpoint CB2                              100.00%       100        -    Covered
        covered/total bins:                     5             5          -
        missing/total bins:                     0             5          -
        % Hit:                                  100.00%       100        -
        bin B_data_0                            9111          1          -    Covered
        bin B_data_max                          7576          1          -    Covered
        bin B_data_min                          5739          1          -    Covered
        bin B_data_walkingones[1]               586           1          -    Covered
        bin B_data_walkingones[2]               567           1          -    Covered
        default bin B_data_default              16654                    -    Occurred
    Coverpoint A_M                              100.00%       100        -    Covered
        covered/total bins:                     2             2          -
        missing/total bins:                     0             2          -
        % Hit:                                  100.00%       100        -
        bin Bins_arith[2]                       7720          1          -    Covered
        bin Bins_arith[3]                       7660          1          -    Covered
    Coverpoint CB3                              100.00%       100        -    Covered
        covered/total bins:                     7             7          -
        missing/total bins:                     0             7          -
        % Hit:                                  100.00%       100        -
        illegal_bin Bins_invalid                79                       -    Occurred
        bin Bins_shift[4]                       8639          1          -    Covered
        bin Bins_shift[5]                       8725          1          -    Covered
        bin Bins_arith[2]                       7720          1          -    Covered
        bin Bins_arith[3]                       7660          1          -    Covered
        bin Bins_bitwise[0]                     8589          1          -    Covered
        bin Bins_bitwise[1]                     8588          1          -    Covered
        bin Bins_trans                          10            1          -    Covered
    Cross corner_case                           100.00%       100        -    Covered
        covered/total bins:                     18            18         -
        missing/total bins:                     0             18         -
        % Hit:                                  100.00%       100        -
        Auto, Default and User Defined Bins:
            bin <Bins_arith[3],B_data_min,A_data_min>
                                                95            1          -    Covered
            bin <Bins_arith[2],B_data_min,A_data_min>
                                                94            1          -    Covered
            bin <Bins_arith[3],B_data_min,A_data_max>
                                                107           1          -    Covered
```

Activate Windows
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File Edit Format View Help

```
        Auto, Default and User Defined Bins:
            bin <Bins_arith[3],B_data_min,A_data_min>
                                                95            1          -    Covered
            bin <Bins_arith[2],B_data_min,A_data_min>
                                                94            1          -    Covered
            bin <Bins_arith[3],B_data_min,A_data_max>
                                                107           1          -    Covered
            bin <Bins_arith[3],B_data_min,A_data_0>
                                                104           1          -    Covered
            bin <Bins_arith[2],B_data_min,A_data_max>
                                                111           1          -    Covered
            bin <Bins_arith[2],B_data_min,A_data_0>
                                                101           1          -    Covered
            bin <Bins_arith[3],B_data_max,A_data_min>
                                                119           1          -    Covered
            bin <Bins_arith[3],B_data_0,A_data_min>
                                                124           1          -    Covered
            bin <Bins_arith[2],B_data_max,A_data_min>
                                                110           1          -    Covered
            bin <Bins_arith[2],B_data_0,A_data_min>
                                                130           1          -    Covered
            bin <Bins_arith[3],B_data_max,A_data_max>
                                                939           1          -    Covered
            bin <Bins_arith[3],B_data_0,A_data_max>
                                                116           1          -    Covered
            bin <Bins_arith[3],B_data_max,A_data_0>
                                                144           1          -    Covered
            bin <Bins_arith[3],B_data_0,A_data_0>       950           1          -    Covered
            bin <Bins_arith[2],B_data_max,A_data_max>
                                                970           1          -    Covered
            bin <Bins_arith[2],B_data_0,A_data_max>
                                                109           1          -    Covered
            bin <Bins_arith[2],B_data_max,A_data_0>
                                                140           1          -    Covered
            bin <Bins_arith[2],B_data_0,A_data_0>       994           1          -    Covered
        Illegal and Ignore Bins:
            ignore_bin walkingB                 129                      -    Occurred
            ignore_bin walkingA                 108                      -    Occurred
    Cross Addation                              100.00%       100        -    Covered
        covered/total bins:                     2             2          -
        missing/total bins:                     0             2          -
        % Hit:                                  100.00%       100        -
        Auto, Default and User Defined Bins:
```

Activate Windows
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coverage_report - Notepad

File Edit Format View Help

```
        bin <Bins_arith[2],B_data_0,A_data_max>
                                              109          1          -    Covered
        bin <Bins_arith[2],B_data_max,A_data_0>
                                              140          1          -    Covered
        bin <Bins_arith[2],B_data_0,A_data_0>  994          1          -    Covered
    Illegal and Ignore Bins:
        ignore_bin walkingB                   129                     -    Occurred
        ignore_bin walkingA                   108                     -    Occurred
 Cross Addation                           100.00%        100          -    Covered
    covered/total bins:                         2          2          -
    missing/total bins:                         0          2          -
    % Hit:                                100.00%        100          -
    Auto, Default and User Defined Bins:
        bin Add_cin0                         3867          1          -    Covered
        bin Add_cin1                         3853          1          -    Covered
 Cross shift                              100.00%        100          -    Covered
    covered/total bins:                         2          2          -
    missing/total bins:                         0          2          -
    % Hit:                                100.00%        100          -
    Auto, Default and User Defined Bins:
        bin shift_Si0                        4340          1          -    Covered
        bin shift_Si1                        4340          1          -    Covered
 Cross shift_rotate                       100.00%        100          -    Covered
    covered/total bins:                         2          2          -
    missing/total bins:                         0          2          -
    % Hit:                                100.00%        100          -
    Auto, Default and User Defined Bins:
        bin shu_rot_d0                       8846          1          -    Covered
        bin shu_rot_d1                       8518          1          -    Covered
 Cross walkingones                        100.00%        100          -    Covered
    covered/total bins:                         2          2          -
    missing/total bins:                         0          2          -
    % Hit:                                100.00%        100          -
    Auto, Default and User Defined Bins:
        bin arithA                            104          1          -    Covered
        bin arithB                             47          1          -    Covered
 Cross invalidation                       100.00%        100          -    Covered
    covered/total bins:                         2          2          -
    missing/total bins:                         0          2          -
    % Hit:                                100.00%        100          -
    Auto, Default and User Defined Bins:
        bin ROpA_notXoR                      6314          1          -    Covered
        bin ROpB_notXoR                      6422          1          -    Covered
```

Activate Windows
Go to Settings to activate Windows.

Ln 1846, Col 67 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
    bin auto[1]                              7150          1          -    Covered
    bin auto[2]                              4225          1          -    Covered
    bin auto[3]                              5207          1          -    Covered
    bin auto[4]                              1326          1          -    Covered
    bin auto[5]                               461          1          -    Covered
    bin auto[6]                              1318          1          -    Covered
    bin auto[7]                               174          1          -    Covered
    bin auto[8]                               429          1          -    Covered
    bin auto[9]                               577          1          -    Covered
    bin auto[10]                               53          1          -    Covered
    bin auto[11]                                6          1          -    Covered
    bin auto[12]                              423          1          -    Covered
    bin auto[13]                               54          1          -    Covered
    bin auto[14]                               56          1          -    Covered
    bin auto[15]                              152          1          -    Covered
    bin auto[16]                              326          1          -    Covered
    bin auto[17]                               32          1          -    Covered
    bin auto[18]                               57          1          -    Covered
    bin auto[19]                               32          1          -    Covered
    bin auto[20]                                5          1          -    Covered
    bin auto[21]                                3          1          -    Covered
    bin auto[22]                                1          1          -    Covered
    bin auto[23]                                6          1          -    Covered
    bin auto[24]                               79          1          -    Covered
    bin auto[25]                               23          1          -    Covered
    bin auto[26]                               48          1          -    Covered
    bin auto[27]                               18          1          -    Covered
    bin auto[28]                               86          1          -    Covered
    bin auto[29]                              242          1          -    Covered
    bin auto[30]                              640          1          -    Covered
    bin auto[31]                             1089          1          -    Covered
    bin auto[32]                              996          1          -    Covered
    bin auto[33]                              712          1          -    Covered
    bin auto[34]                               71          1          -    Covered
    bin auto[35]                               55          1          -    Covered
    bin auto[36]                               68          1          -    Covered
    bin auto[37]                               13          1          -    Covered
    bin auto[38]                               26          1          -    Covered
    bin auto[39]                               27          1          -    Covered
    bin auto[40]                               25          1          -    Covered
    bin auto[41]                               50          1          -    Covered
    bin auto[42]                                8          1          -    Covered
    bin auto[43]                               16          1          -    Covered
```

Activate Windows
Go to Settings to activate Windows.

Ln 1846, Col 67 | 100% | Windows (CRLF) | UTF-8

coverage_report - Notepad

File Edit Format View Help

```
        % Hit:                              100.00%        100          -
    Coverpoint serial_in                      0.00%        100          -    ZERO
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Coverpoint special                        0.00%        100          -    ZERO
        covered/total bins:                       6          6          -
        missing/total bins:                       0          6          -
        % Hit:                              100.00%        100          -
    Coverpoint CB1                          100.00%        100          -    Covered
        covered/total bins:                       5          5          -
        missing/total bins:                       0          5          -
        % Hit:                              100.00%        100          -
    Coverpoint CB2                          100.00%        100          -    Covered
        covered/total bins:                       5          5          -
        missing/total bins:                       0          5          -
        % Hit:                              100.00%        100          -
    Coverpoint A_M                          100.00%        100          -    Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Coverpoint CB3                          100.00%        100          -    Covered
        covered/total bins:                       7          7          -
        missing/total bins:                       0          7          -
        % Hit:                              100.00%        100          -
    Cross corner_case                       100.00%        100          -    Covered
        covered/total bins:                      18         18          -
        missing/total bins:                       0         18          -
        % Hit:                              100.00%        100          -
    Cross Addation                          100.00%        100          -    Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Cross shift                             100.00%        100          -    Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Cross shift_rotate                      100.00%        100          -    Covered
        covered/total bins:                       2          2          -
        missing/total bins:                       0          2          -
        % Hit:                              100.00%        100          -
    Cross walkingones                       100.00%        100          -    Covered
        covered/total bins:                       2          2          -
```


Ln 2365, Col 51 | 100% | Windows (CRLF) | UTF-8